

8

7

6

5

4

3

2

1

VERTEX
SYSTEM 4.0

01/22/16

04-S28-2014

D

C

B

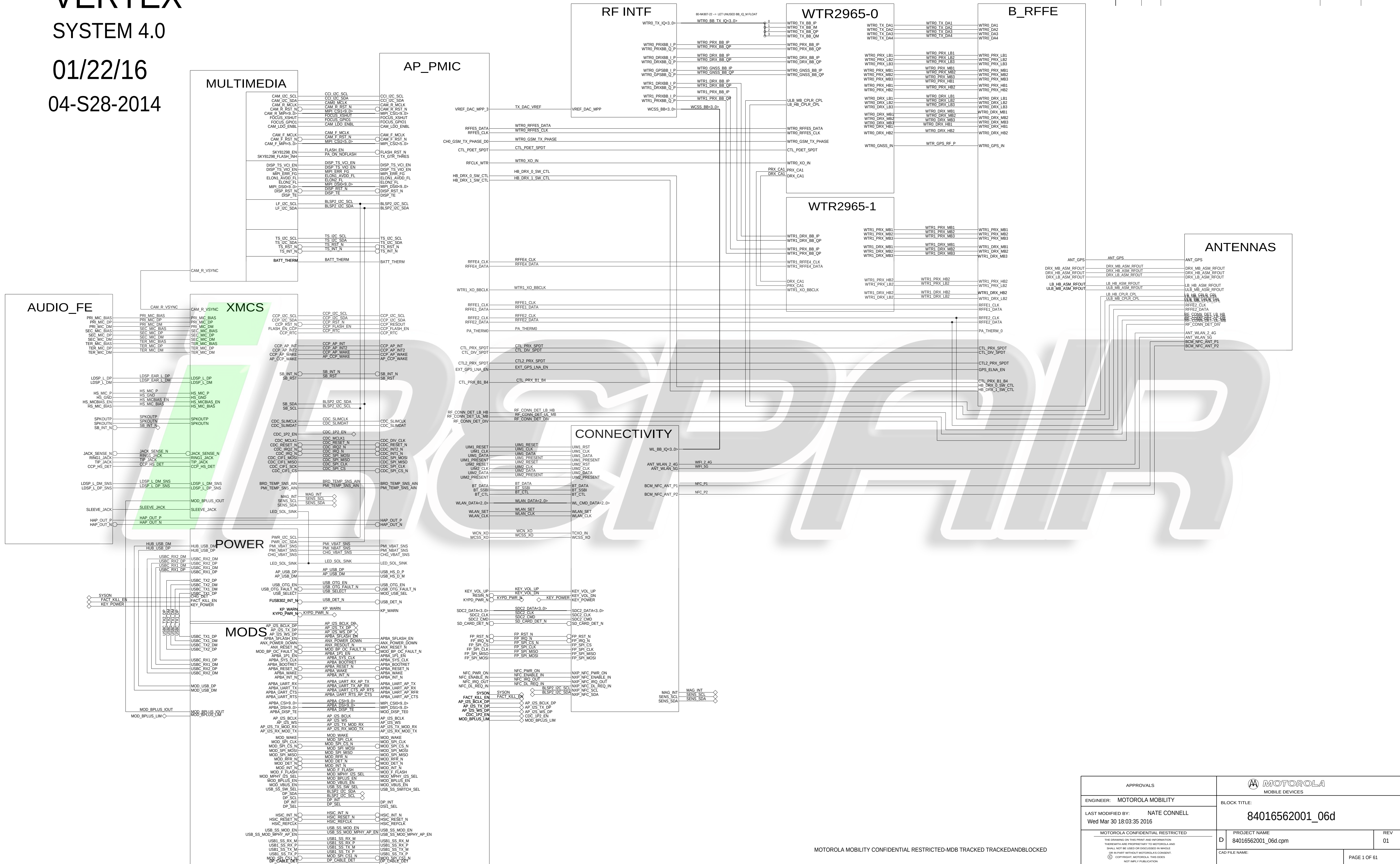
A

D

C

B

A



8

7

6

5

4

3

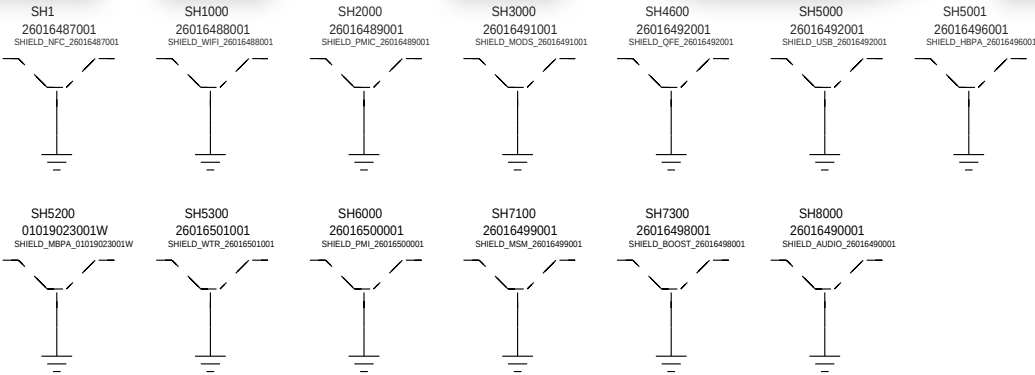
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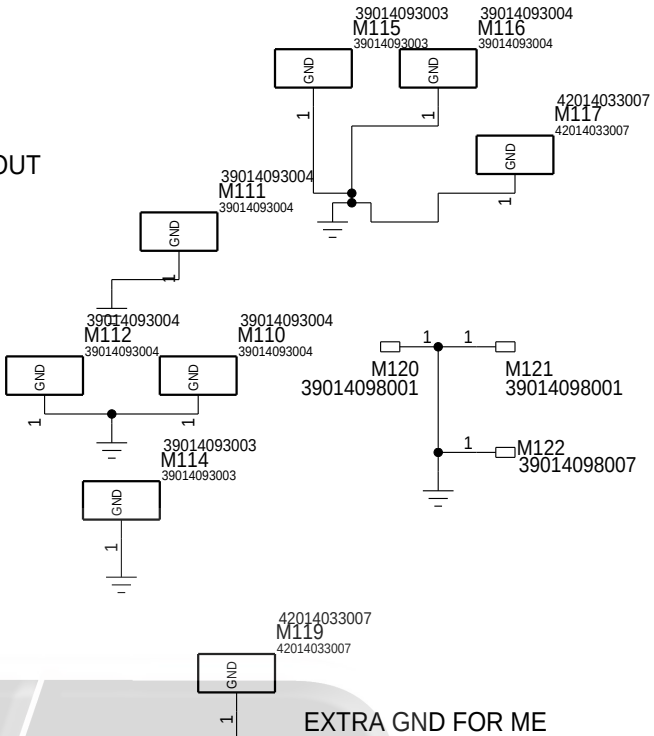
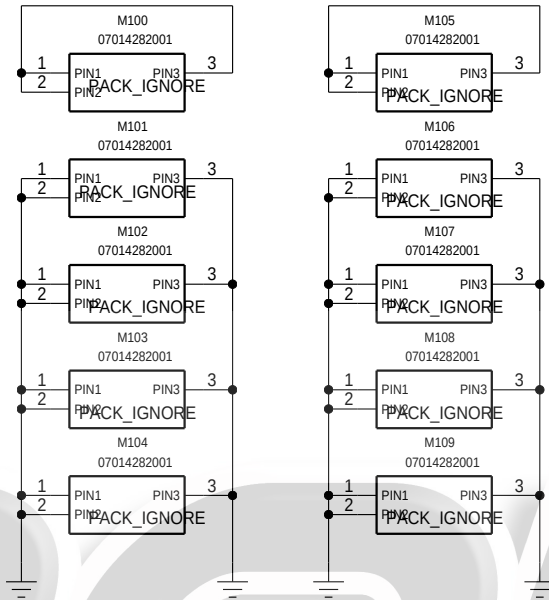
VERTEX ME/SHIELDS

JANUS REFERENCE DESIGNATOR RANGES BY BLOCK

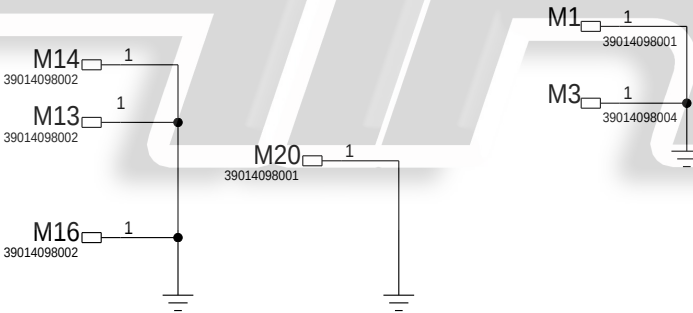
- 000 - 499: ANTENNA
- 500 - 599: USER INPUT
- 600 - 799: POWER
- 1000-2999: MSM8852 + PM8952
- 3000-3049, 3100-3149: WTR2955
- 3050-3099, 3150-3199: CA_WTR2955
- 3200-3999: MSM8952 + PM8952
- 4000-4499: CONNECTIVITY
- 4500-4999: AUDIO FE
- 5000-5499: DISPLAY
- 5500-5999: CAMERA
- 6000-6999: XMCS
- 7000-8999: RFFE



PLACE M100 AND M105 NEAR DIVERSITY ANTENNA KEEP OUT



UNIVERSAL GROUND CONTACTS



MOTOROLA MOBILITY CONFIDENTIAL RESTRICTED-MDB TRACKED TRACKEDANDBLOCKED

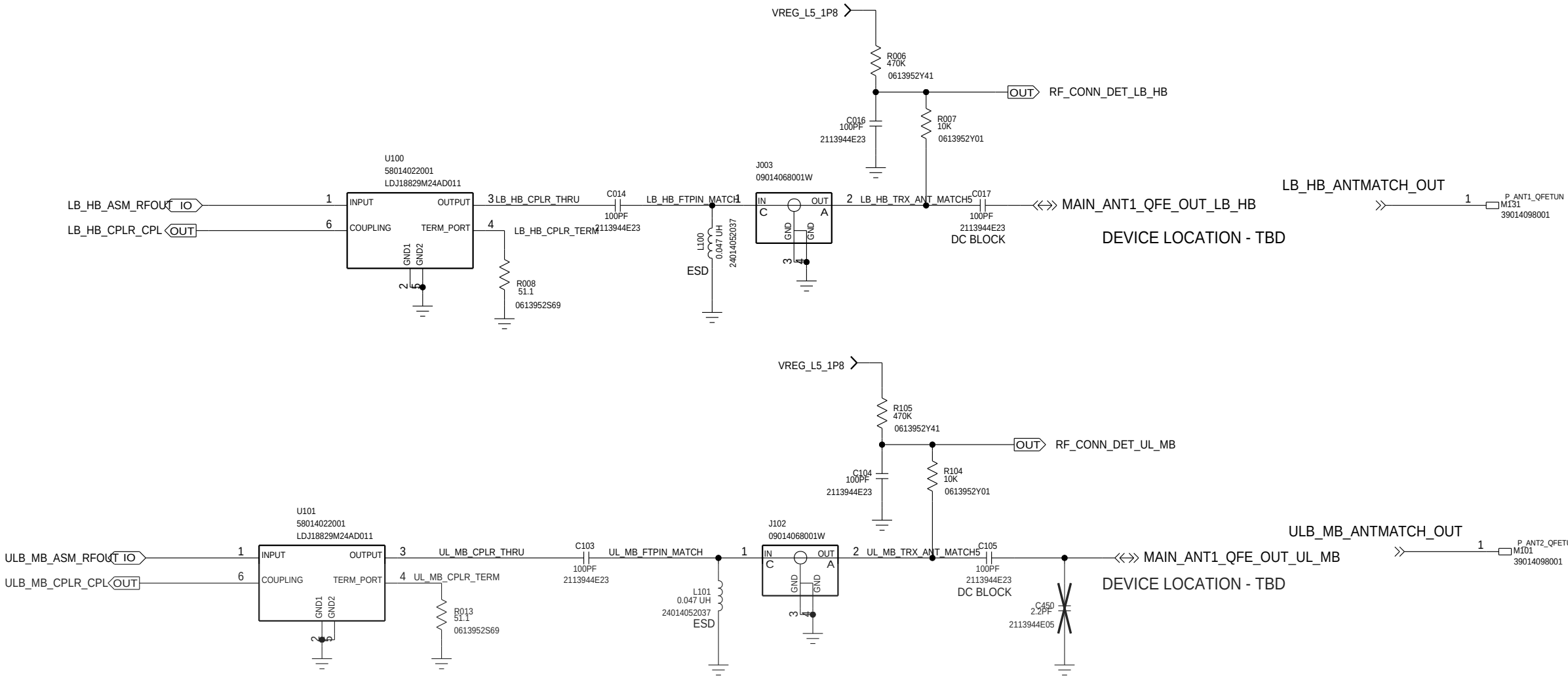
APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY:		84016562001_06d	
MOTOROLA CONFIDENTIAL RESTRICTED		PROJECT NAME	REV
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		CAD FILE NAME:	PAGE 2 OF 61

ANT: ANTENNA
REF:100-130

REVISIONS				
ZONE	LTR	DESCRIPTION	DATE	APPR.

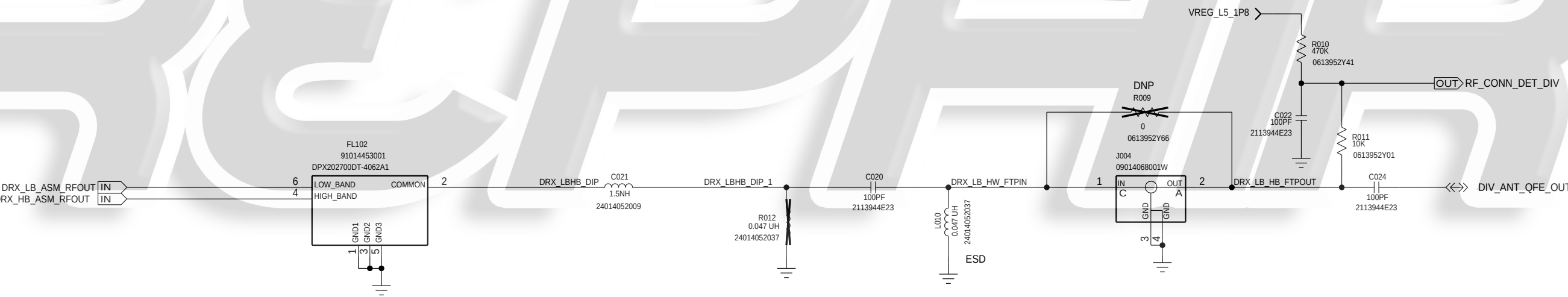
MAIN ANTENNA

DEVICE LOCATION - BOTTOM RIGHT
(GSM/UTMS/LTE)

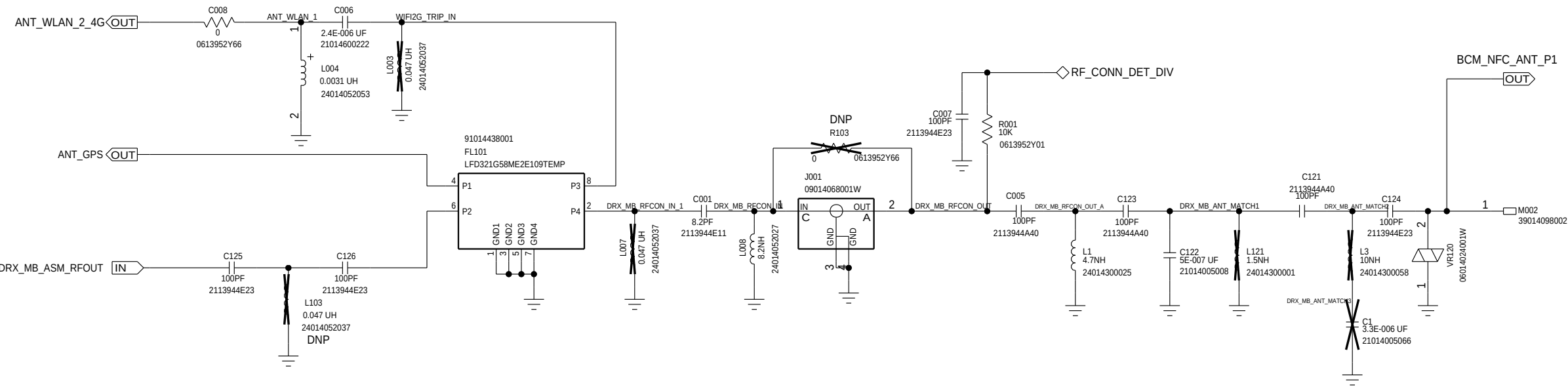


DRX LOW / HIGH BAND ANTENNA

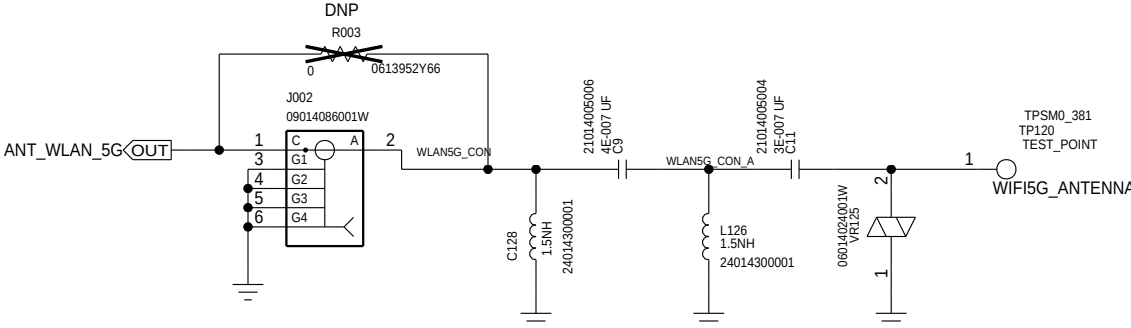
DEVICE LOCATION - TOP LEFT
(GSM/UTMS/LTE)



DRX MID BAND/GPS/BT/WLAN ANTENNA



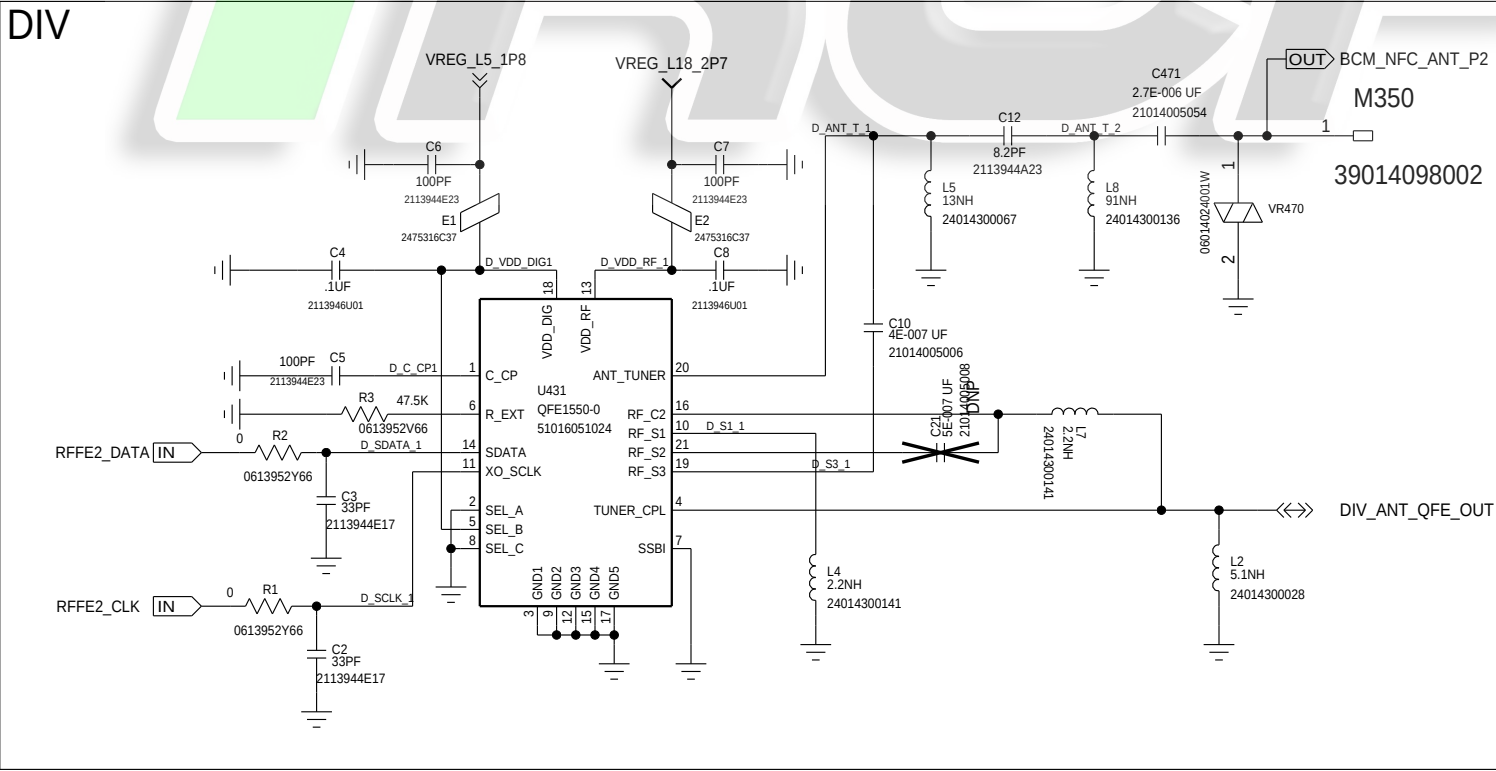
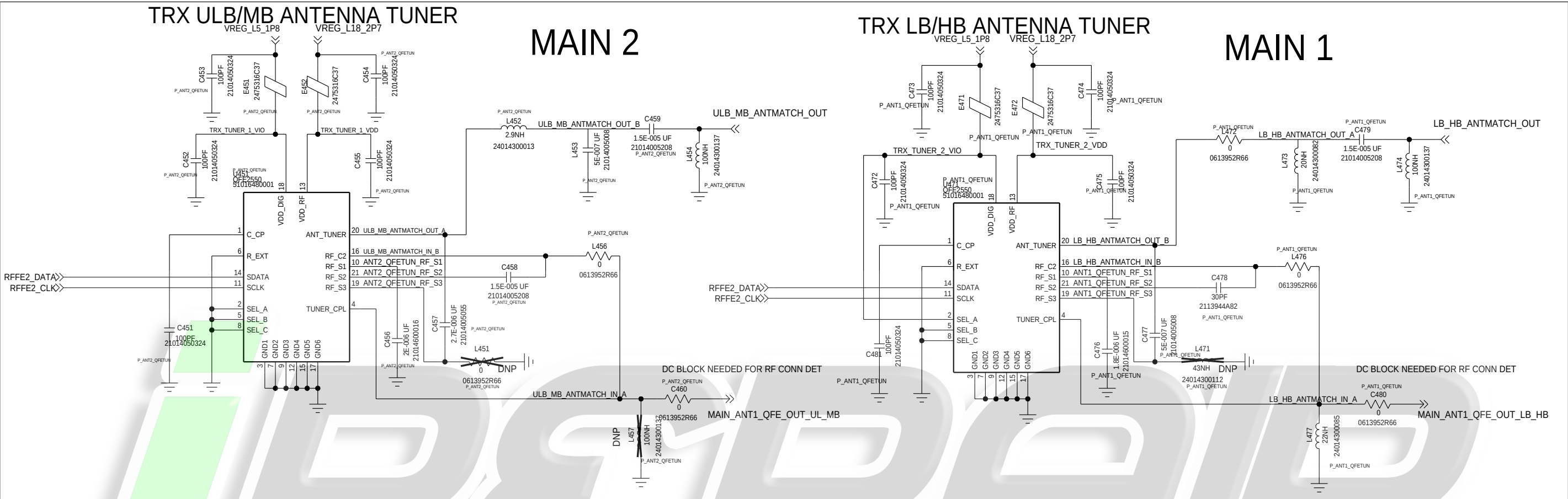
WIFI 5G ANTENNA



APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER:	ADDISON ANTENNA TEAM	BLOCK TITLE: antennas	
LAST MODIFIED BY:	LEO Mon Apr 18 13:32:11 2016	PROJECT NAME C 84016562001_06d.cpm	
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ANT: QCOM TUNERS

REFDES: 400-499

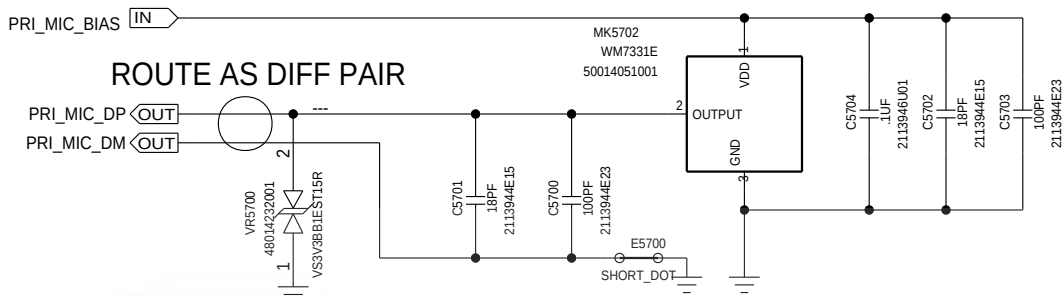


APPROVALS		 MOTOROLA MOBILE DEVICES		
ENGINEER: ADDISON ANTENNA TEAM		BLOCK TITLE: antennas		
LAST MODIFIED BY: LEO Mon Apr 18 13:12:55 2016				
MOTOROLA CONFIDENTIAL RESTRICTED THE DRAWING ON THIS PRINT AND INFORMATION THEREWITH ARE PROPRIETARY TO MOTOROLA AND SHALL NOT BE USED OR DISCUSSED IN WHOLE OR IN PART WITHOUT MOTOROLA'S CONSENT. © COPYRIGHT, MOTOROLA. THIS DOES NOT IMPLY PUBLICATION		PROJECT NAME		REV
		B 84016562001_06d.cpm		01
		CAD FILE NAME:		PAGE 4 OF 61

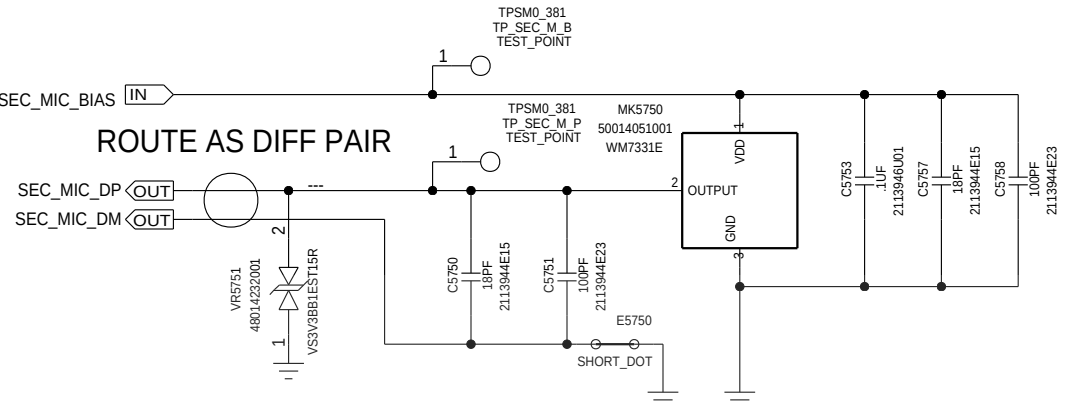
AFE: MICS

REF: 5700-5799

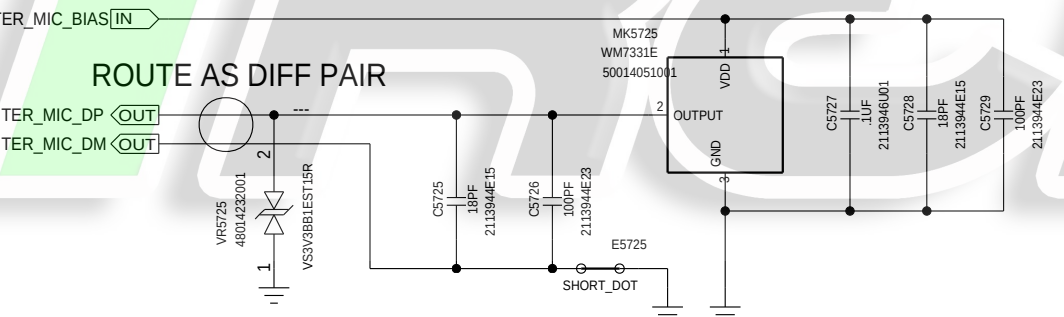
PRIMARY MIC
REF: 5700-5709



SECONDARY MIC
REF:5750-5759



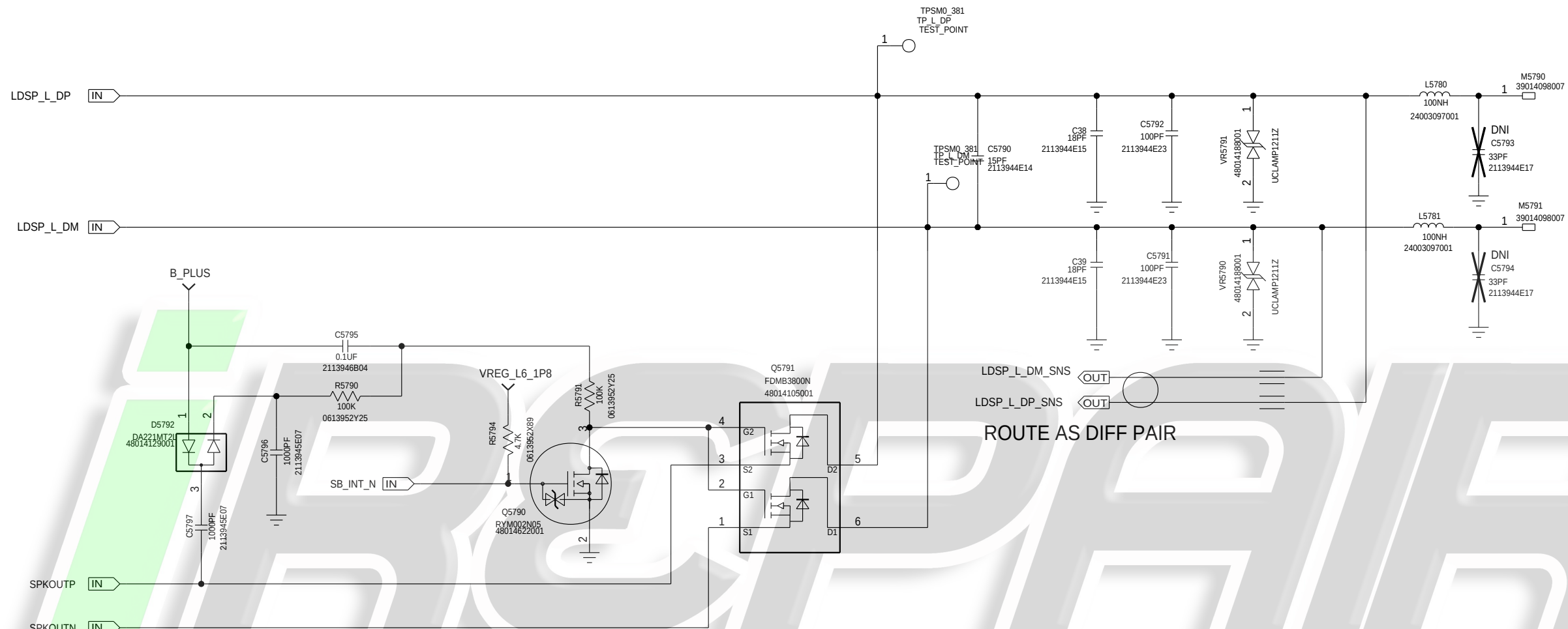
TERTIARY MIC
REF: 5725-5730



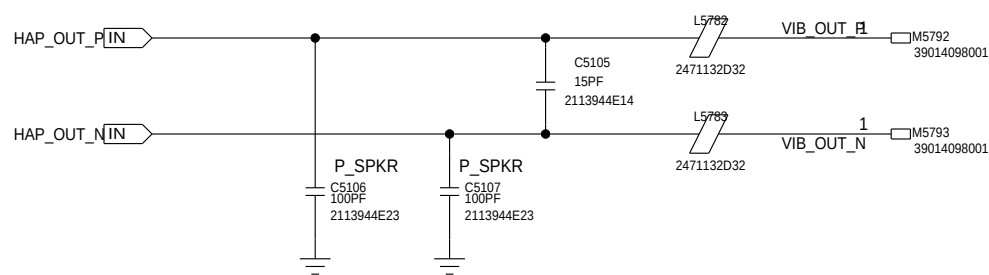
APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: DENNIS BUDNICK		BLOCK TITLE:	
LAST MODIFIED BY: NATE CONNELL Tue Feb 23 12:16:56 2016		audio_fe	
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		CAD FILE NAME: PAGE 5 OF 61	

AFE: SPEAKERS

LOUDSPEAKER/EAR PIECE
REF:5790-5799



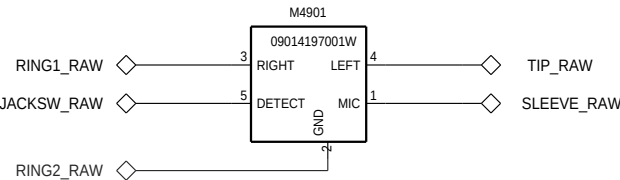
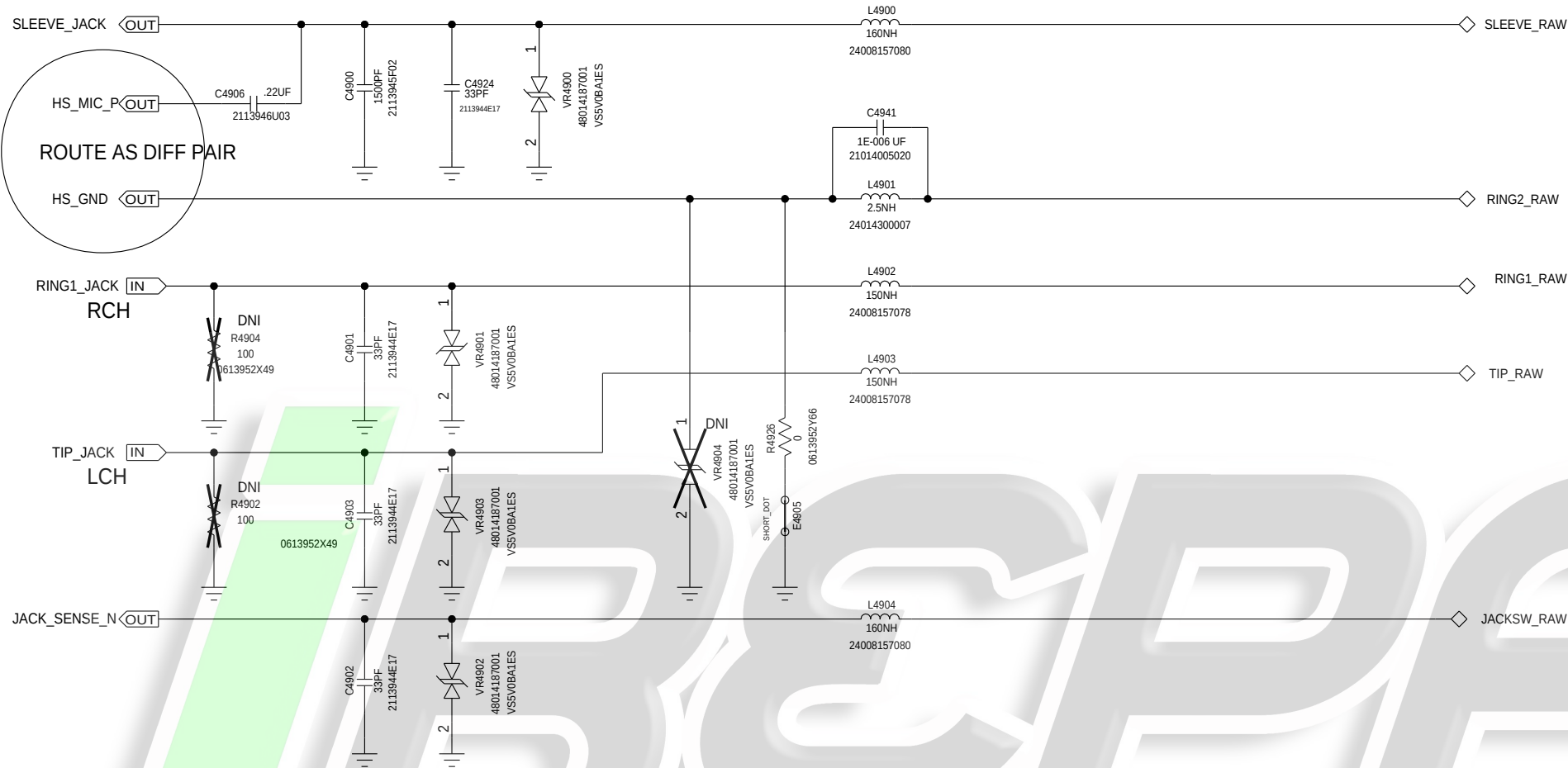
LINEAR VIBRATOR
REF 5100-5119



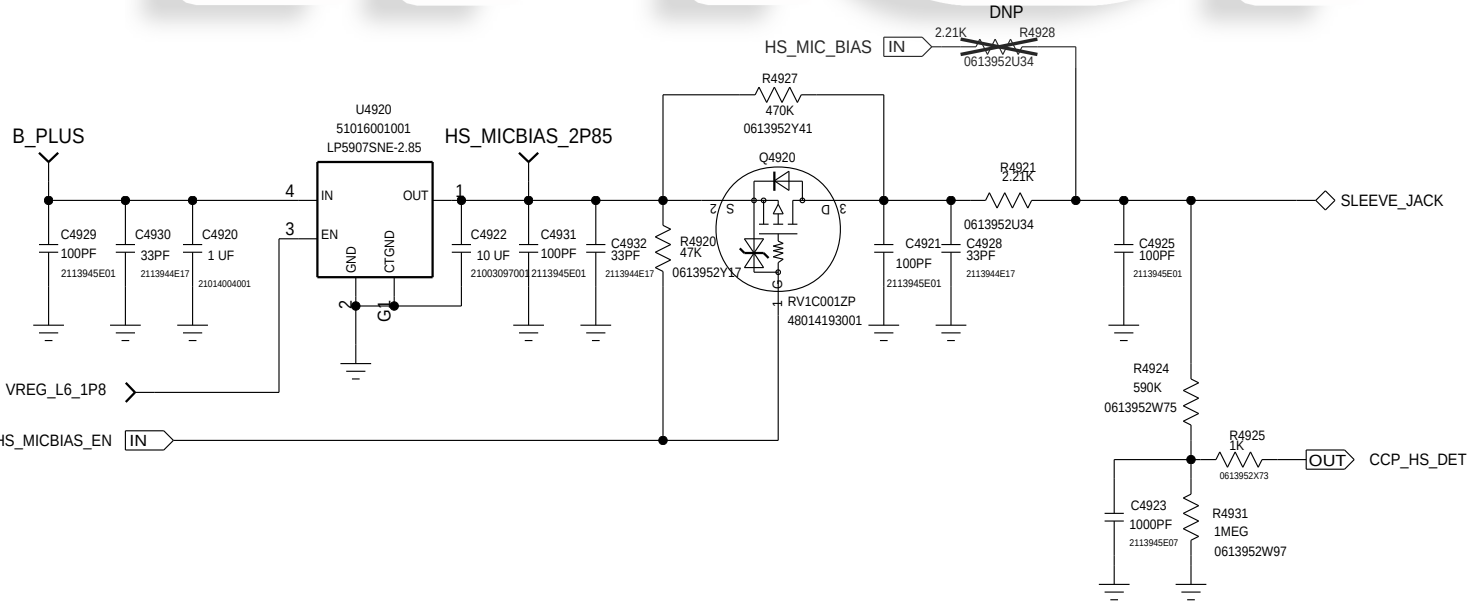
APPROVALS		 MOTOROLA MOBILE DEVICES		
ENGINEER: DENNIS BUDNICK		BLOCK TITLE: audio_fe		
LAST MODIFIED BY: TIM MILLER Mon Apr 11 11:44:26 2016				
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		CAD FILE NAME:		PAGE 6 OF 61

AFE: 3.5MM HEADSET

3.5MM HSJ
REF:4900-4949



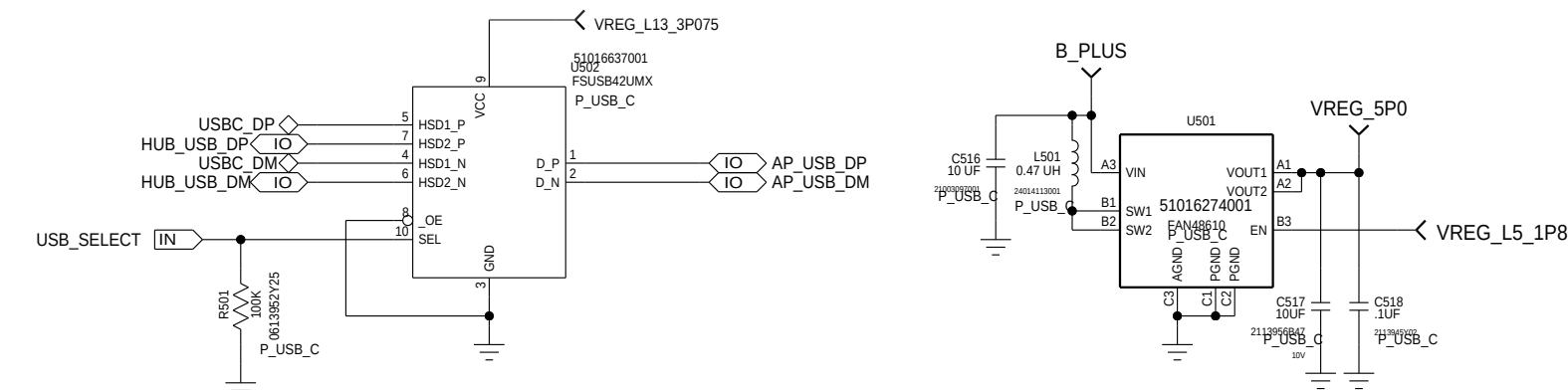
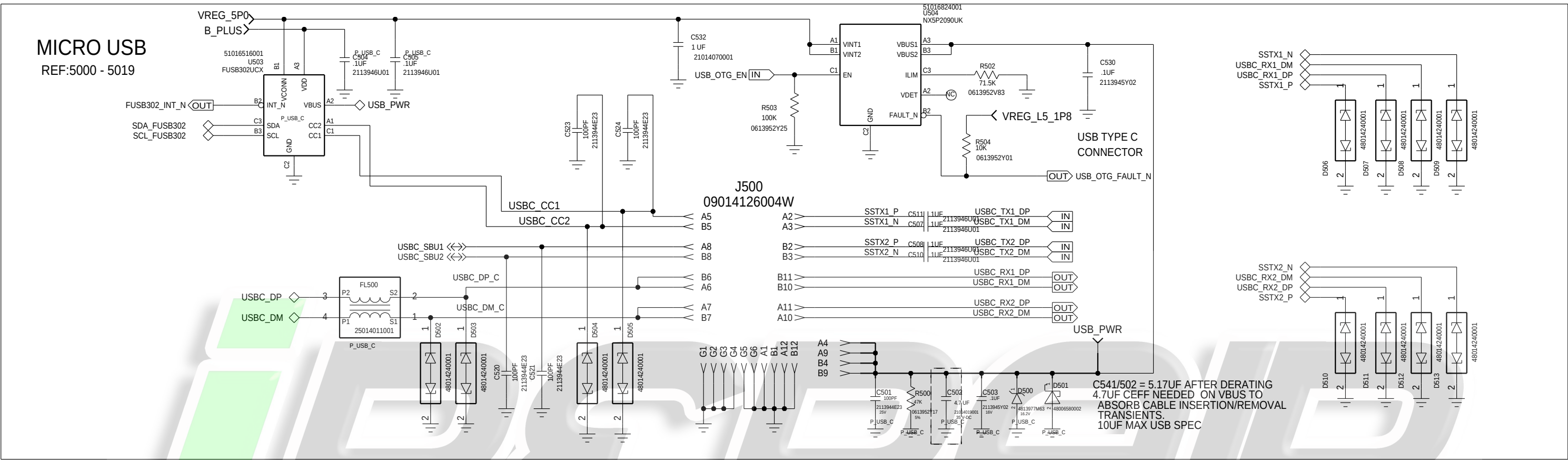
SENSOR HUB HEADSET DETECTION CIRCUIT



APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: MMI		BLOCK TITLE: audio_fe	
LAST MODIFIED BY: MMI Thu Mar 03 15:48:18 2016		PROJECT NAME 84016562001_06d.cpm	
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		PAGE 7 OF 61	

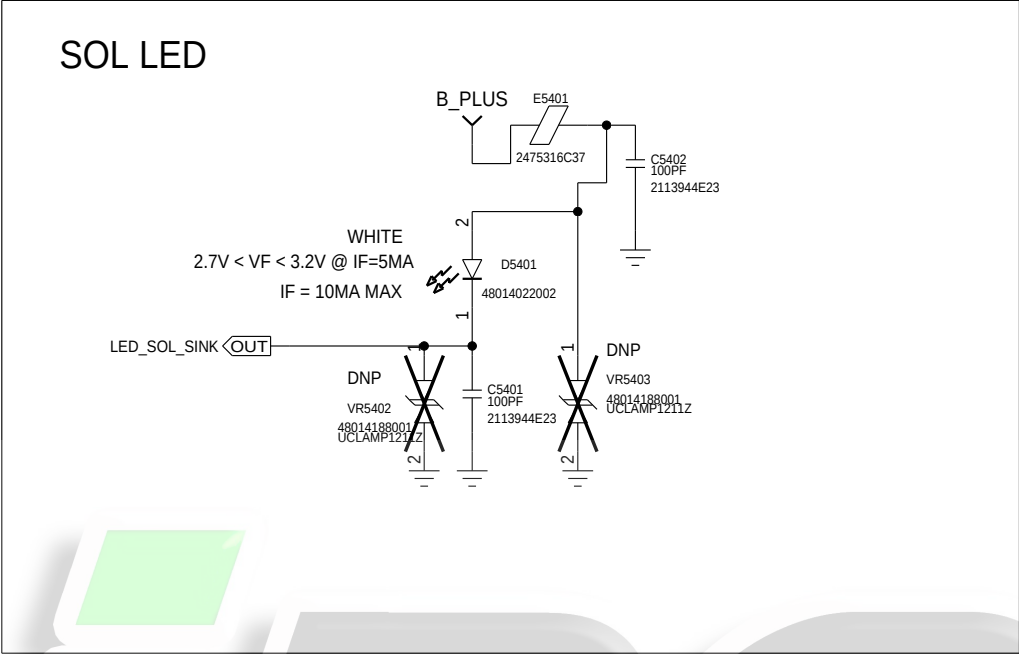
PWR: USB-C

REF:5000 - 5219



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: RICH NG		BLOCK TITLE: power	
LAST MODIFIED BY: NATE CONNELL Wed Mar 30 18:17:00 2016		PROJECT NAME 84016562001_06d.cpm	
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		PAGE 8 OF 61	

PWR: LIGHTING



REPAIR

MOTOROLA MOBILITY CONFIDENTIAL RESTRICTED-MDB TRACKED TRACKEDANDBLOCKED

APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY: TIM MILLER Fri Feb 26 11:36:26 2016		power	
MOTOROLA CONFIDENTIAL RESTRICTED THE DRAWING ON THIS PRINT AND INFORMATION THEREWITH ARE PROPRIETARY TO MOTOROLA AND SHALL NOT BE USED OR DISCUSSED IN WHOLE OR IN PART WITHOUT MOTOROLA'S CONSENT. © COPYRIGHT, MOTOROLA. THIS DOES NOT IMPLY PUBLICATION		PROJECT NAME	
		B	84016562001_06d.cpm
		CAD FILE NAME:	
		REV 01	
		PAGE 9 OF 61	

PWR: BATT

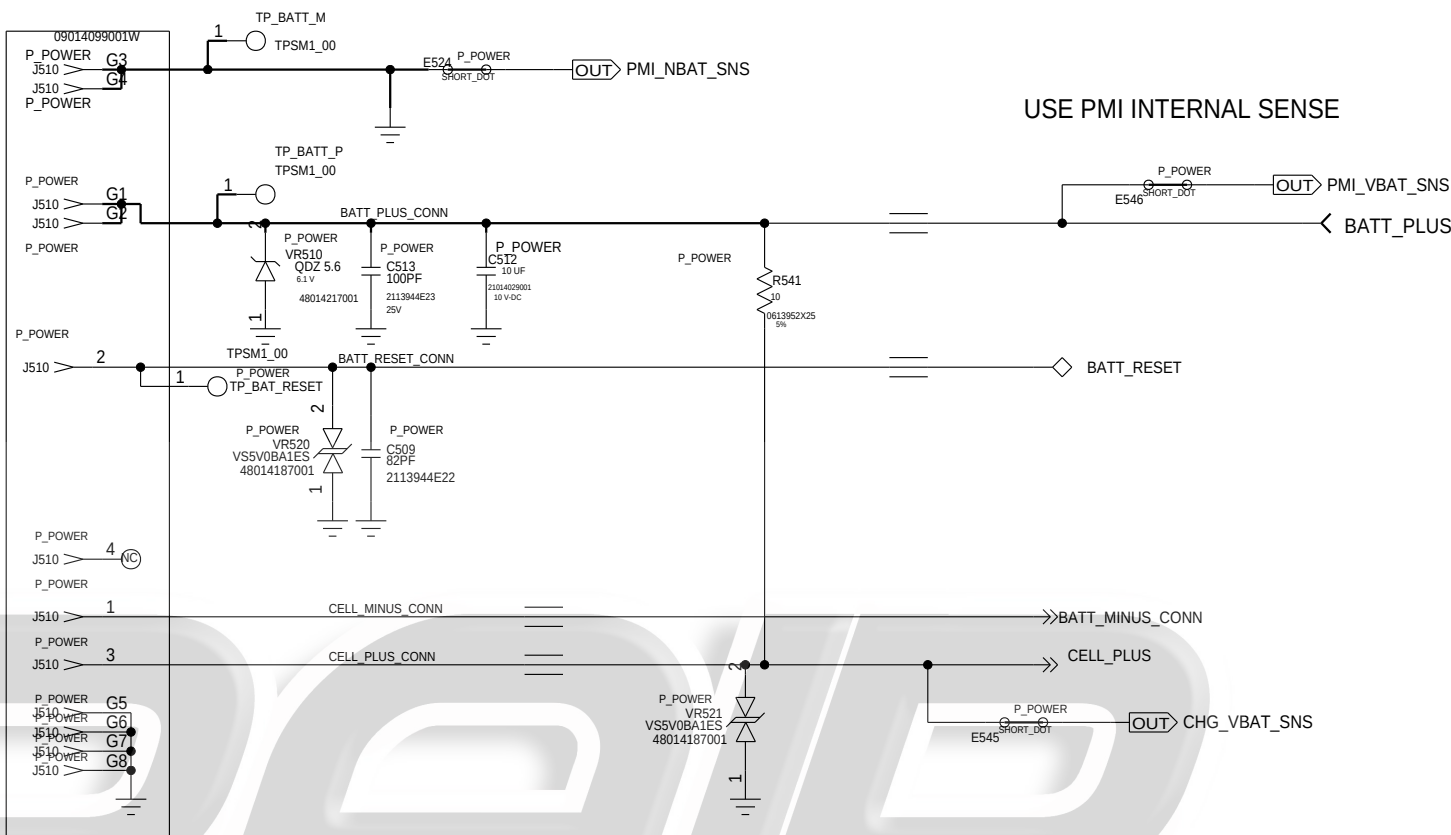
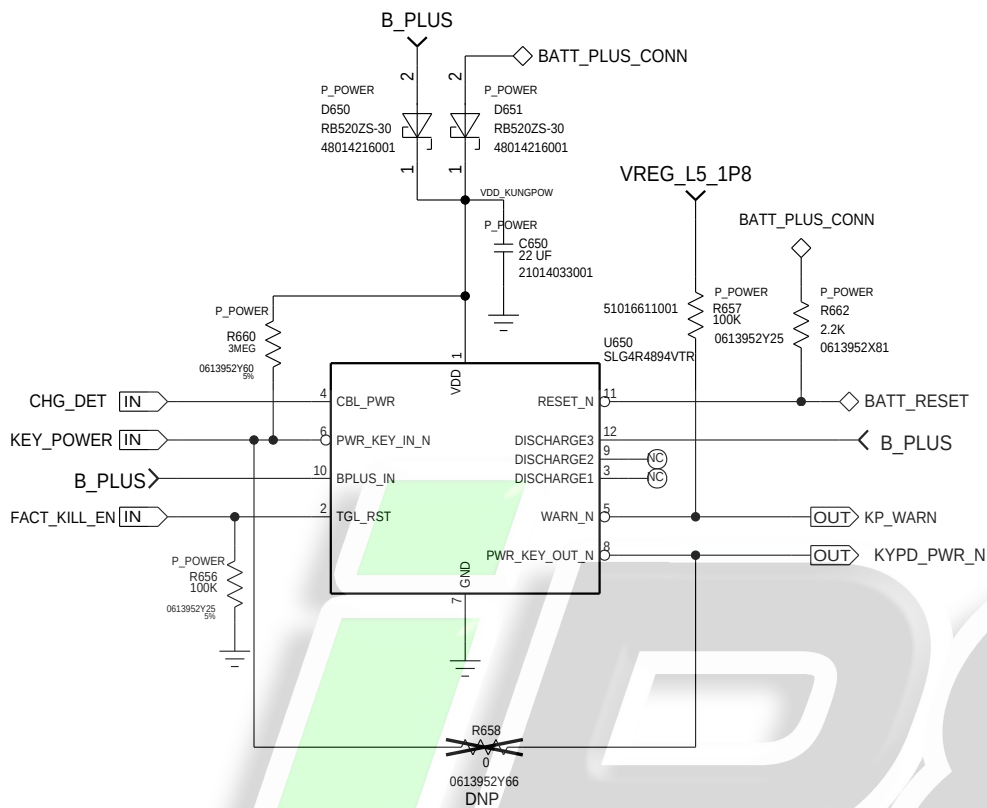
REF:5000 - 5219

BATTERY BLOCK

REF:5200 - 5209

PARALLEL BATTERY BLOCK FOR BATTERY ELIMINATOR

KUNG KOW LITE BATTERY DISCONNECT



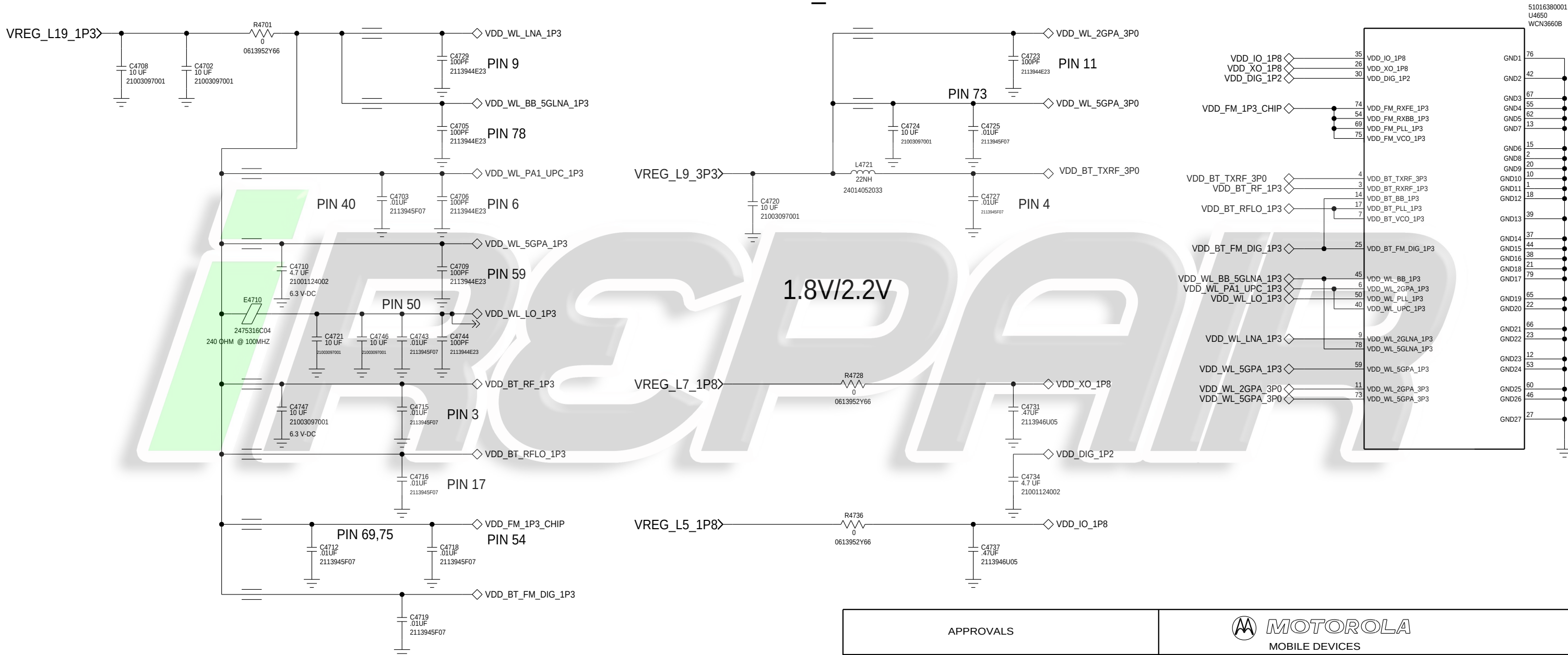
APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: RICH NG		BLOCK TITLE: power	
LAST MODIFIED BY: NATE CONNELL Wed Feb 24 17:25:07 2016		PROJECT NAME B 84016562001_06d.cpm	
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WCN3660B POWER
REF 4700-4799

NEED TO SCRUB ALL SUPPLY SEQUENCE/VOLTAGE/CURRENT NUMBERS FOR WCN3660B

BT_FM_WLAN_1P3

WLBT_3.0V



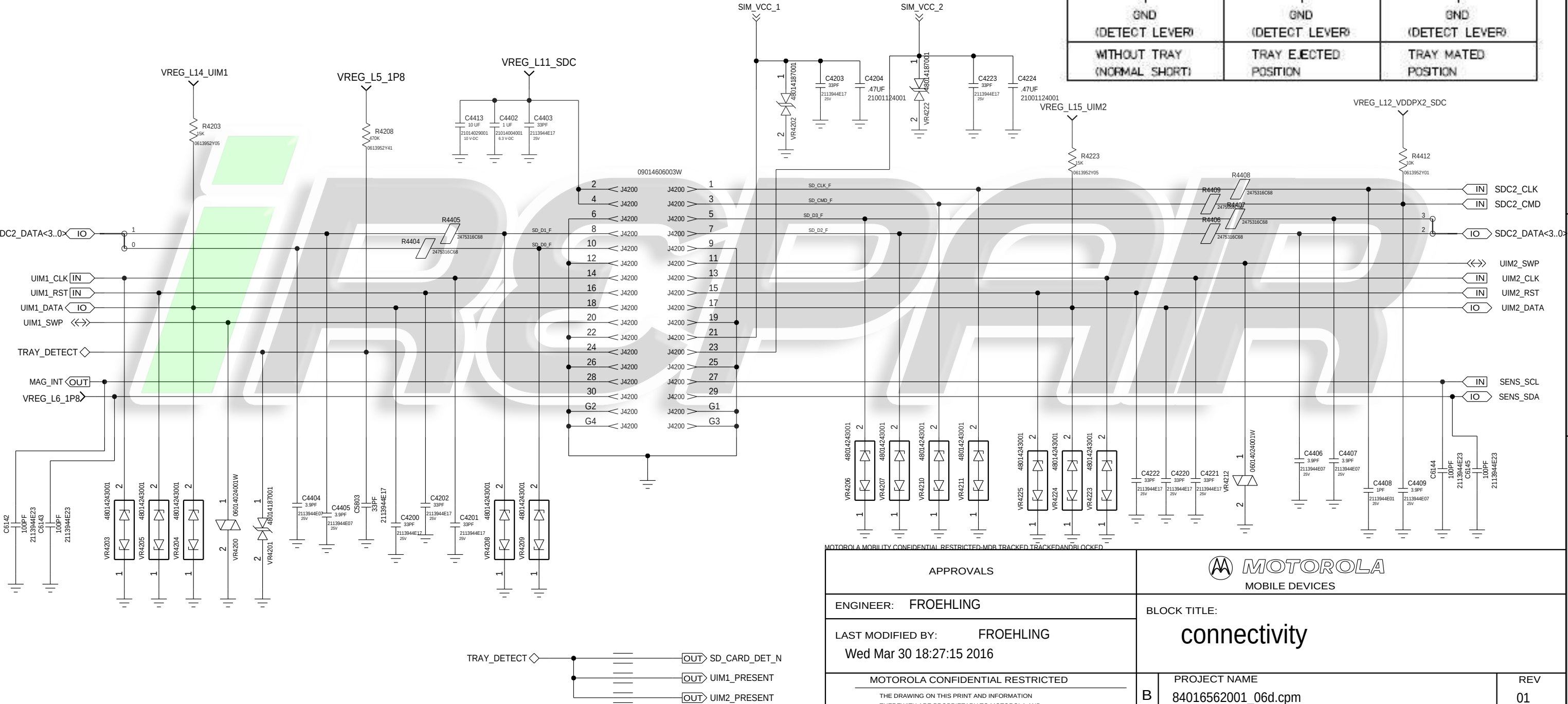
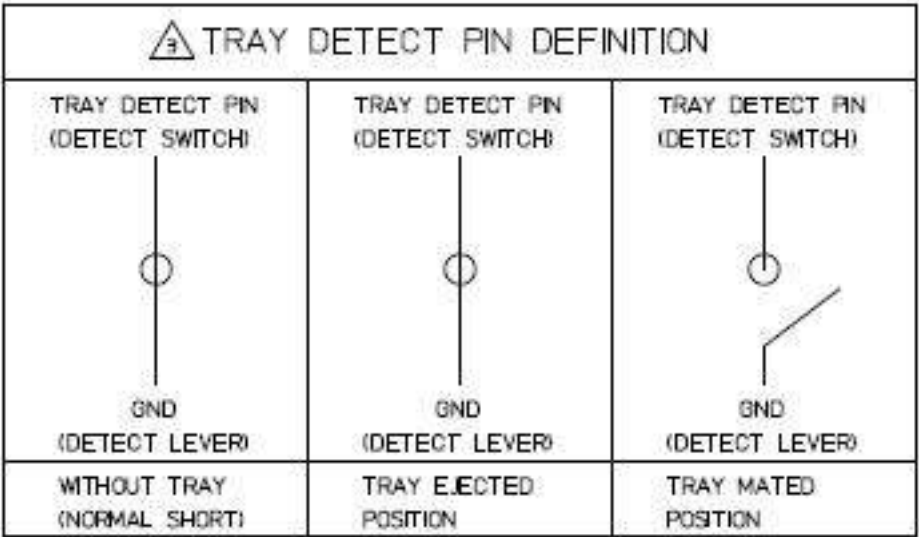
NOTE: RESERVE C4717 AND C4718 FOR 0.01UF BYPASSES

BASED ON 80-WL006-44 REV C

APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY: FROEHLING Tue Feb 23 12:17:04 2016		connectivity	
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		B 84016562001_06d.cpm	REV 01
		CAD FILE NAME:	
		PAGE 12 OF 61	

CONN: UIM1/UIM2/SDCARD

REF:4200-4219



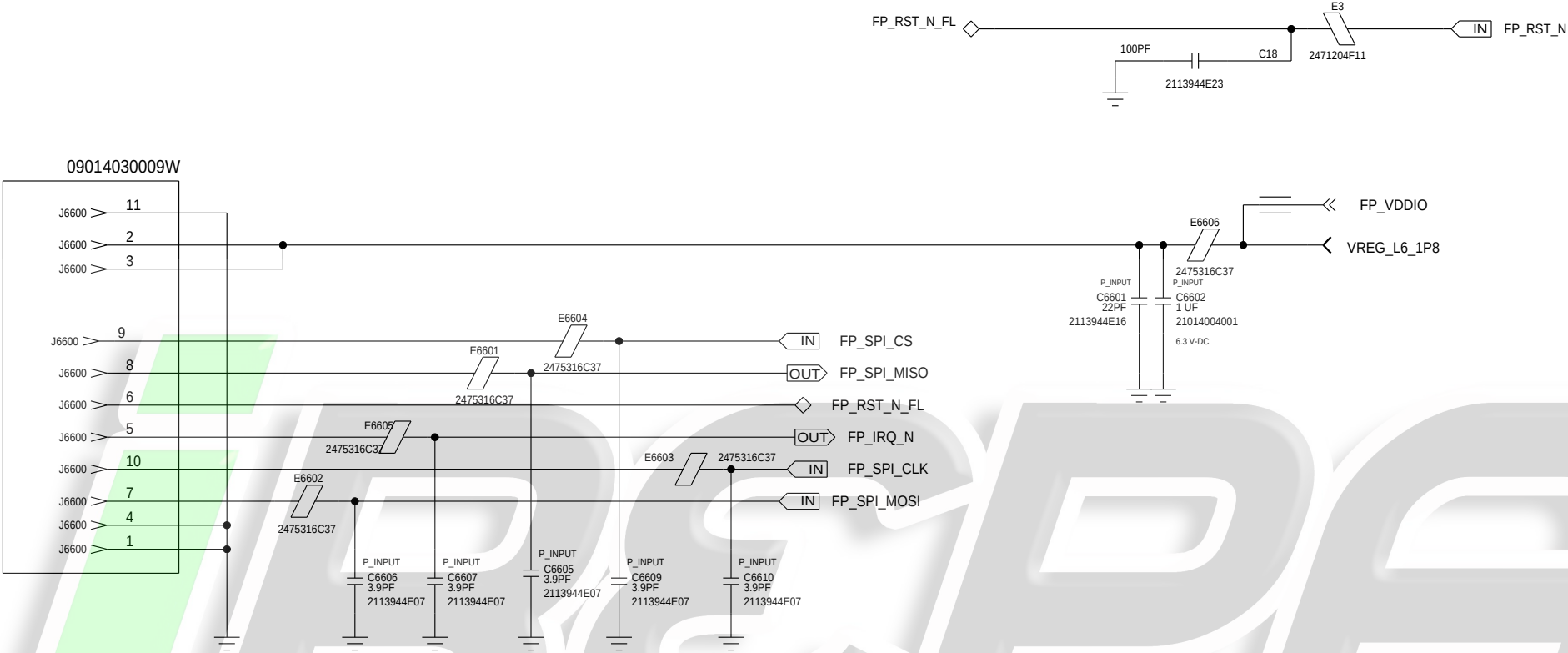
NOTE 1: USE 24.9K PULL-UP TO SUPPORT HOT SWAP WITH FREEZE-IO FEATURE.

NOTE 2: UIM_PWR BYPASS OF 0.33UF RECOMMENDED BY NFC IC SUPPLIER, PLACE NEAR CONNECTOR.

APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: FROEHLING		BLOCK TITLE:	
LAST MODIFIED BY: FROEHLING Wed Mar 30 18:27:15 2016		connectivity	
MOTOROLA CONFIDENTIAL RESTRICTED		PROJECT NAME	REV
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		CAD FILE NAME:	PAGE 13 OF 61

USR: FPC

REF: 6600 - 6699



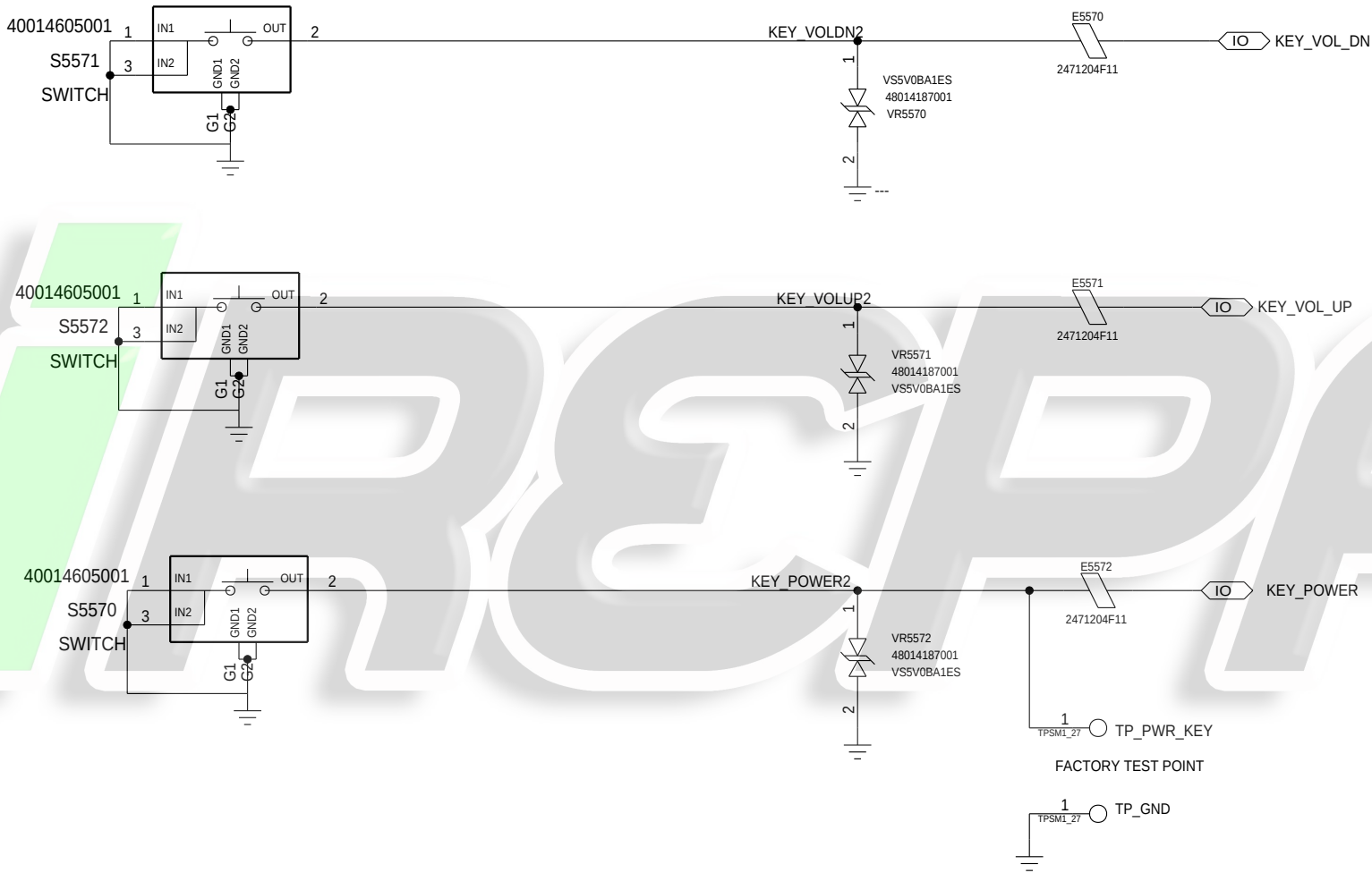
APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: NATE CONNELL		BLOCK TITLE: connectivity	
LAST MODIFIED BY: NATE CONNELL Tue Feb 23 12:17:07 2016		PROJECT NAME B 84016562001_06d.cpm	
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		CAD FILE NAME: PAGE 14 OF 61	

SIDE KEY & BATT THERM CONN

REF:5570-5579

MECHANICAL KEYS

VOL_UP/DOWN/GND TPS NEED P0 BRINGUP ONLY

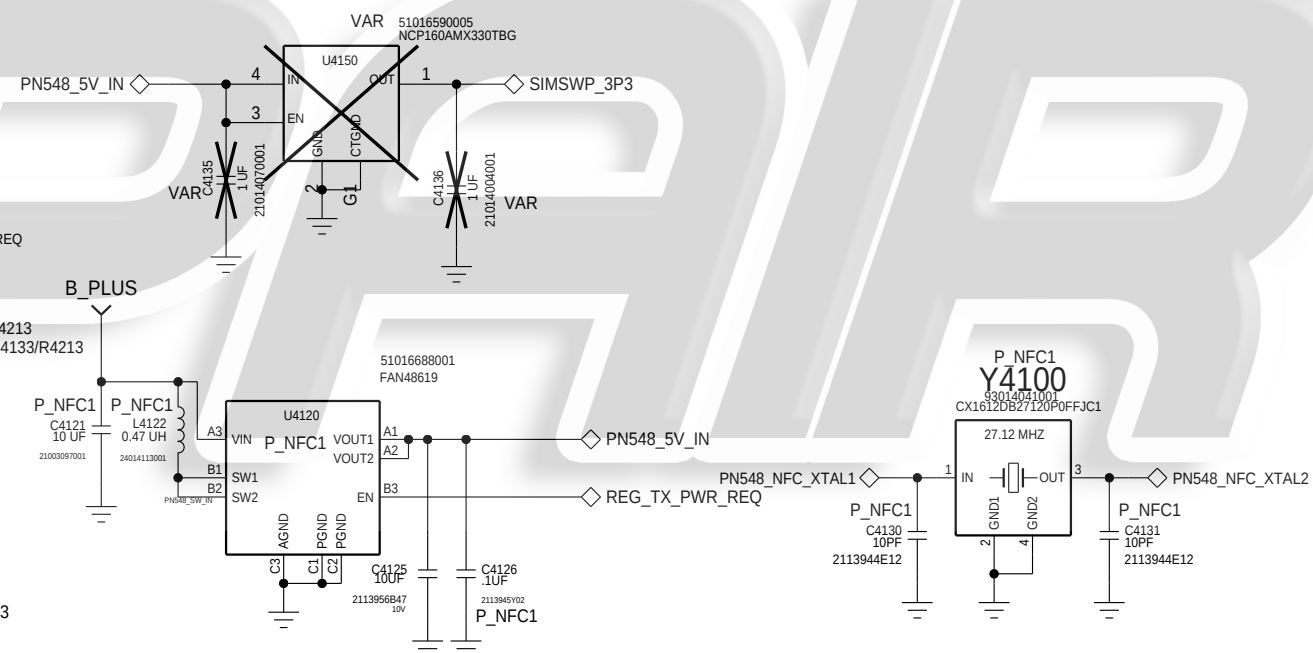
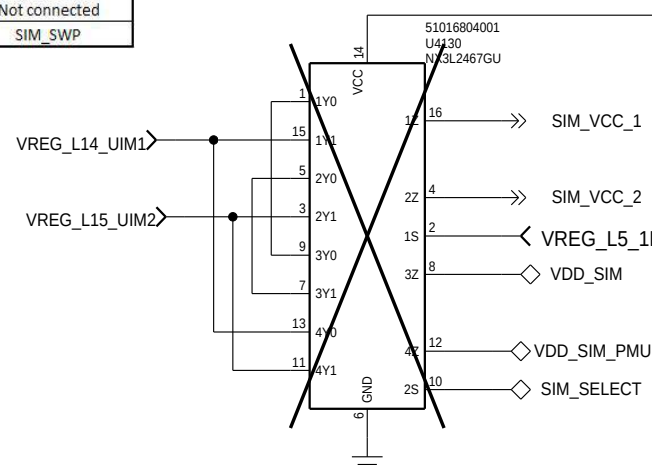
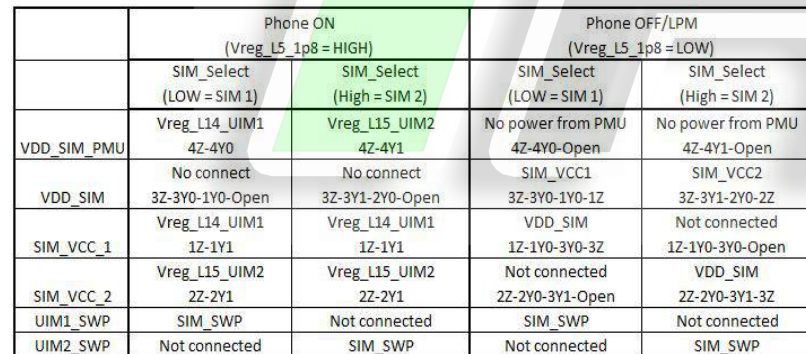


APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: NATE CONNELL		BLOCK TITLE:	
LAST MODIFIED BY: NATE CONNELL Tue Feb 23 12:17:12 2016		connectivity	
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		CAD FILE NAME:	PAGE 15 OF 61

REF: 4100-4199

HIF2 HIF1 WRITE READ

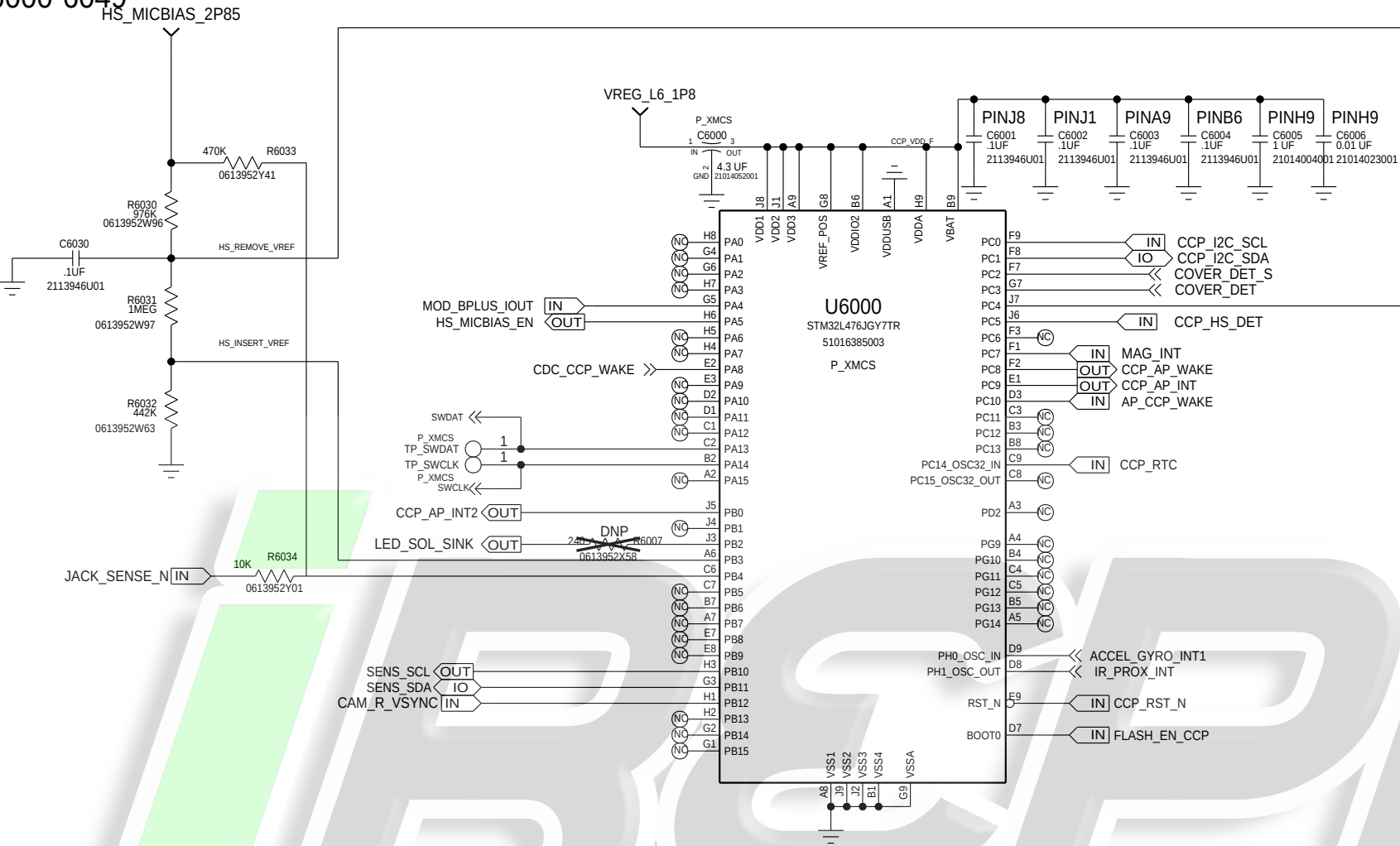
HIF2	HIF1	WRITE	READ
LOW	LOW	0X50	0X51
LOW	HIGH	0X52	0X53
HIGH	LOW	0X54	0X55
HIGH	HIGH	0X56	0X57



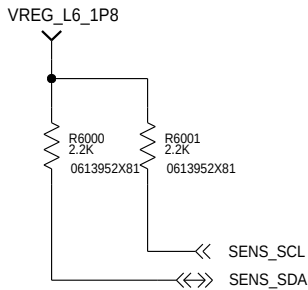
APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: FROEHLING		BLOCK TITLE: connectivity	
LAST MODIFIED BY: FROEHLING Thu Apr 14 14:55:54 2016			
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		CAD FILE NAME:	PAGE 16 OF 61

XMCS: CCP

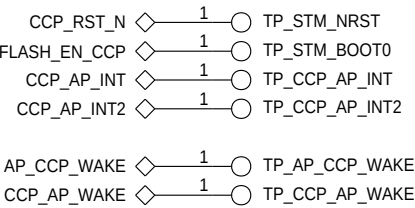
REF: 6000-6049



I2C SUPPORT

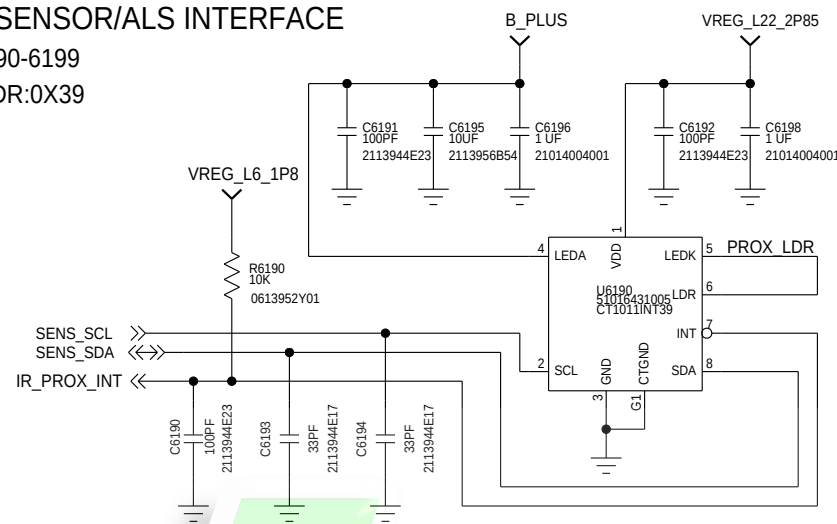
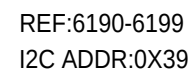


TESTPOINTS

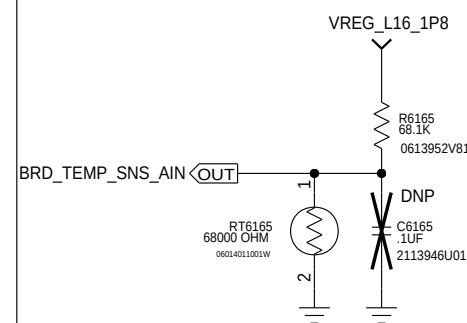


APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: JOHN GORSICA		BLOCK TITLE: xmcs	
LAST MODIFIED BY: NATE CONNELL Fri Apr 01 13:41:30 2016		PROJECT NAME 84016562001_06d.cpm	
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		PAGE 17 OF 61	

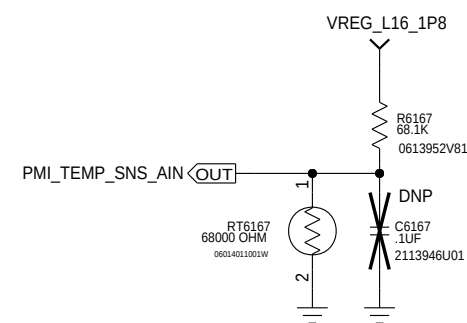
REF: 6100-6199



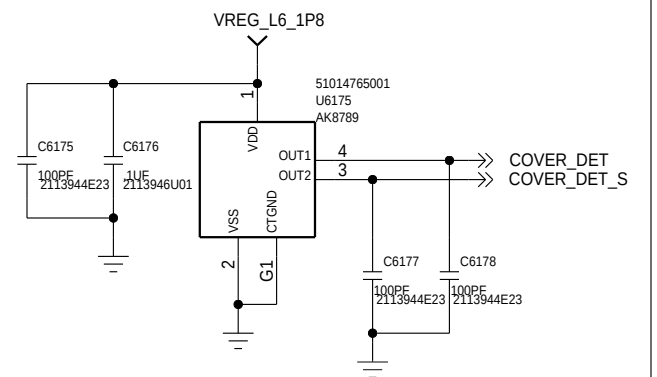
BOARD TEMP SENSOR



CHRG TEMP SENSOR

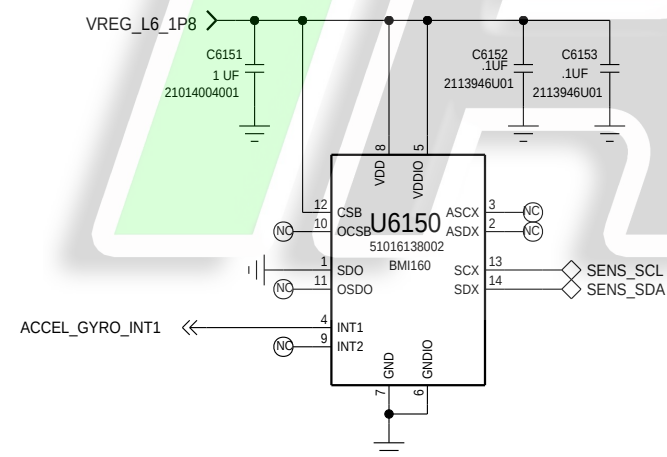


HALL EFFECT SENSOR



REF:6150-6159

I2C ADDR: 0X??



PRODUCTS MUST FOLLOW BUNKER HILL PCB ORIENTATION

REF:6140-6149

I2C ADDR: 0X0C

MOVED TO SIM FLEX PCB FROM DVT2

JOHN GORSICA/NATE CONNELL
APPROVALS

ENGINEER: HUNG PHAM

LAST MODIFIED BY: HUNG PHAM
Tue Mar 29 14:27:55 2016

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BLOCK TITLE:

xmcs

	PROJECT NAME
B	84016562001_06d.cpm

CAD FILE NAME:

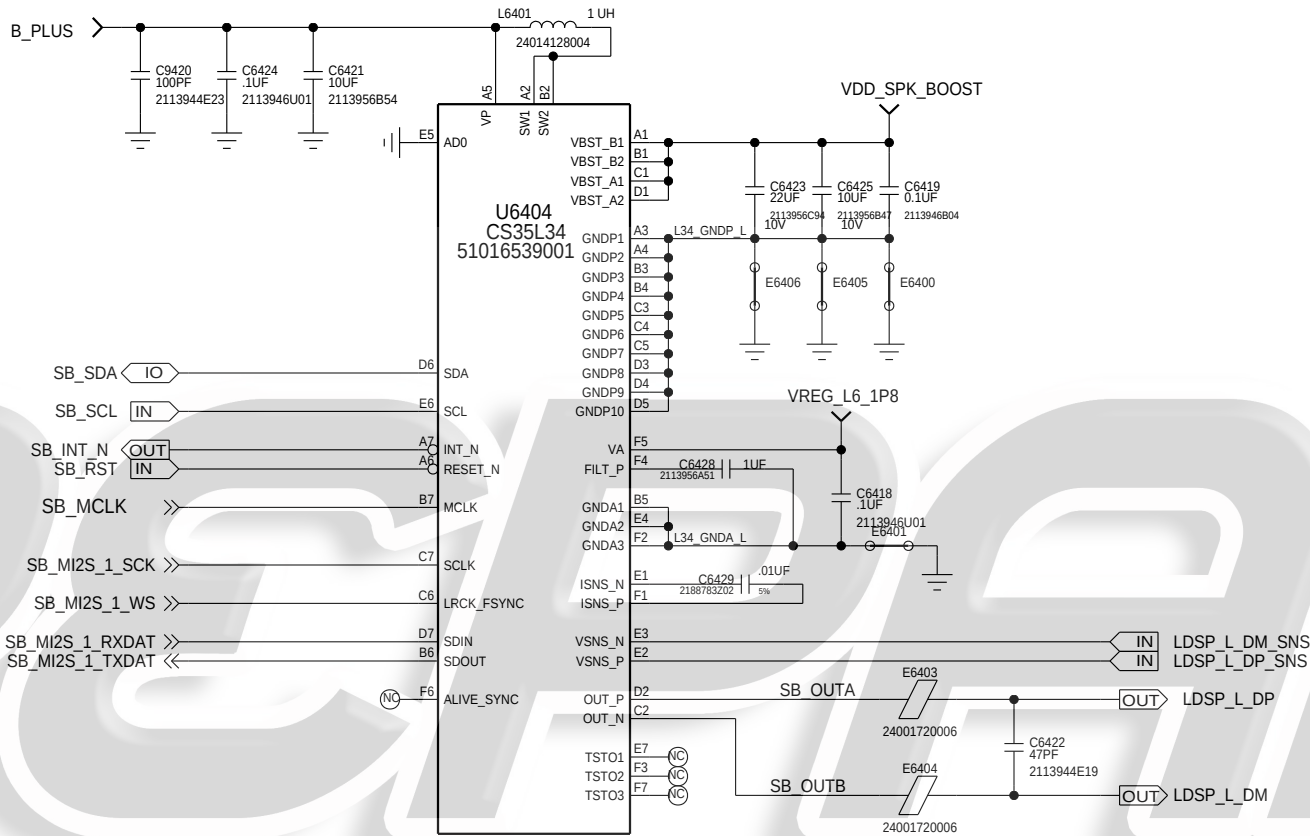
REV
1

PAGE 18 OF 61

XMCS: SMART BOOST WATTSON

REF 6400-6499

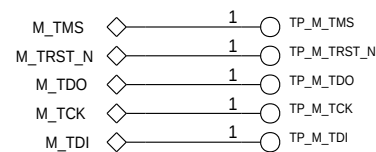
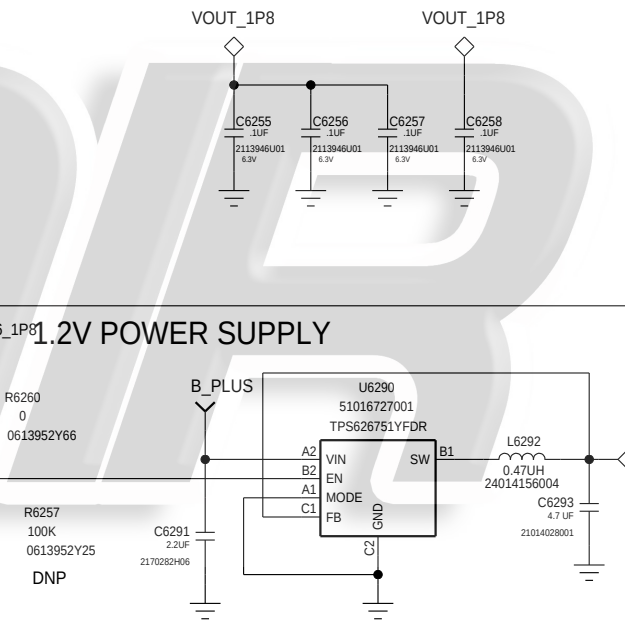
*** BOOST IC I2C ADD. 0X80



SB_MI2S_1_SCK	◇	1	○	TP_SB_SCK
SB_MI2S_1_WS	◇	1	○	TP_SB_WS
SB_MI2S_1_TXDAT	◇	1	○	TP_SB_TX
SB_MI2S_1_RXDAT	◇	1	○	TP_SB_RX
SB_INT_N	◇	1	○	TP_SB_INT
SB_RST	◇	1	○	TP_SB_RST

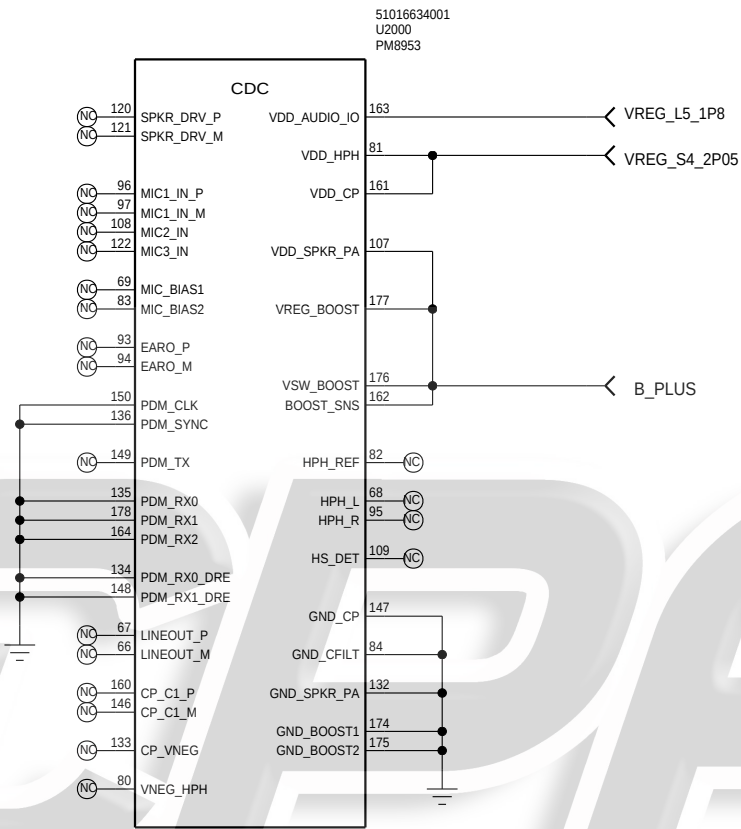
APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY: Tue Feb 23 12:17:24 2016		xmcs	
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		CAD FILE NAME:	
		REV 01	
		PAGE 19 OF 61	

*** CODEC I2C ADD. 0X34

PAGE 20 OF 61

XMCS: PMIC CODEC

REF 2500-2579



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY: FAN021 Tue Feb 23 12:17:30 2016		xmcs	
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		CAD FILE NAME: PAGE 21 OF 61	

87654321

D

C

B

A

87654321

XMCS: IR GESTURES

REF 6500-6549

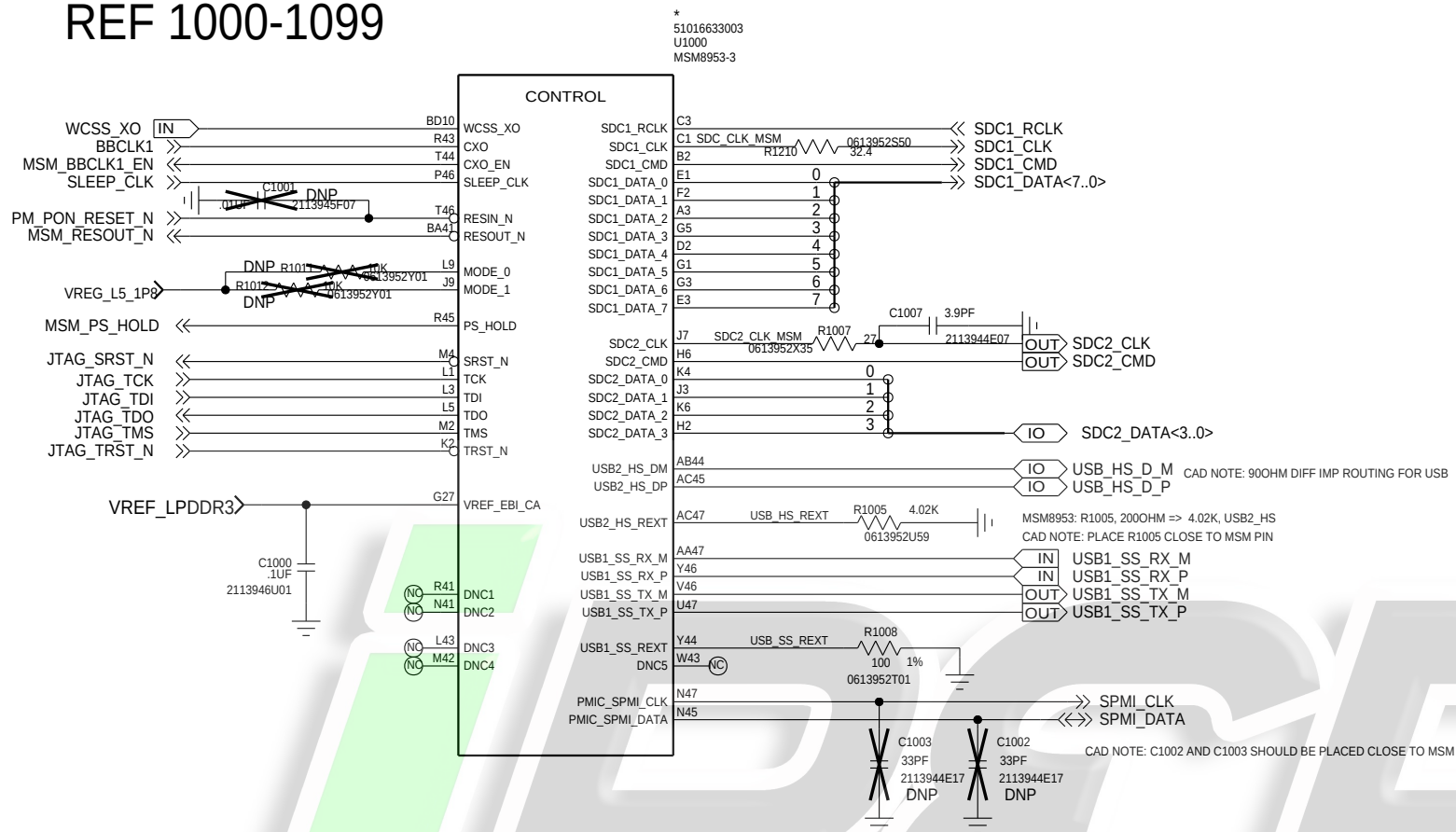


MOTOROLA MOBILITY CONFIDENTIAL RESTRICTED-MDB TRACKED TRACKEDANDBLOCKED

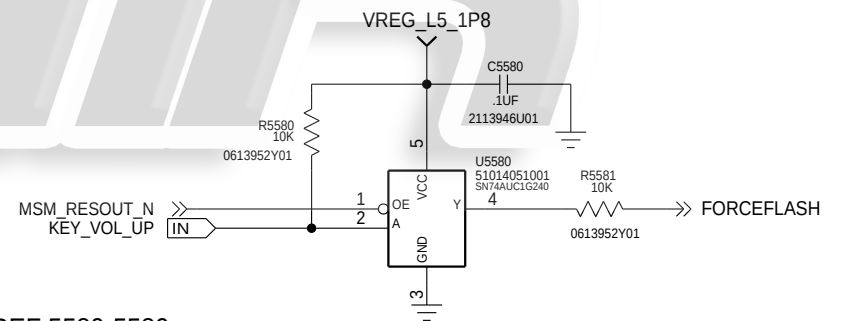
APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MATT MOTTIER		BLOCK TITLE:	
LAST MODIFIED BY: UDI ELRAN Mon Dec 28 15:04:31 2015		xmcs	
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		CAD FILE NAME:	
		REV 01	
		PAGE 22 OF 61	

MSM: CTRL

REF 1000-1099



FORCE FLASH SUPPORT WITH VOL UP KEY (DNI FOR PRODUCTION)



REF:5580-5589

APPROVALS

ENGINEER:

LAST MODIFIED BY: FAN021
Thu Apr 14 15:01:35 2016

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BLOCK TITLE:

ap_pmic

PROJECT NAME

84016562001_06d.cpm

REV

01

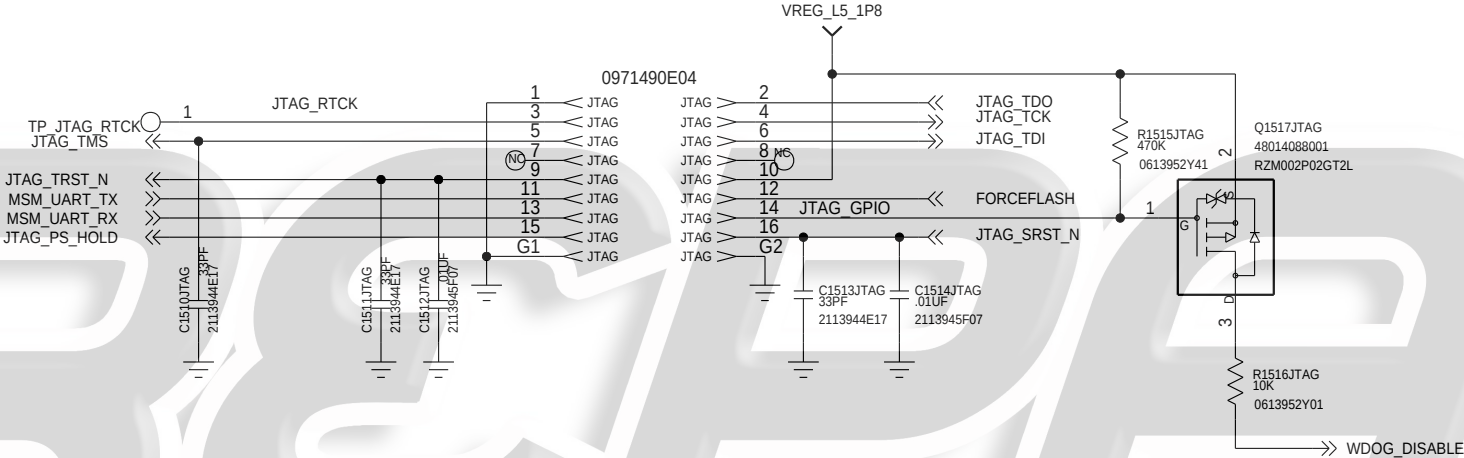
CAD FILE NAME:

PAGE 23 OF 61

MSM: DEBUG

REF 1000-1099

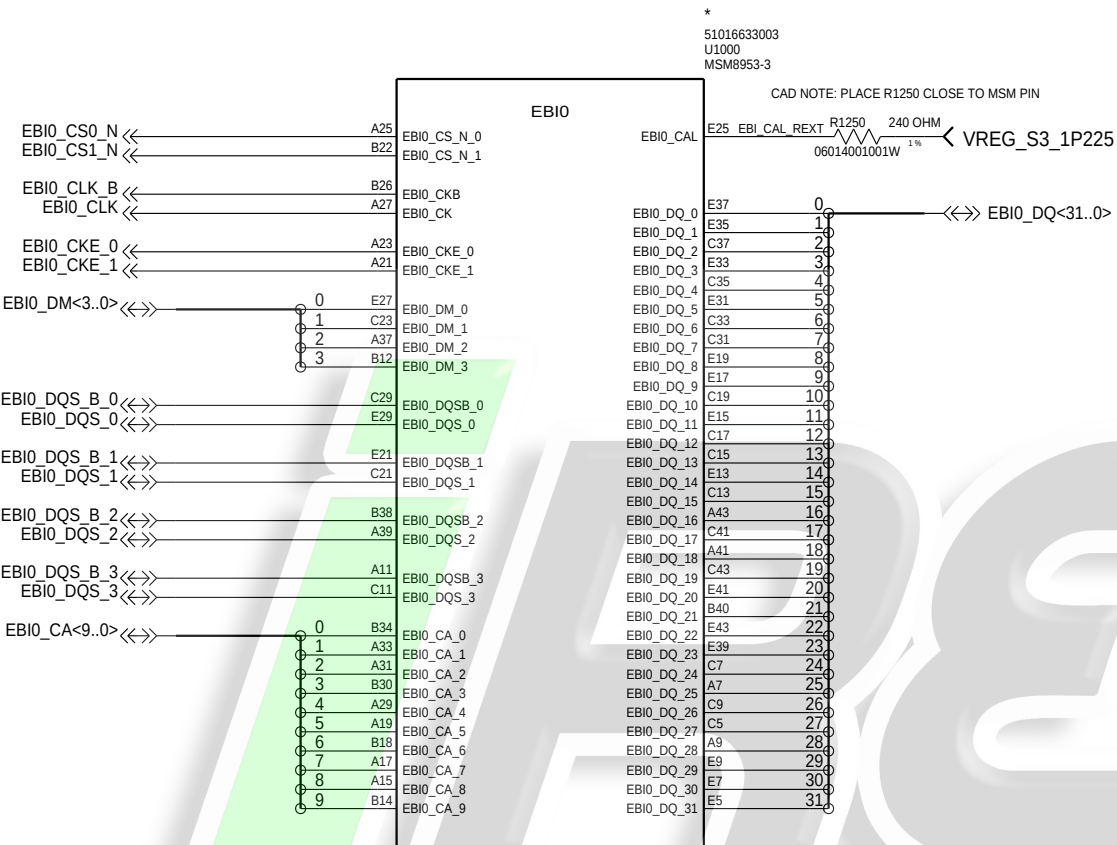
JTAG:MINI



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY: FAN021 Tue Feb 23 12:17:44 2016		ap_pmic	
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		B 84016562001_06d.cpm	
		REV 01	
		CAD FILE NAME:	
		PAGE 24 OF 61	

MSM: MIPI/DSI/CSI/EBIO

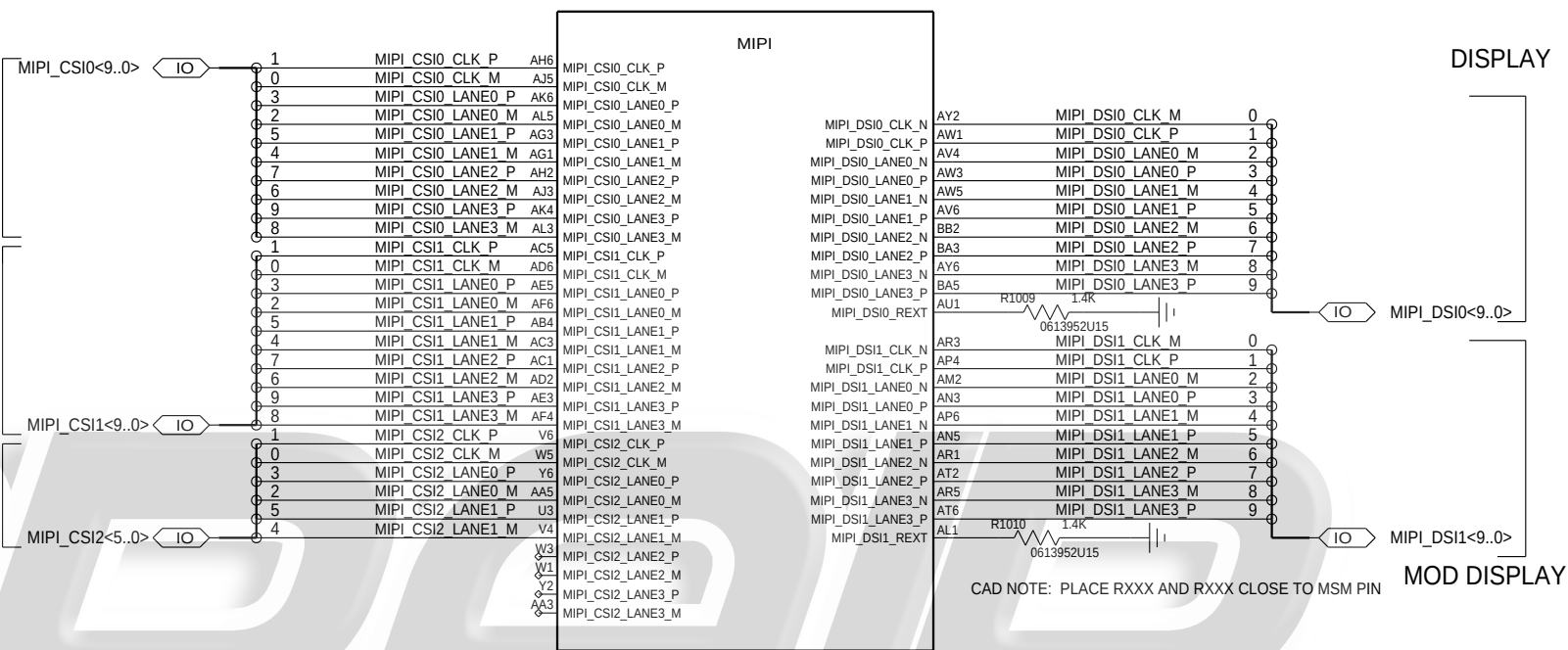
REF 1200-1299



MODS CAMERA

REAR CAMERA

FRONT CAMERA



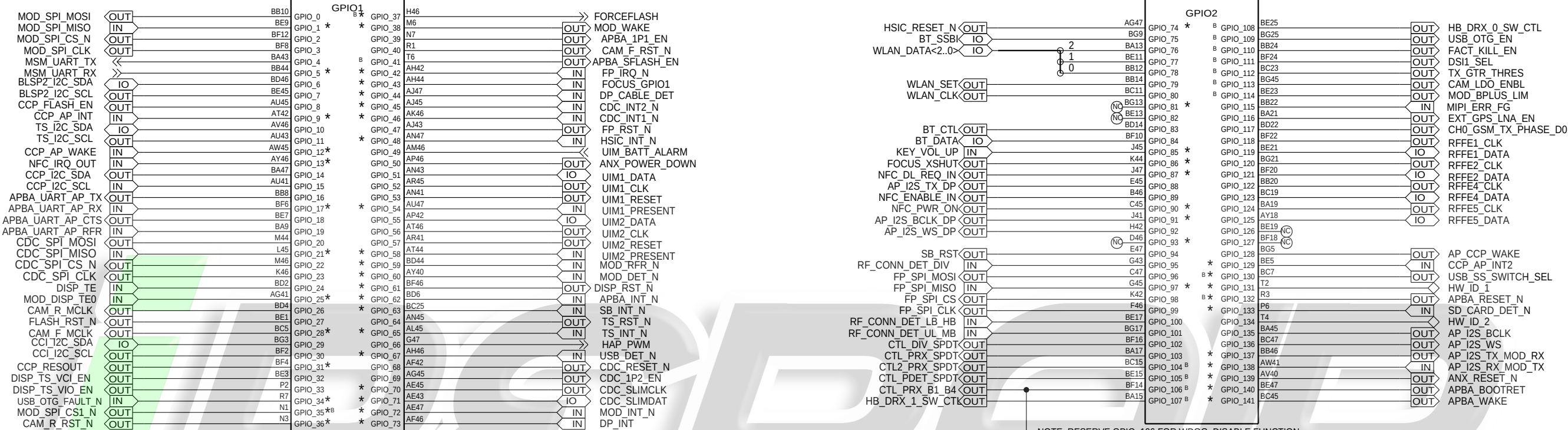
APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY: Thu Apr 14 15:01:33 2016		ap_pmic	
MOTOROLA CONFIDENTIAL RESTRICTED THE DRAWING ON THIS PRINT AND INFORMATION THEREWITH ARE PROPRIETARY TO MOTOROLA AND SHALL NOT BE USED OR DISCUSSED IN WHOLE OR IN PART WITHOUT MOTOROLA'S CONSENT. © COPYRIGHT, MOTOROLA. THIS DOES NOT IMPLY PUBLICATION		PROJECT NAME	
		B 84016562001_06d.cpm	REV 01
		CAD FILE NAME:	
		PAGE 25 OF 61	

MSM: GPIO (0-141)

REF 1200-1299

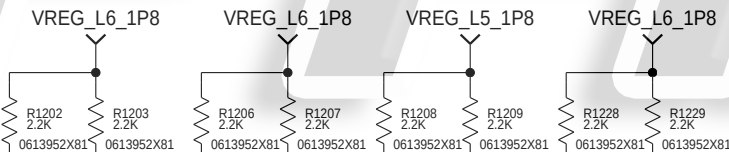
NOTE: DO NOT HAVE PULL-UPS ON THE FOLLOWING GPIOs UNLESS INTEND FOR BOOT OR SECURE BOOT RELATED CONFIGURATIONS:
GPIO_8, 9, 91, 107, 108, 109

NOTE: ASTERISK(*) INDICATE MODEM POWER MANAGEMENT (MPM) WAKE-UP PINS.



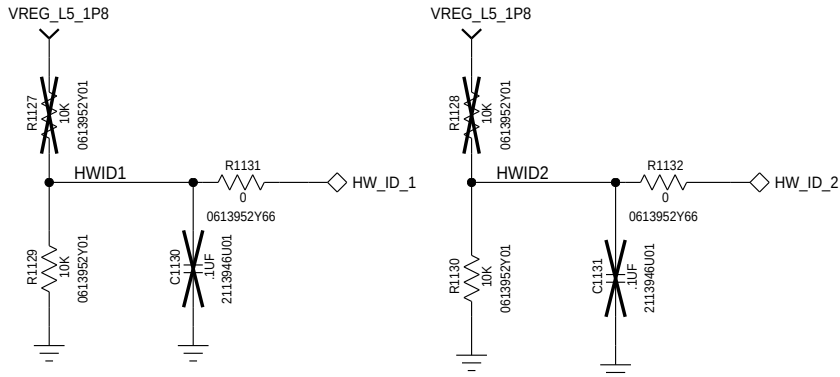
WAITING FOR GPIO ASSIGNMENT
NEED WAKABLE

I2C PULL-UP RESISTORS



MOT BOARD ID

FOR RUNTIME REVISION INFORMATION, REFER TO:
<https://sites.google.com/a/motorola.com/runtime-product-ids>



APPROVALS

ENGINEER:

LAST MODIFIED BY: FAN021
Thu Apr 14 15:01:31 2016

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MOTOROLA
MOBILE DEVICES

BLOCK TITLE:

ap_pmic

PROJECT NAME

84016562001_06d.cpm

REV

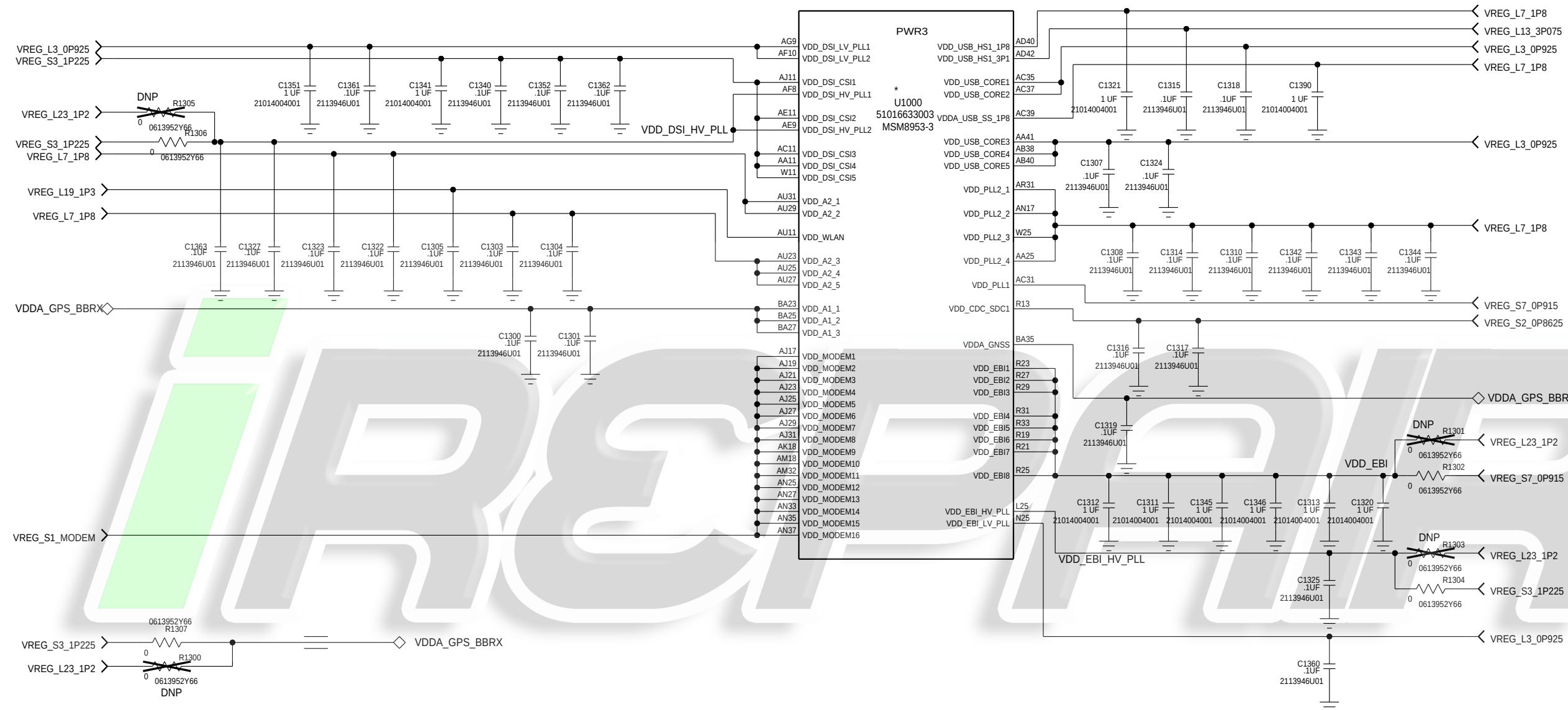
01

CAD FILE NAME:

PAGE 26 OF 61

MSM: PWR1
REF 1300-1399

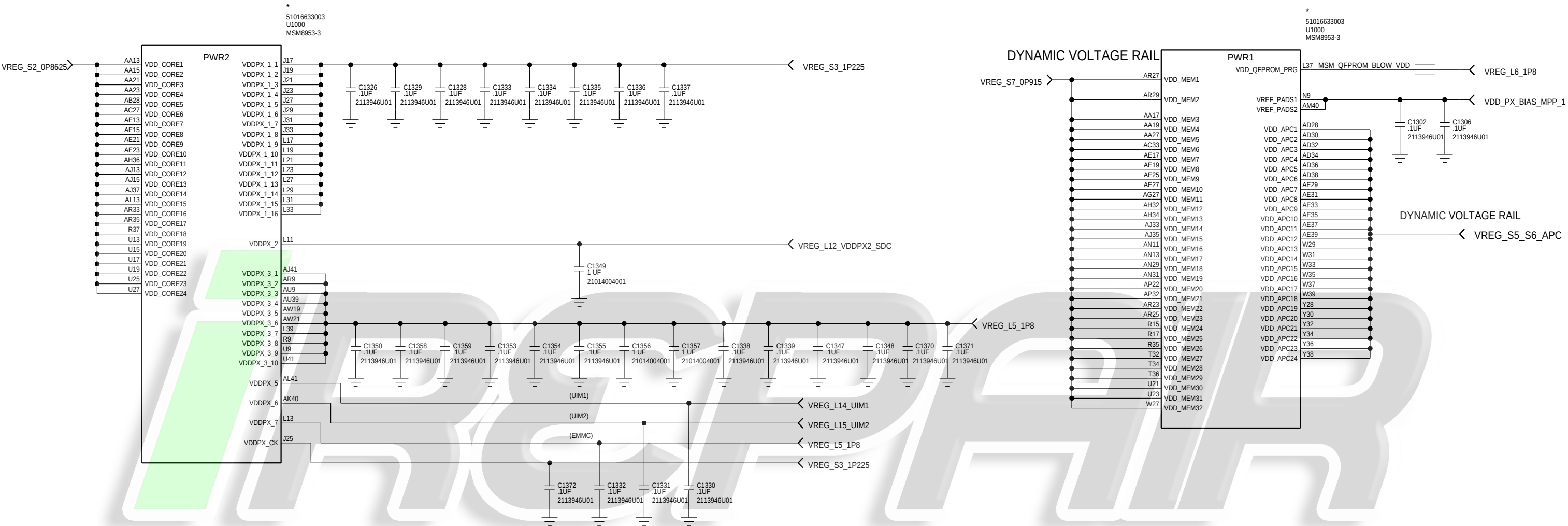
CAD NOTE: PLACE ONE DECAP NEAR PINS V5, V6 AND
THE OTHER DECAP NEAR PIN R5



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:	BLOCK TITLE: ap_pmic		
LAST MODIFIED BY: FAN021 Thu Apr 14 15:01:27 2016			
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		CAD FILE NAME:	PAGE 27 OF 61

MSM: PWR2

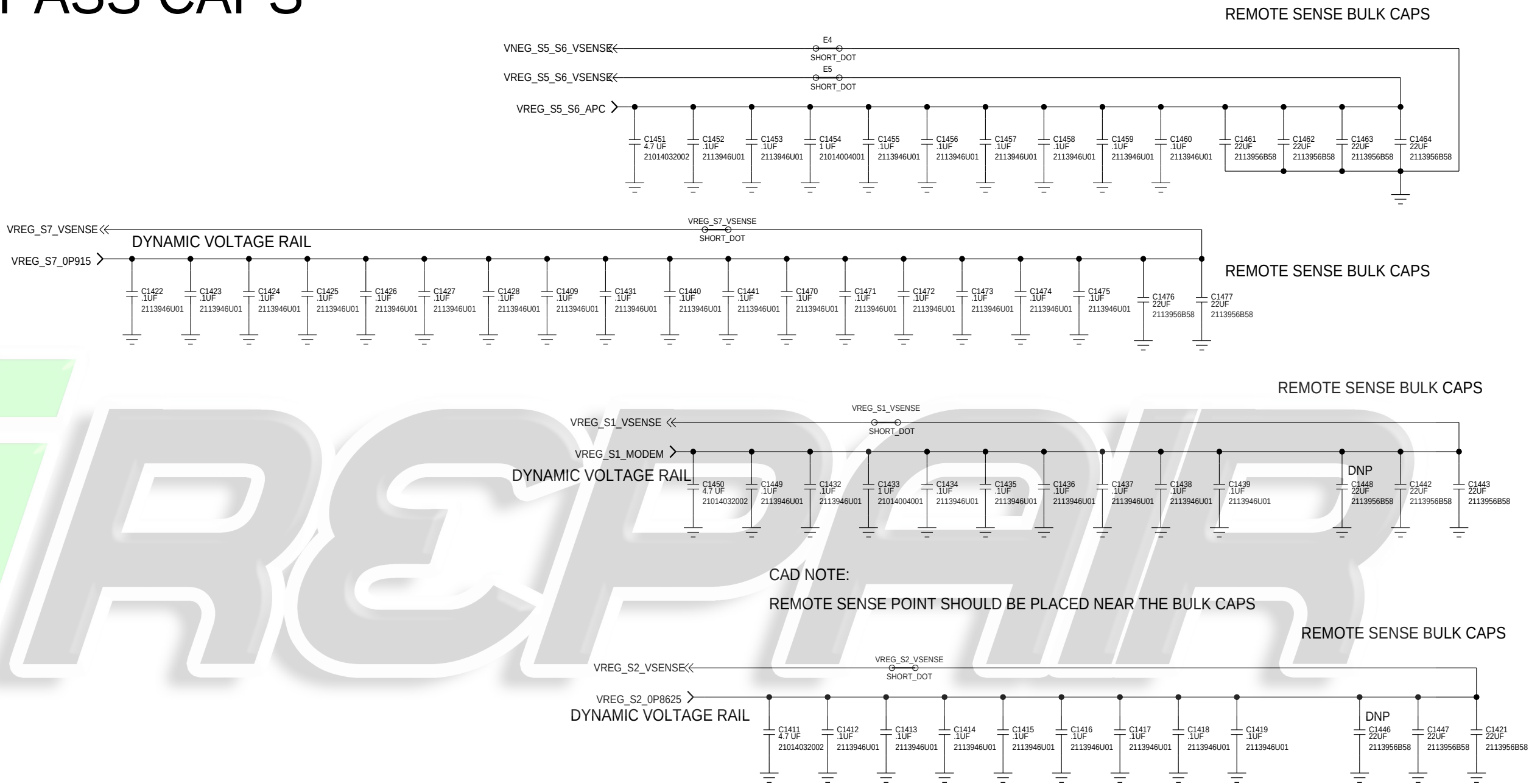
REF 1300-1399



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY: FAN021 Thu Apr 14 15:01:26 2016		ap_pmic	
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		B 84016562001_06d.cpm	01
CAD FILE NAME:			PAGE 28 OF 61

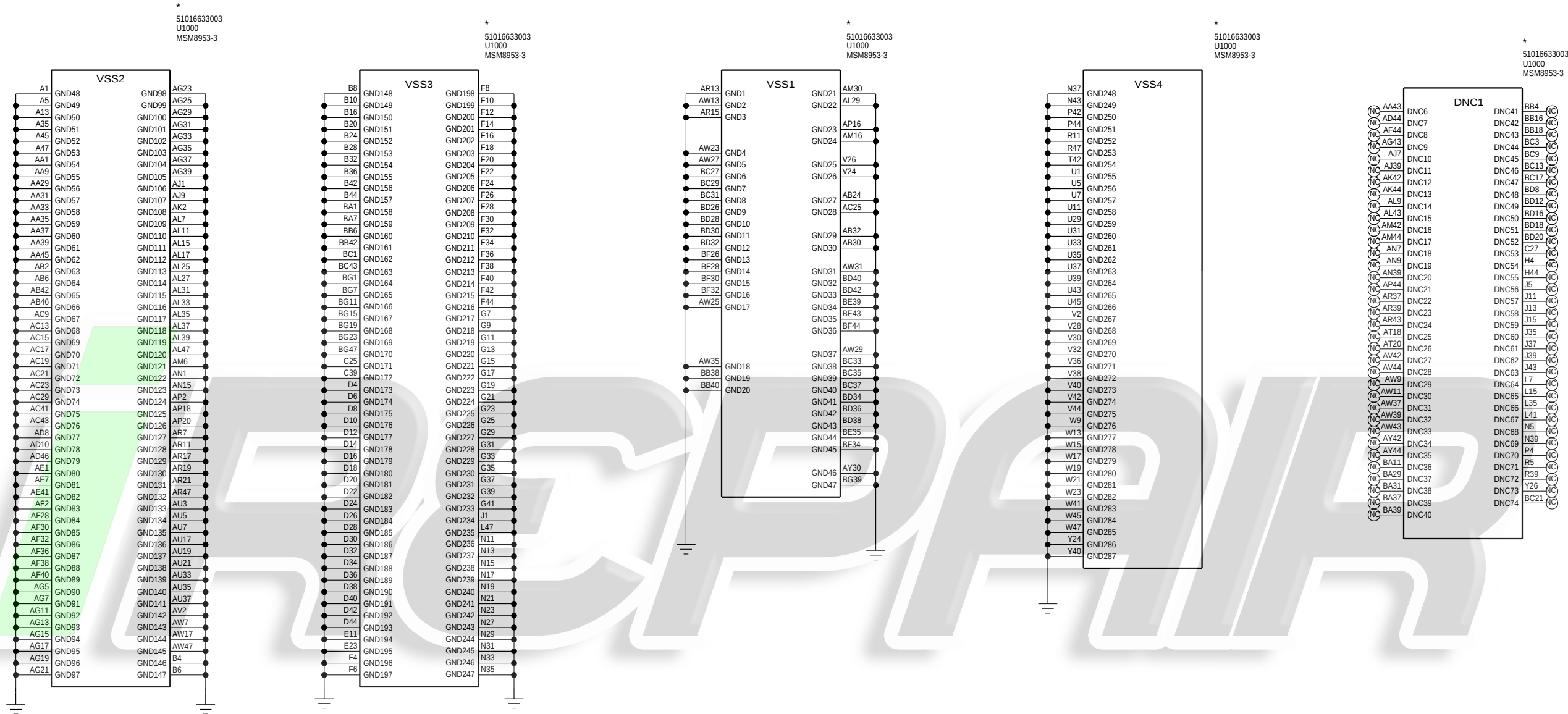
MSM: BYPASS CAPS

REF 1400-1499



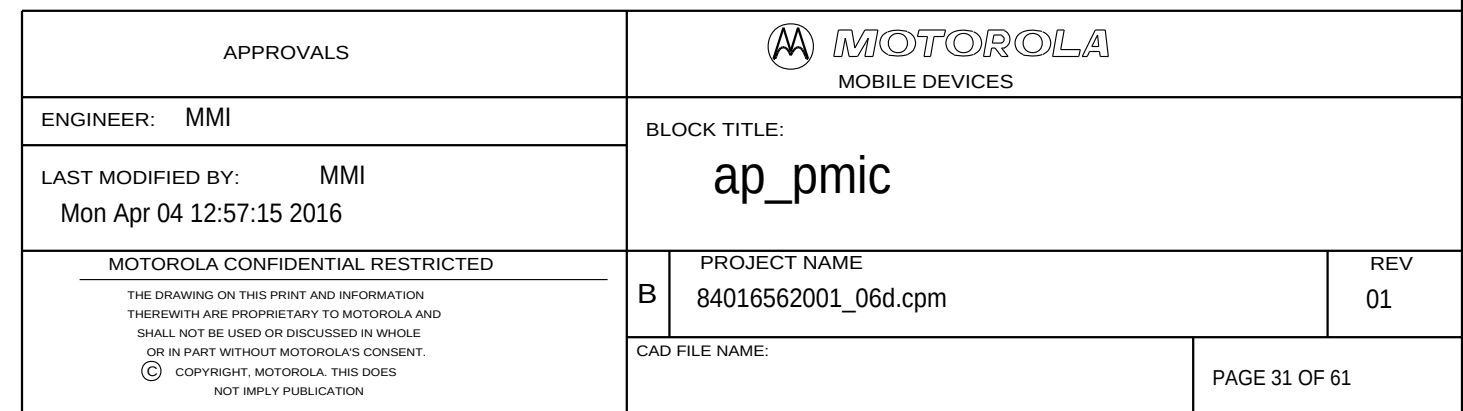
APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY: FAN021 Tue Feb 23 12:18:03 2016		ap_pmic	
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		B 84016562001_06d.cpm	REV 01
		CAD FILE NAME:	
		PAGE 29 OF 61	

MSM: GROUND/DNC



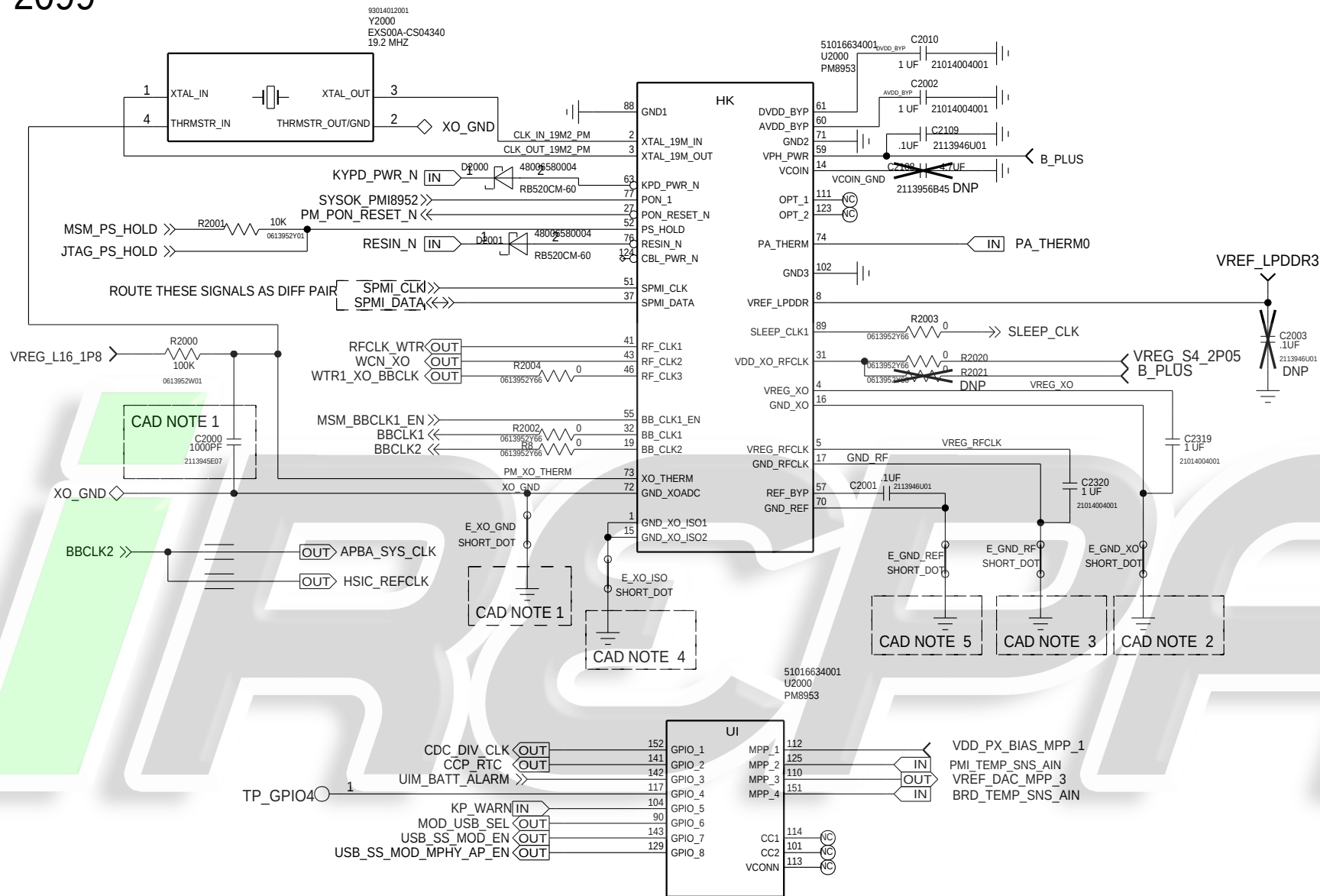
APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY: FAN021 Thu Apr 14 15:01:29 2016		ap_pmic	
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		B 84016562001_06d.cpm	01
		CAD FILE NAME:	
		PAGE 30 OF 61	

REF: 1600-1699 MCP: 16GB EMMC, 2GB LPDDR3



PM8953: HK/GHRG/LED/GPIO

REF: 2000-2099

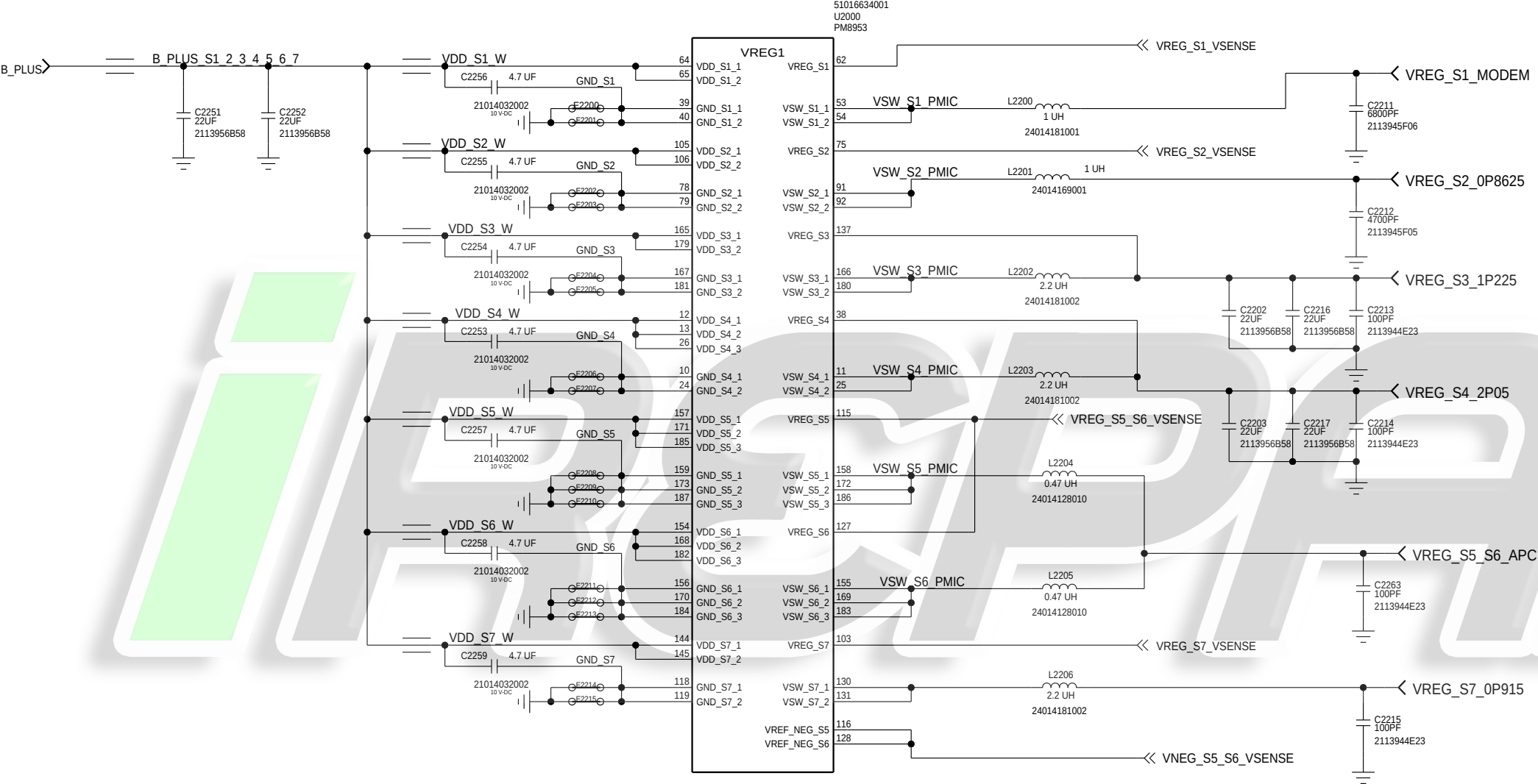


CAD NOTE 1 : PLACE C2000 CLOSER TO Y2000 AND CONNECT GND_XOADC TO CAP GND AND THEN CONNECT TO MAIN GND PLANE THROUGH DEDICATED VIA
CAD NOTE 2 : CONNECT GND_XO TO C2319 CAP GND AND THEN CONNECT TO MAIN GND USING DEDICATED VIA.
CAD NOTE 3 : CONNECT GND_RFCLK TO C2320 CAP GND AND THEN CONNECT TO MAIN GND PLANE USING DEDICATED VIA.
CAD NOTE 4 : CONNECT GND_XO_ISO TO MAIN GND PLANE USING DEDICATED VIA. DO NOT SHARE THIS GND WITH OTHER GROUNDS.
CAD NOTE 5 : CONNECT GND_REF TO CAPACITOR C2001 GND AND THEN CONNECT TO MAIN GND PLANE USING DEDICATED VIA. DON NOT SHARE THIS GND WITH OTHER GND.

APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: MMI		BLOCK TITLE:	
LAST MODIFIED BY: MMI Tue Feb 23 12:17:42 2016		ap_pmic	
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		CAD FILE NAME:	
		PAGE 32 OF 61	

PM8953: SMPS

REF: 2200-2299

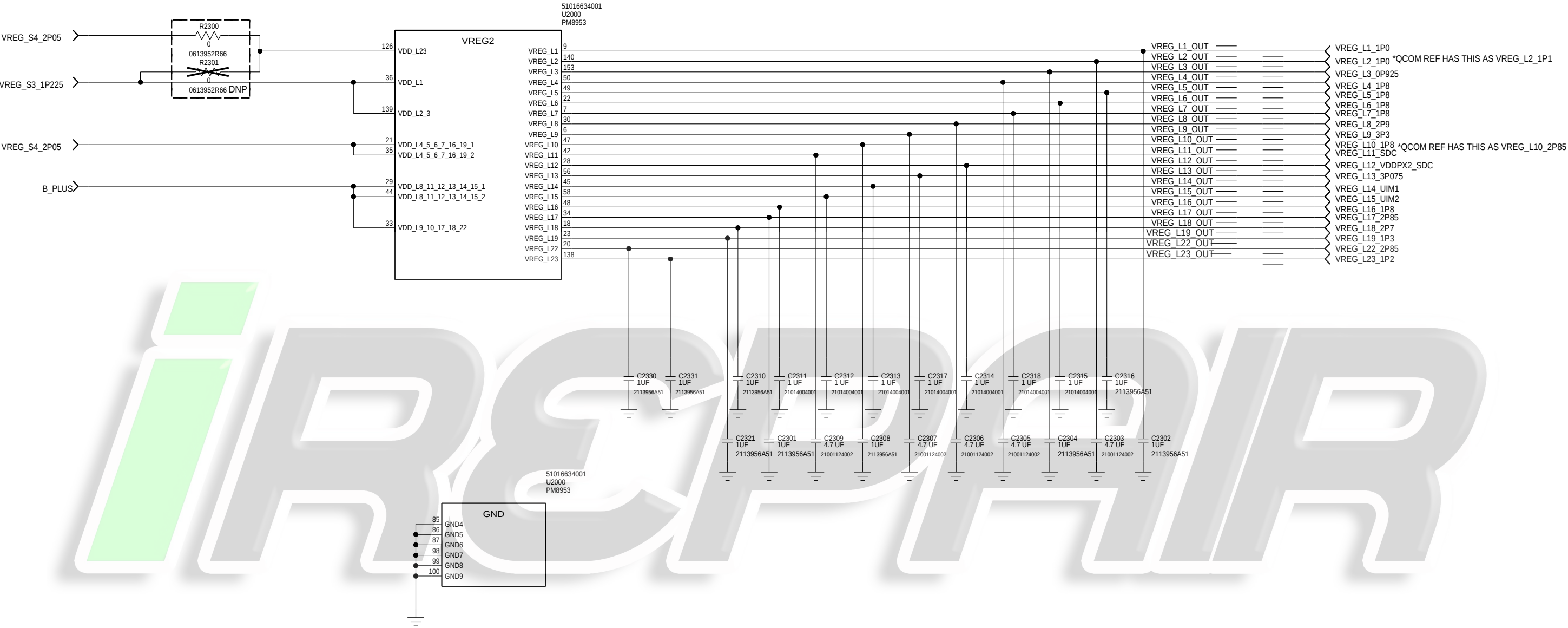


NOTE: PLACE INPUT CAPS CLOSE TO RESPECTIVE VIN PINS
EACH CAP GND SHOULD BE STAR-ROUTED

APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: MMI		BLOCK TITLE:	
LAST MODIFIED BY: MMI Tue Feb 23 12:18:01 2016		ap_pmic	
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		CAD FILE NAME: PAGE 33 OF 61	

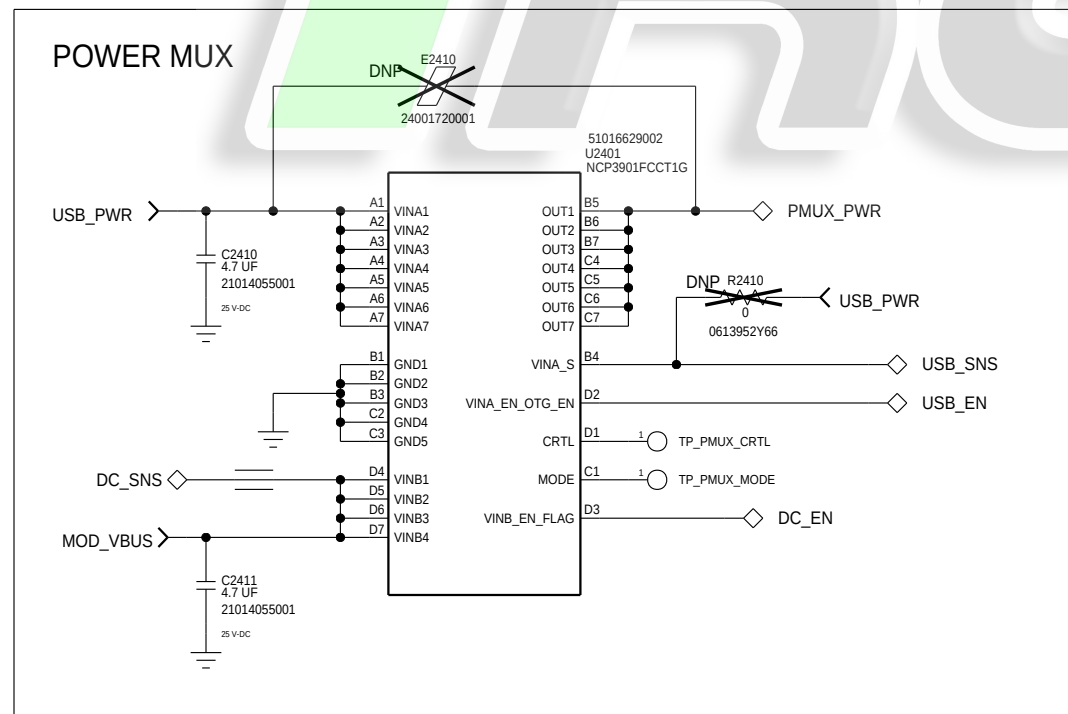
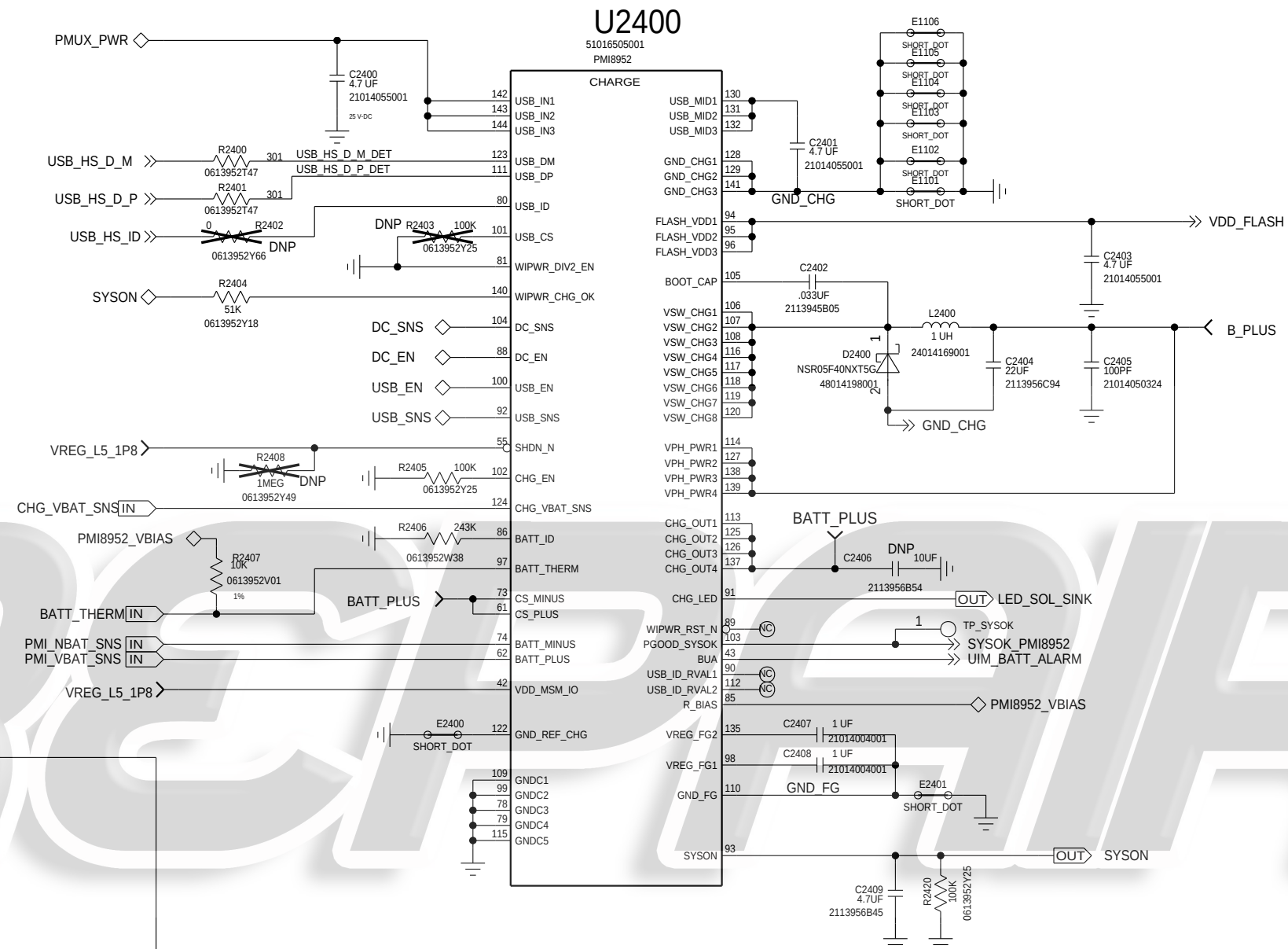
PM8953: VREGS

REF: 2300-2399



APPROVALS		 MOTOROLA MOBILE DEVICES		
ENGINEER: MMI		BLOCK TITLE: ap_pmic		
LAST MODIFIED BY: MMI Tue Feb 23 12:17:56 2016				
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		B 84016562001_06d.cpm		01
		CAD FILE NAME:		PAGE 34 OF 61

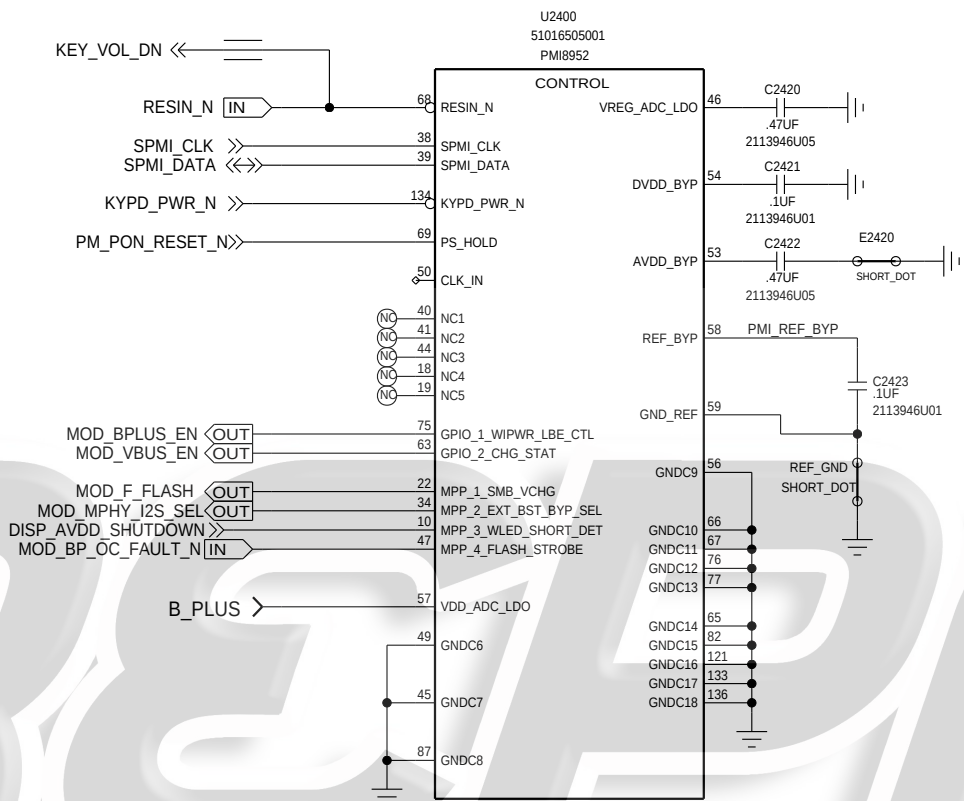
PMI8952: CHARGING
REF: 2400-2419



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MMI	BLOCK TITLE: ap_pmic		
LAST MODIFIED BY: MMI Tue Feb 23 12:17:37 2016			
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		CAD FILE NAME:	PAGE 35 OF 61

PMI8952: IO_REF

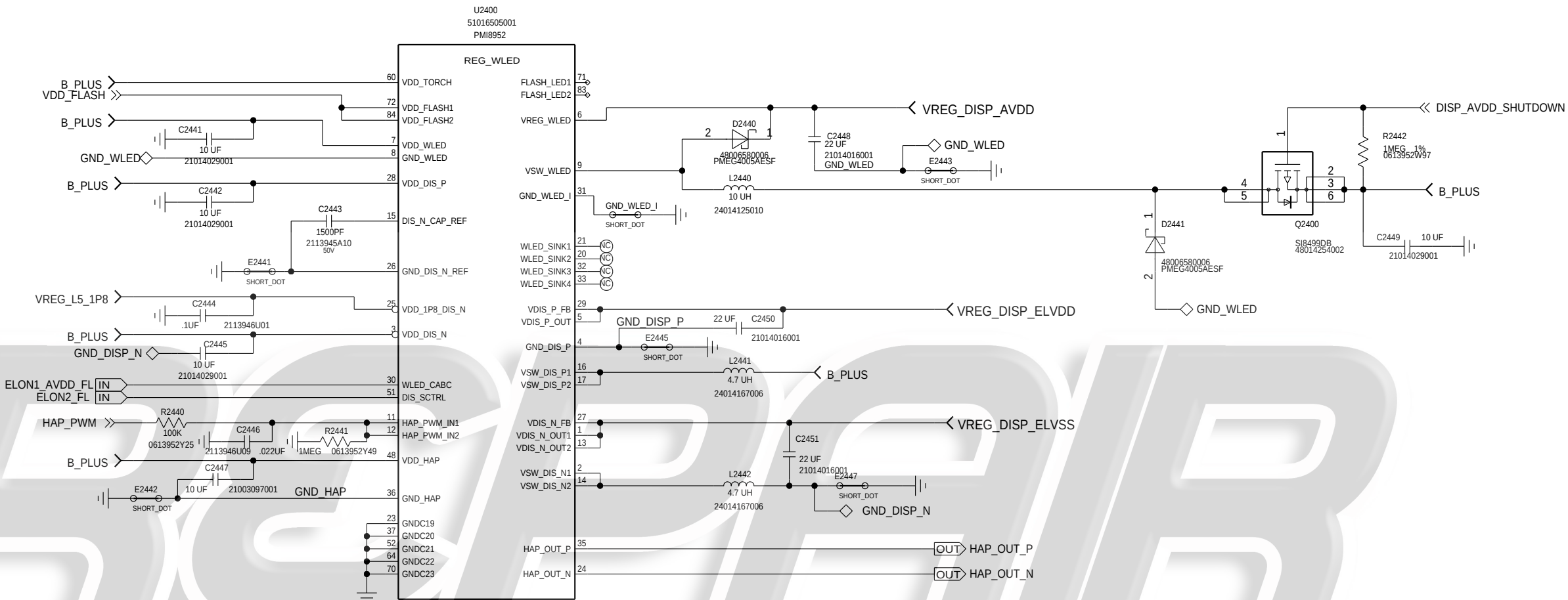
REF: 2420-2439



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MMI		BLOCK TITLE:	
LAST MODIFIED BY: MMI Wed Mar 30 12:37:06 2016		ap_pmic	
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		B 84016562001_06d.cpm	REV 01
		CAD FILE NAME:	
		PAGE 36 OF 61	

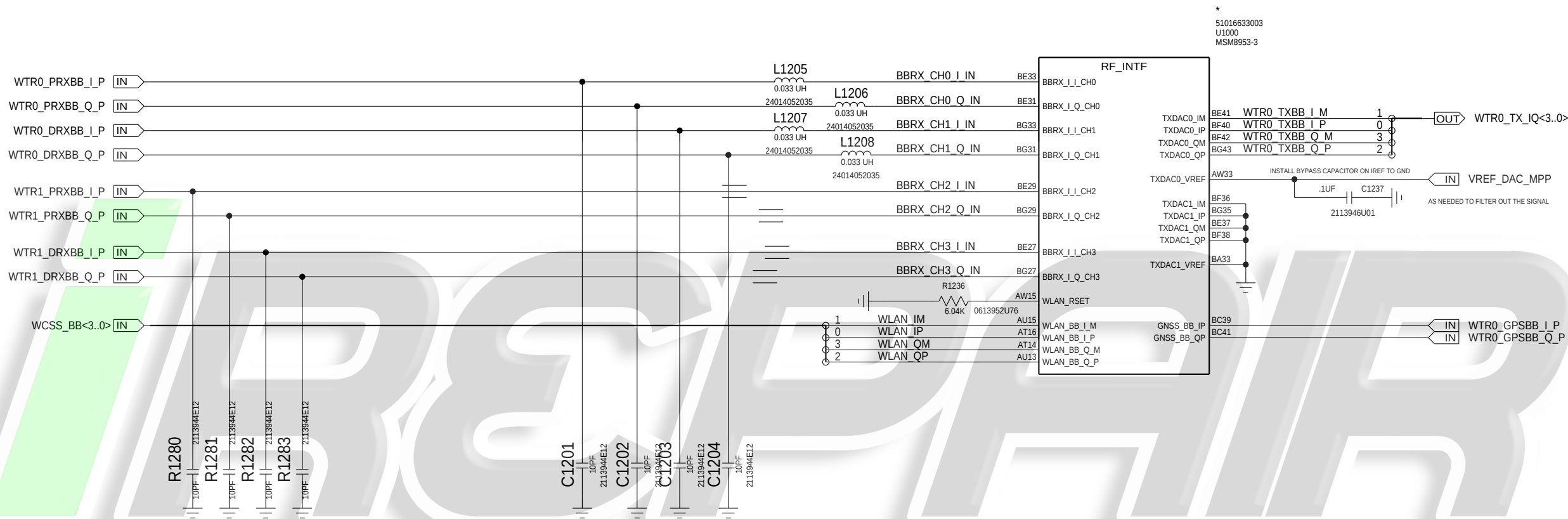
PMI8952: REG_WLED

REF: 2440-2459



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MMI		BLOCK TITLE:	
LAST MODIFIED BY: MMI Fri Feb 26 14:40:38 2016		ap_pmic	
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		REV 01 PAGE 37 OF 61	

MSM: RF INTERFACE
REF 1200-1299



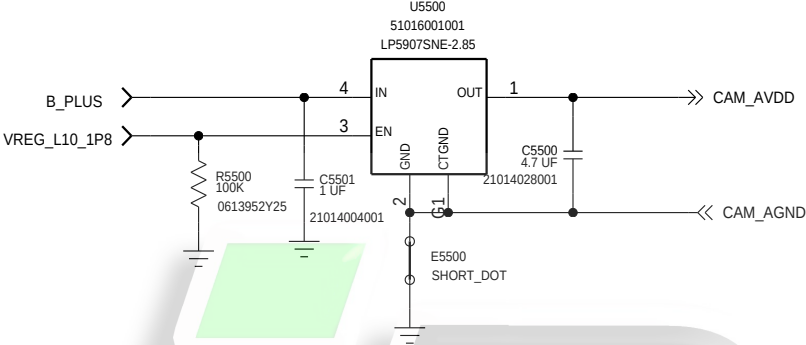
NEED TO APPLY LC FILTER FOR REDUCE B40 RX SPURS(QC 80-NP237-56)
LC FILTER COULD BE PLACED ANYWHERE BETWEEN WTR BBRX_IQ AND MSM BBRX_IQ PINS
BUT THE GND FOR SHUNT C SHOULD NOT BE SHARED WITH EITHER WTR OR MSM GND ON ANY LAYER.
A DIRECT VIA FROM SHUNT C TO MAIN GND IS RECOMMENDED

APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY: FAN021 Thu Apr 14 15:01:25 2016		block_mdm9x25m_pm8019	
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		B 84016562001_06d.cpm	REV 01
		CAD FILE NAME:	
		PAGE 38 OF 61	

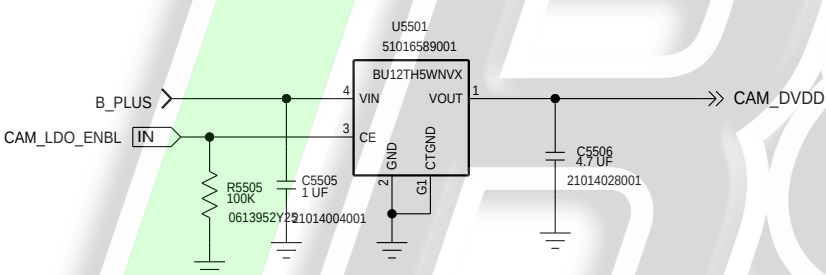
CAM: POWER

REF:5500-5529

ANALOG REGULATOR 2.85V

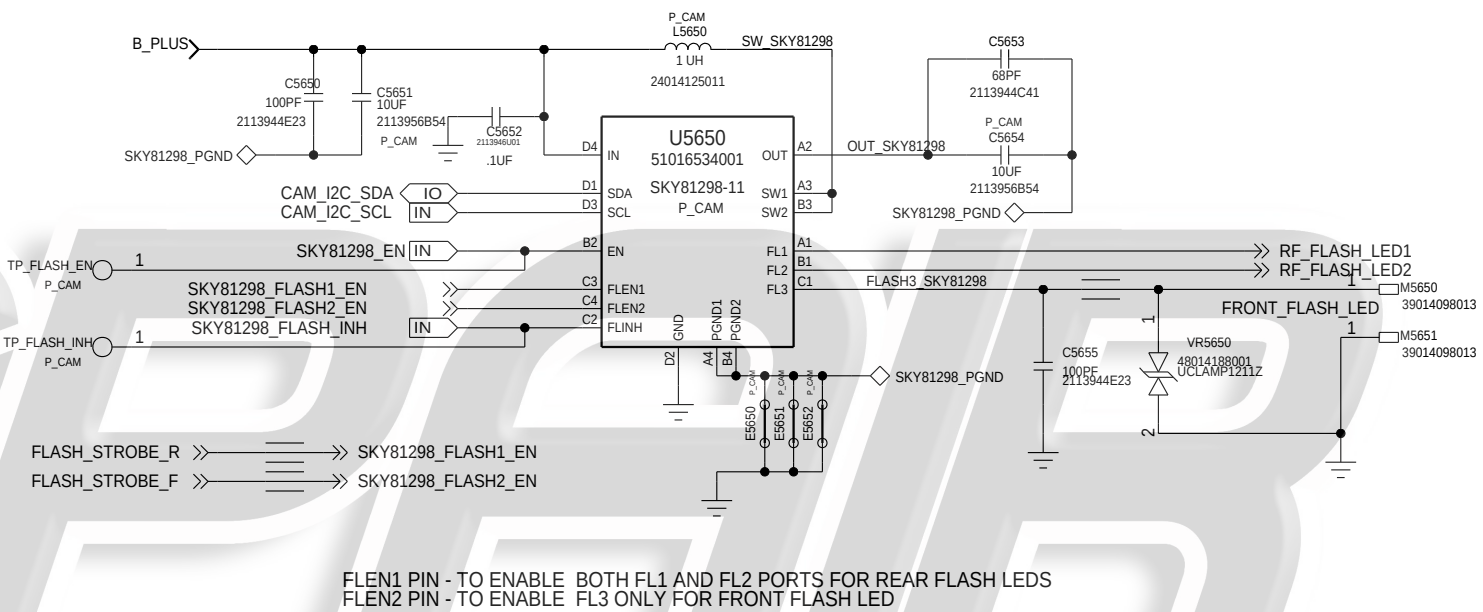


DIGITAL REGULATOR 1.2V



LED FLASH DRIVER FOR REAR AND FRONT IMAGERS

REF:5650-5669



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MMI		BLOCK TITLE:	
LAST MODIFIED BY: MMI Wed Mar 09 11:52:24 2016		multimedia	
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		CAD FILE NAME:	
		REV 01	
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CAM: REAR COMMON

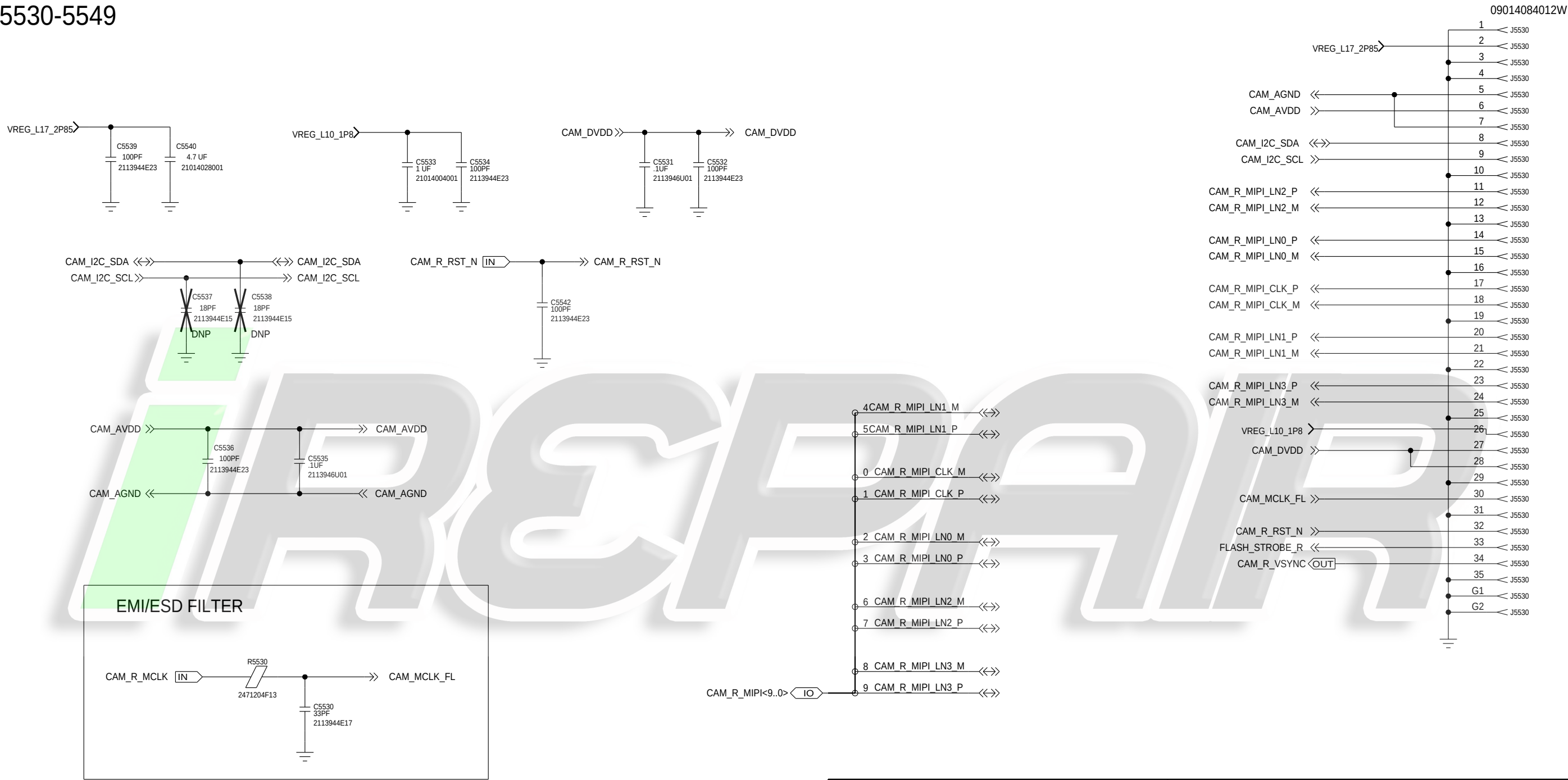
REF:5530-5549



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: JOHN KRAMER		BLOCK TITLE:	
LAST MODIFIED BY: AZIZ NOORI Mon Jan 04 09:46:49 2016		multimedia	
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		CAD FILE NAME:	
		REV 01	
		PAGE 40 OF 61	

CAM: REAR

REF:5530-5549

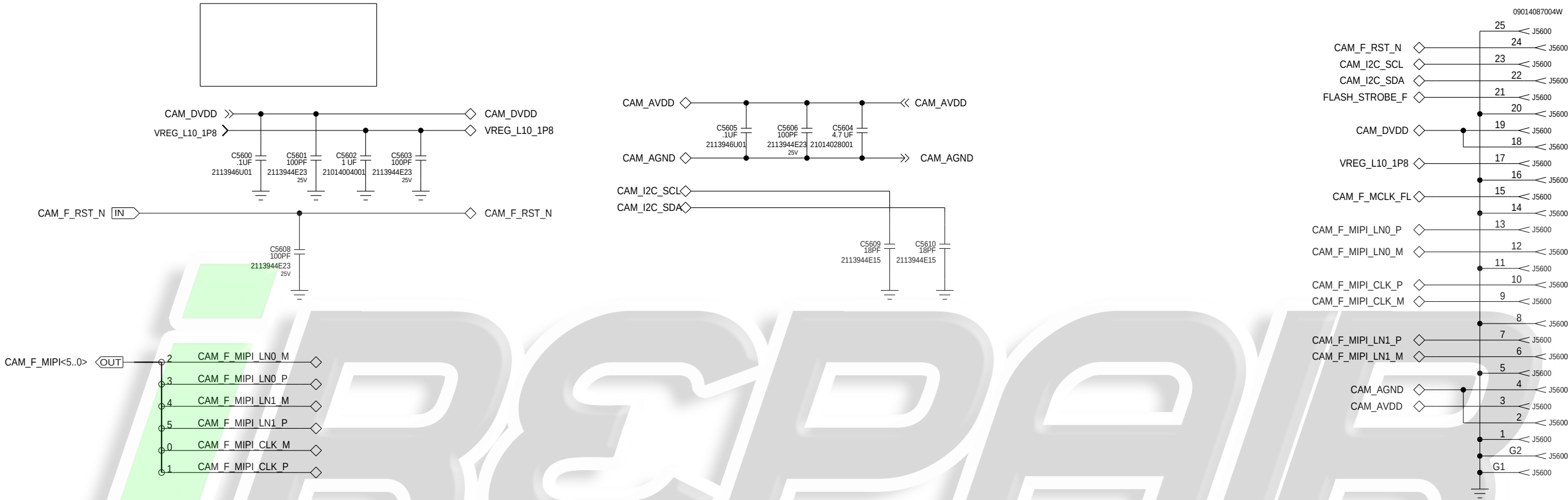


APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: JOHN KRAMER		BLOCK TITLE: multimedia	
LAST MODIFIED BY: AZIZ NOORI Thu Mar 10 11:01:38 2016		PROJECT NAME B 84016562001_06d.cpm	
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		PAGE 41 OF 61	

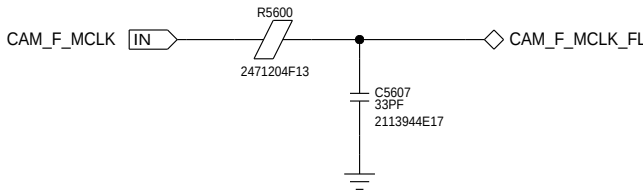
CAM: FRONT

REF 5600-5620

FF CAMERA CONNECTOR FRONT CAM I2C ADDR:



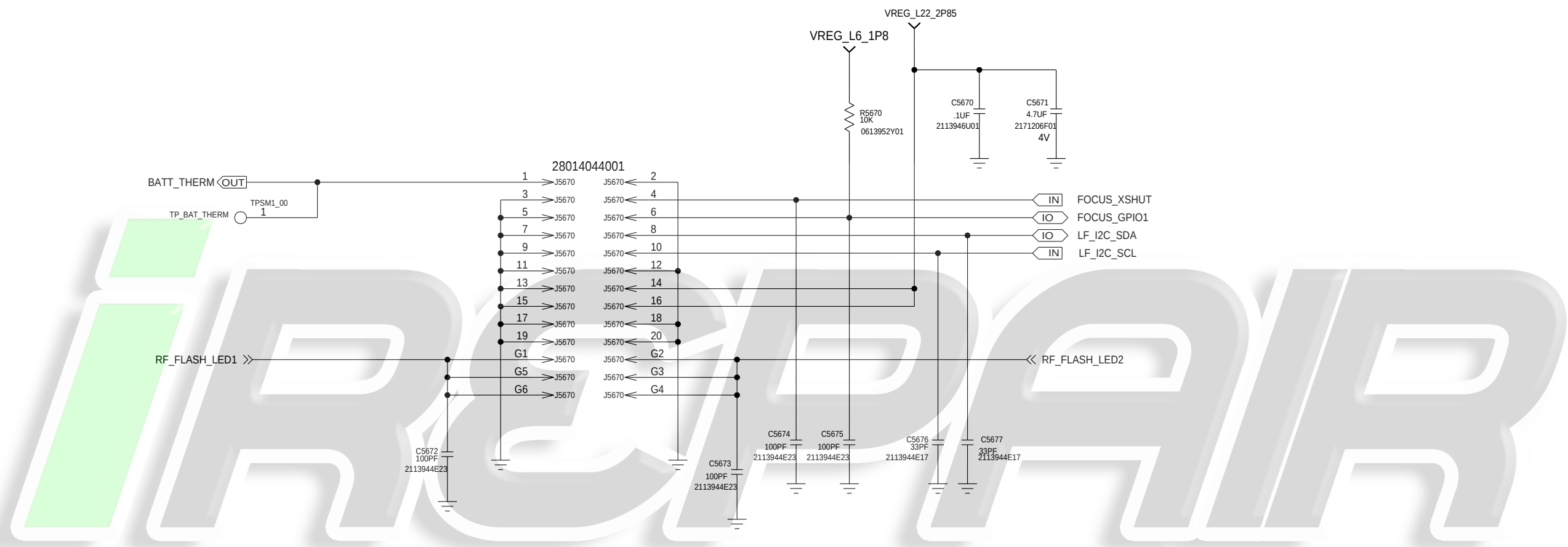
EMI/ESD FILTER



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: JOHN KRAMER		BLOCK TITLE: multimedia	
LAST MODIFIED BY: NATE CONNELL Thu Mar 10 11:01:22 2016		PROJECT NAME B 84016562001_06d.cpm	
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		PAGE 42 OF 61	

LASER FOCUS & FLASH CONN

REF:5670-5679

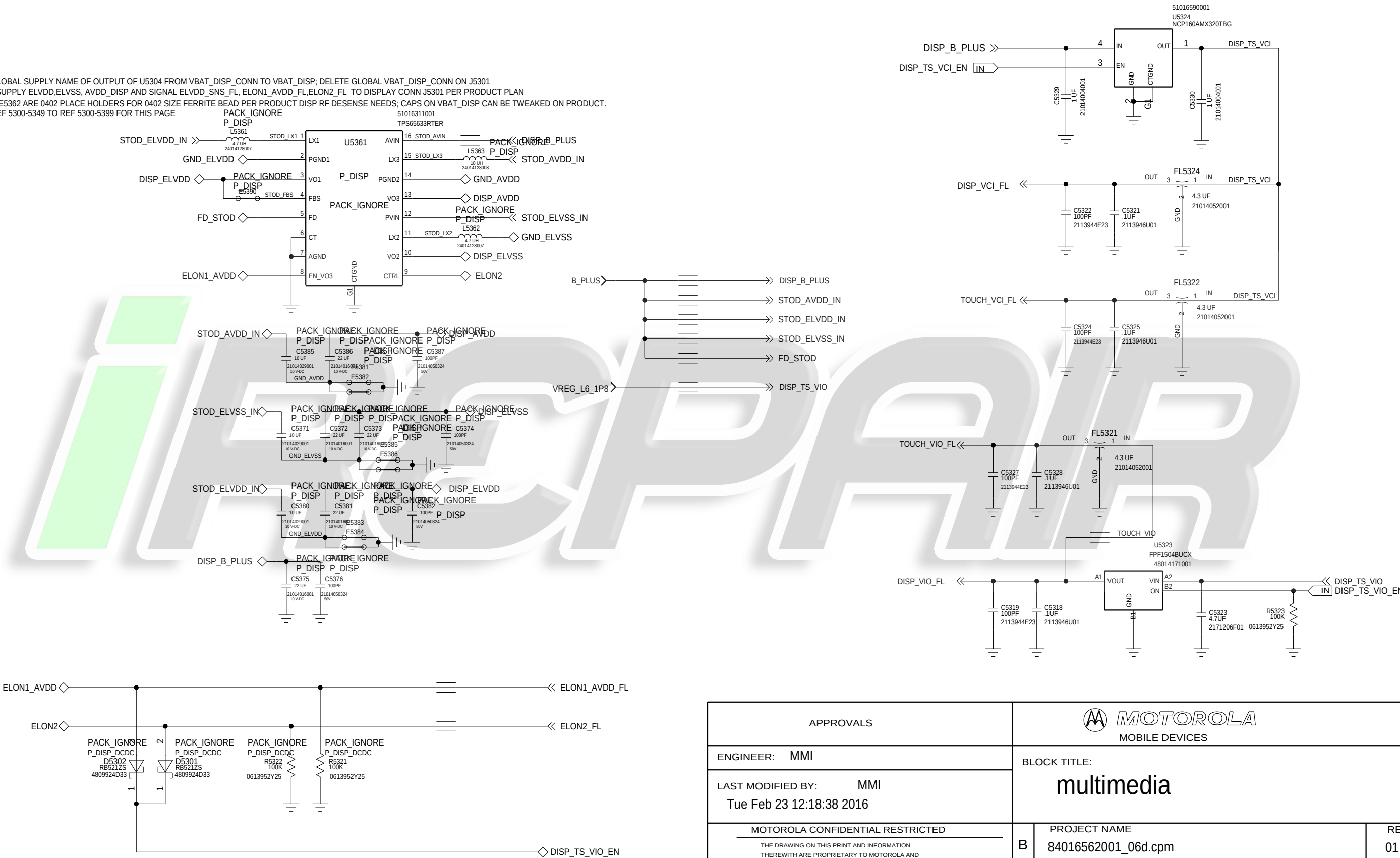


APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: RICH WOLNY		BLOCK TITLE: multimedia	
LAST MODIFIED BY: HUNG PHAM Fri Mar 25 16:05:42 2016			
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		CAD FILE NAME: PAGE 43 OF 61	

DISP: POWER

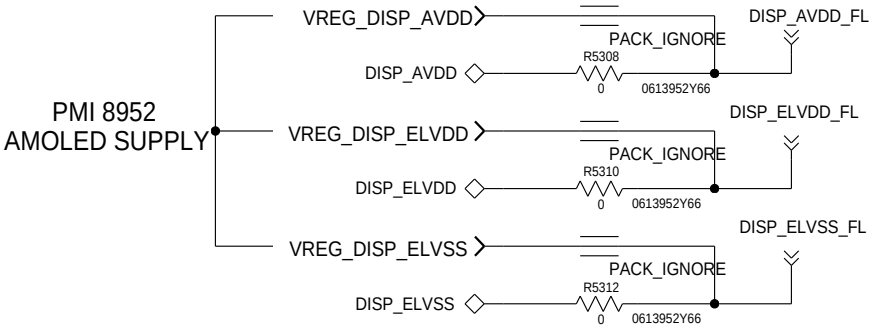
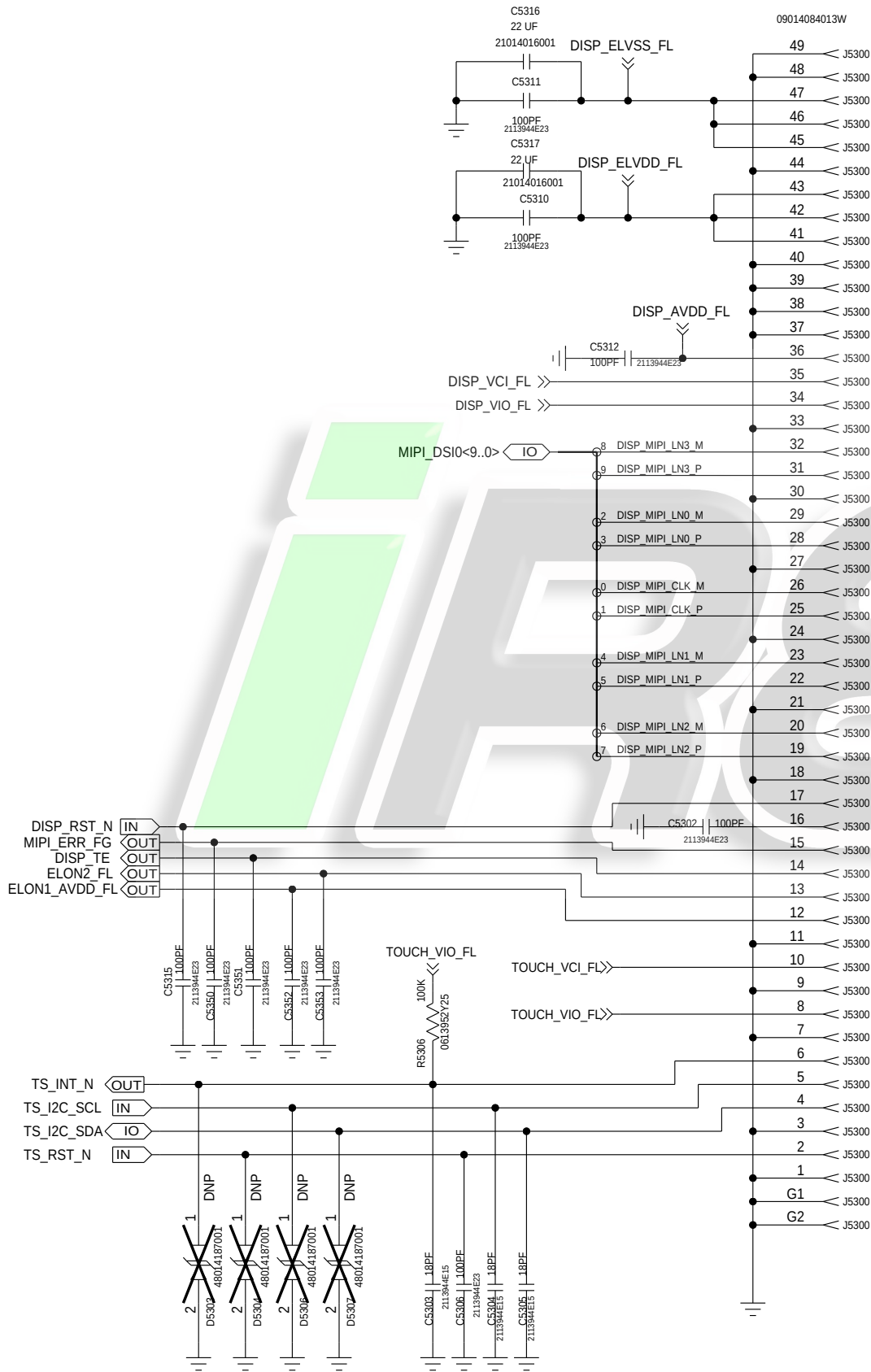
NOTES

- 1. CHANGE GLOBAL SUPPLY NAME OF OUTPUT OF U5304 FROM VBAT_DISP_CONN TO VBAT_DISP; DELETE GLOBAL VBAT_DISP_CONN ON J5301.
- 2. CONNECT SUPPLY ELVDD,ELVSS, AVDD_DISP AND SIGNAL ELVDD_SNS_FL, ELON1_AVDD_FL,ELON2_FL TO DISPLAY CONN J5301 PER PRODUCT PLAN
- 3. E5361 AND E5362 ARE 0402 PLACE HOLDERS FOR 0402 SIZE FERRITE BEAD PER PRODUCT DISP RF DESENSE NEEDS; CAPS ON VBAT_DISP CAN BE TWEAKED ON PRODUCT.
- 4. CHANGE REF 5300-5349 TO REF 5300-5399 FOR THIS PAGE



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MMI		BLOCK TITLE: multimedia	
LAST MODIFIED BY: MMI Tue Feb 23 12:18:38 2016		PROJECT NAME B 84016562001_06d.cpm	
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		PAGE 44 OF 61	

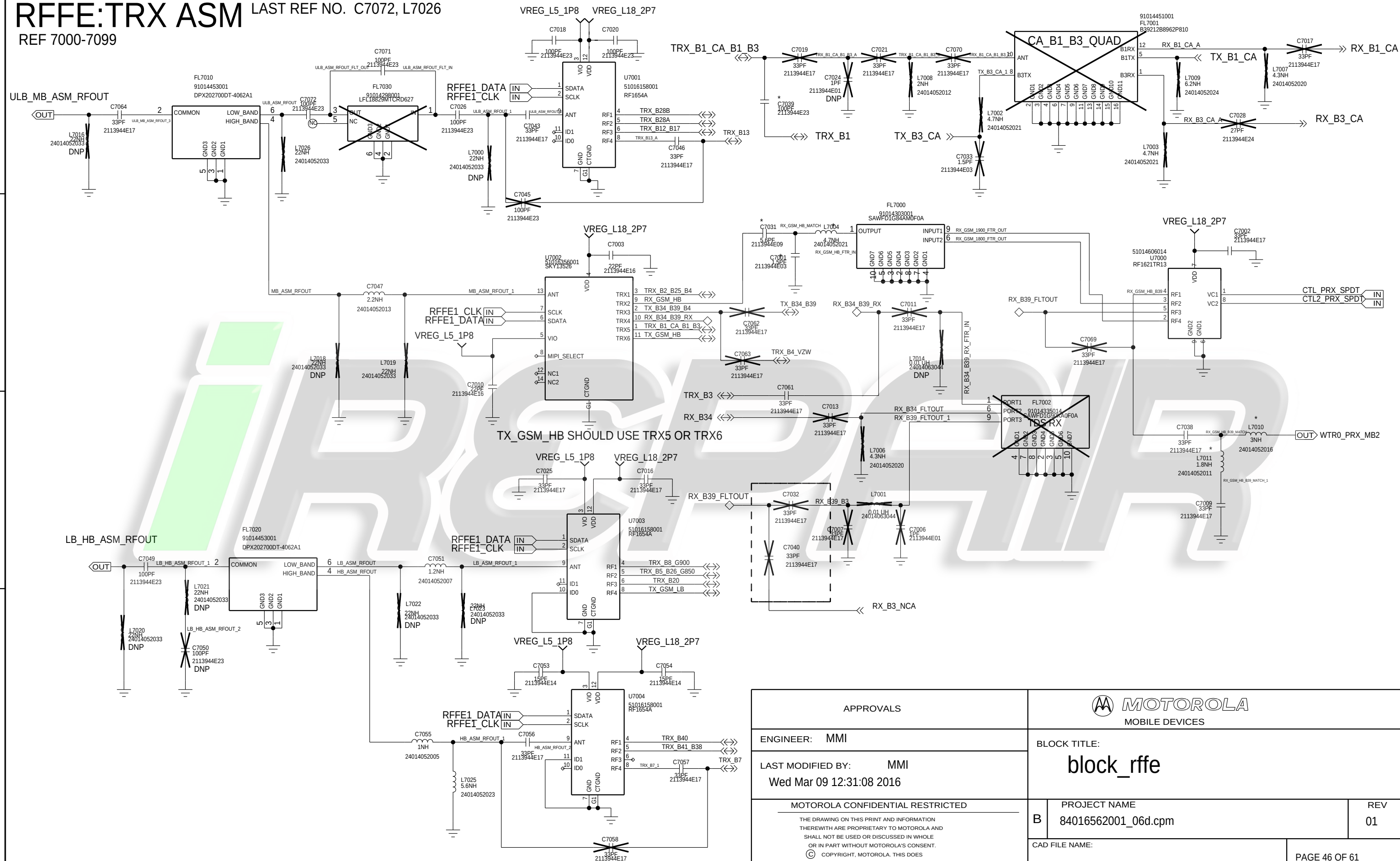
DISP: CONNECTOR



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MMI		BLOCK TITLE: multimedia	
LAST MODIFIED BY: MMI Tue Feb 23 12:18:30 2016		PROJECT NAME 84016562001_06d.cpm	
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		PAGE 45 OF 61	

REF 7000-7099

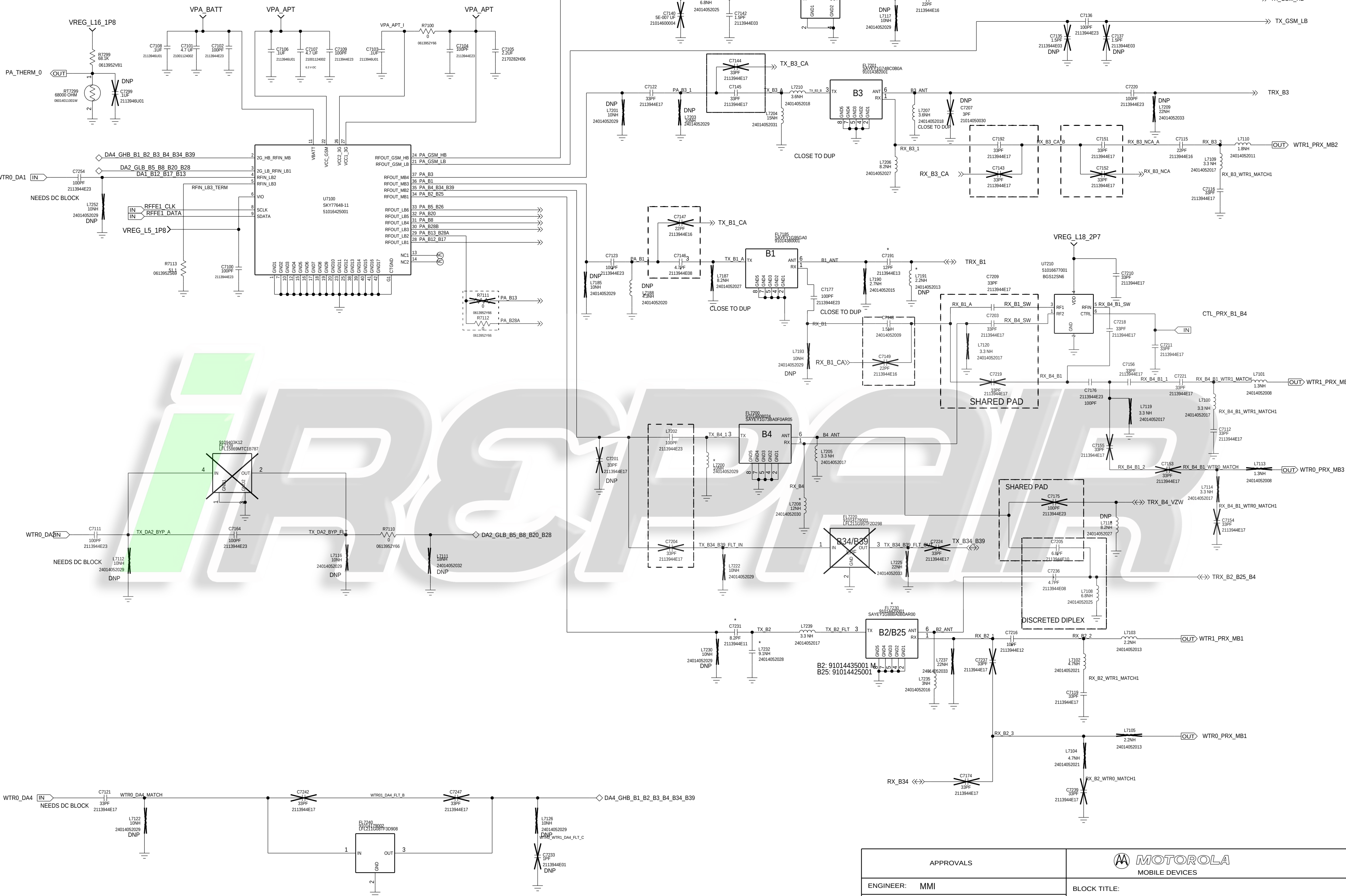
LAST REF NO. C7072, L7026



APPROVALS		 MOTOROLA MOBILE DEVICES	
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LAST MODIFIED BY: MMI Wed Mar 09 12:31:08 2016			
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		CAD FILE NAME:	PAGE 46 OF 61

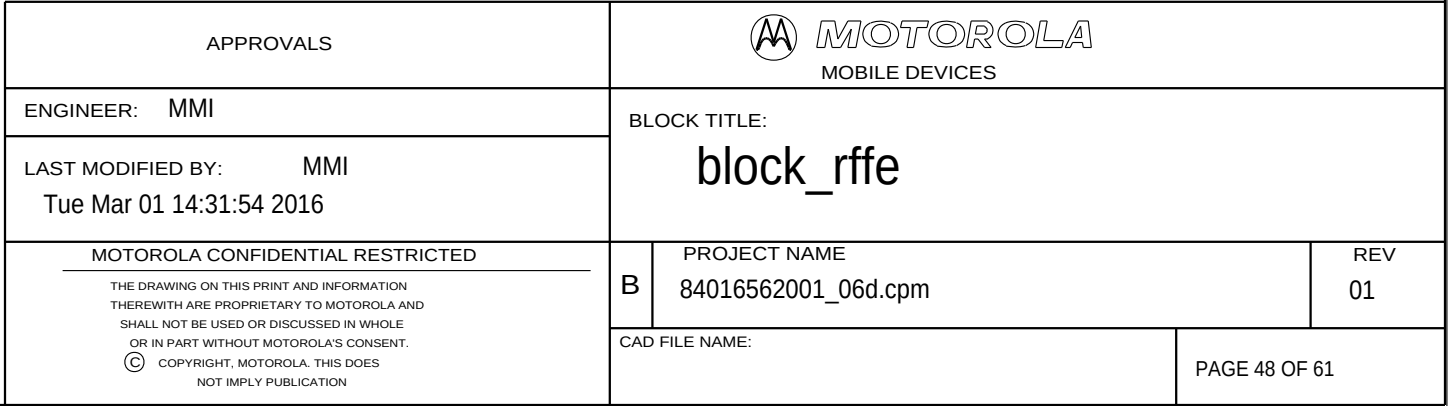
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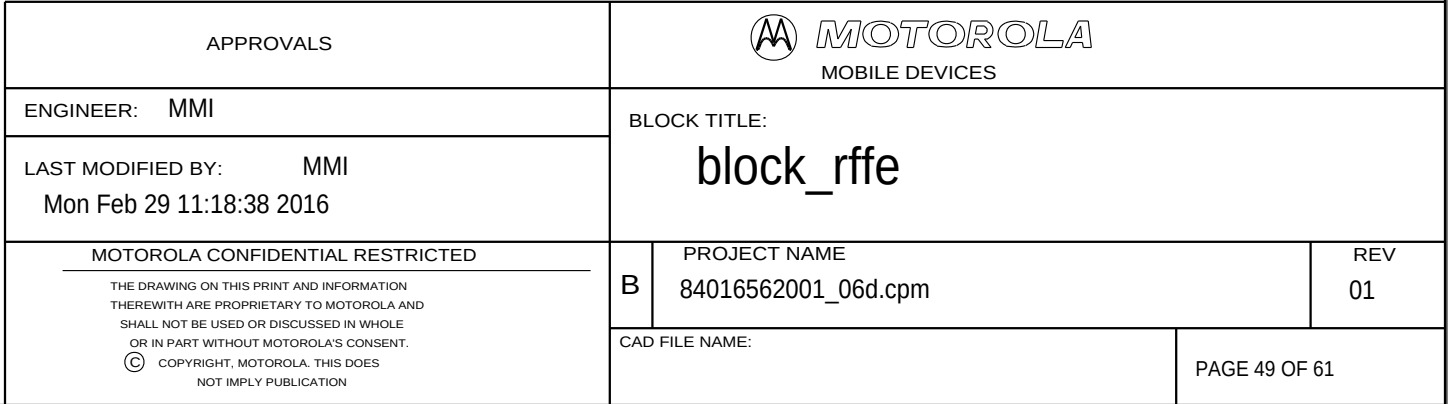


APPROVALS		REVISIONS	
ENGINEER:	MMI	ZONE	LTR
LAST MODIFIED BY:	MMI	DESCRIPTION	DATE
Fri Apr 15 10:20:49 2016			APPR.
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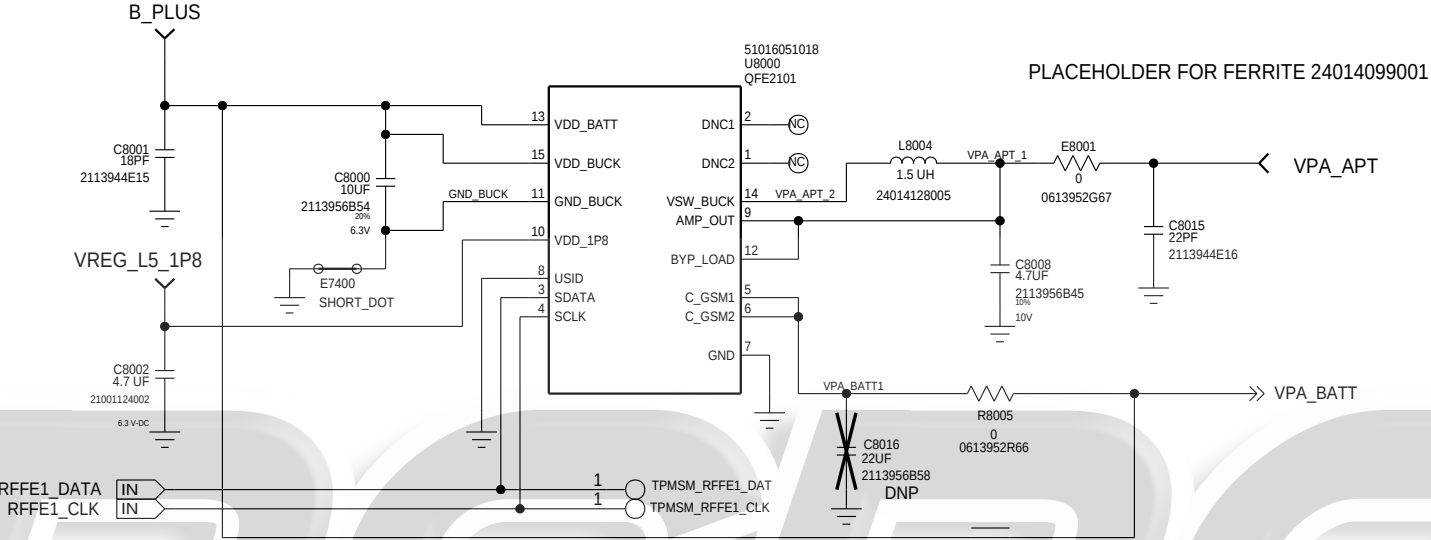
REF 7300-7399
LAST REF NO. C7396, L7365, R7371



LAST REF NO, C7454, L7449, R7416



RFFE:QFE2101
REF 7400-7499



E7400 : DO NOT VIA DIRECTLY TO GROUND PLANE. KEEP ISOLATED ON SURFACE LAYER;

BYPASS OPTION FOR VPA_VBATT
DNP APPROPRIATELY

APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MMI		BLOCK TITLE:	
LAST MODIFIED BY: MMI Tue Feb 23 12:19:02 2016		block_rffe	
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		B 84016562001_06d.cpm	REV 01
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		PAGE 50 OF 61	

REF 7500-7649
LAST REF NO. C7627, L7566

The schematic diagram illustrates the matching networks for the RX and TX paths of the WTR0 and WTR1 transistors.

WTR0 RX Path: The input signal, labeled "RX_B41_MATCH", is connected to the gate of the WTR0 transistor through a series combination of an inductor L7518 (24014052002) and a capacitor C7540 (530 PF). The output of the WTR0 transistor is labeled "OUT" and "WTR0_DRX_HB2".

WTR1 TX Path: The input signal, labeled "DRX_B7_MATCH", is connected to the gate of the WTR1 transistor through a series combination of an inductor L7524 (2.7nH) and a capacitor C7540 (530 PF). The output of the WTR1 transistor is labeled "OUT" and "WTR1_DRX_HB2".

Common Components: A common inductor L7524 (2.7nH) is connected between the output of the WTR0 transistor and the input of the WTR1 transistor. A common capacitor C7540 (530 PF) is connected between the input of the WTR0 transistor and the input of the WTR1 transistor.



NEED TO DECIDE THE NO. OF CAP BASED ON LAYOUT

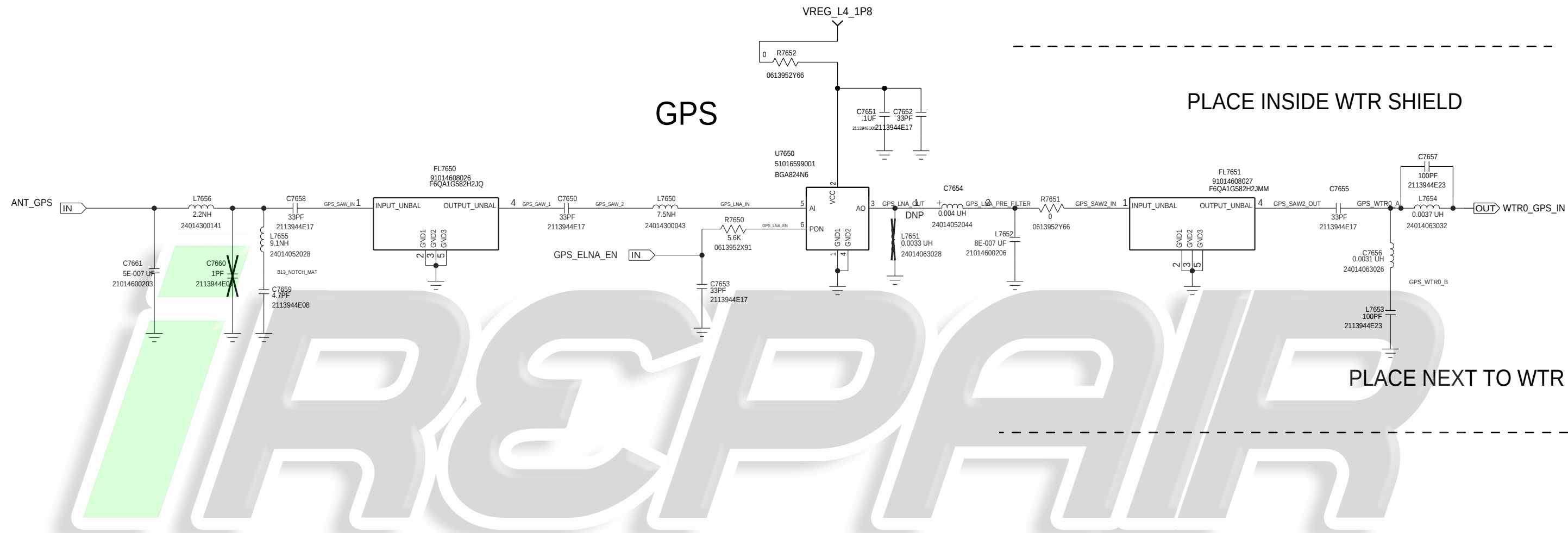


MID BANDS

GE 51 of 61

RFFE: GPS

REF 7650-7699

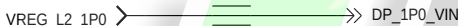
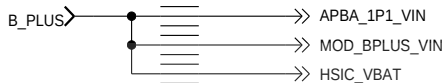


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CAD FILE NAME:			PAGE 52 OF 61

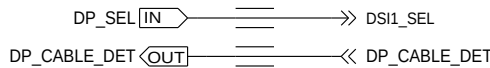
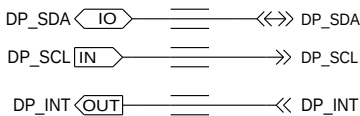
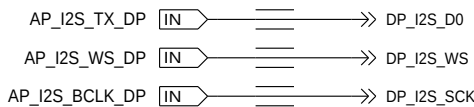
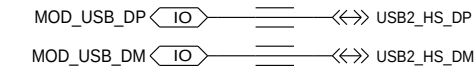
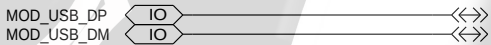
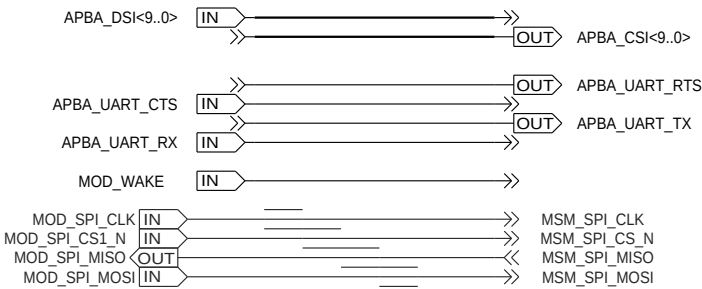
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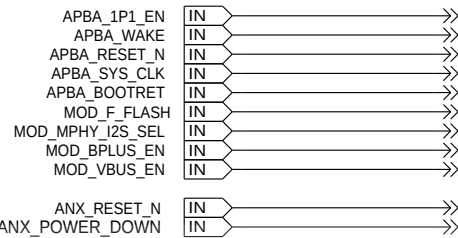
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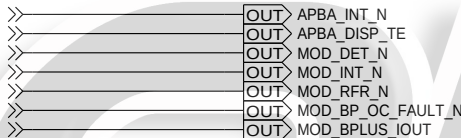
INTERFACES



GPIO - INPUT



GPIO - OUTPUT



APPROVALS

ENGINEER: NATE CONNELL

LAST MODIFIED BY: NATE CONNELL
Thu Apr 07 14:37:18 2016

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BLOCK TITLE:

slice_if

PROJECT NAME
B 84016562001_06d.cpm

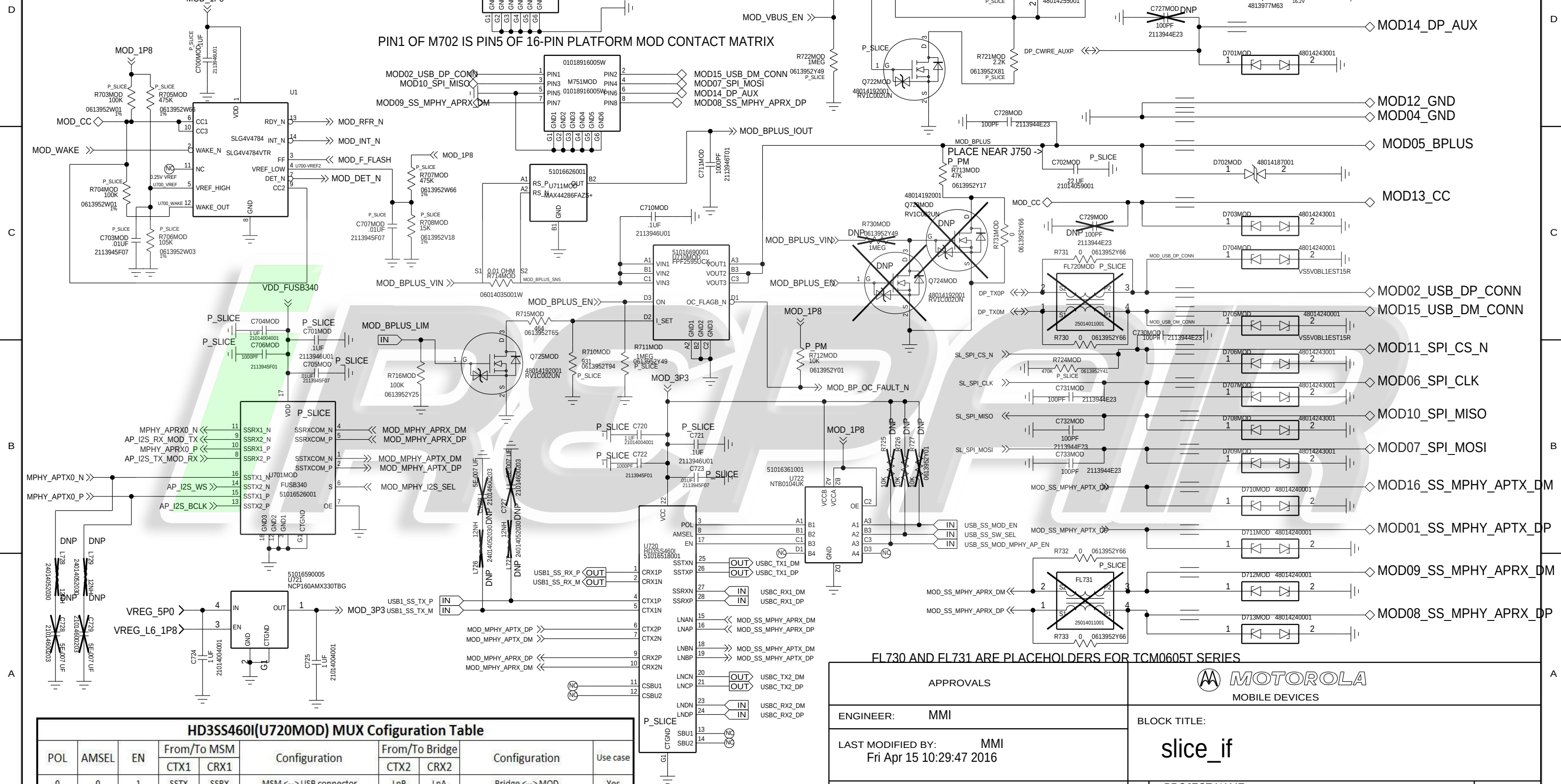
REV
01

CAD FILE NAME:

MODS: I/F

PIN1 OF M701 IS PIN1 OF 16-PIN PLATFORM MOD CONTACT MATRIX

PIN1 OF M702 IS PIN5 OF 16-PIN PLATFORM MOD CONTACT MATRIX



HD3SS460I(U720MOD) MUX Cofiguration Table

POL	AMSEL	EN	From/To MSM		Configuration	From/To Bridge		Configuration	Use case
			CTX1	CRX1		CTX2	CRX2		
0	0	1	SSTX	SSRX	MSM <--> USB connector	LnB	LnA	Bridge <--> MOD	Yes
1	0	1	LnB	LnA	MSM <--> MOD	SSTX	SSRX	Bridge <--> USB connector	Yes
0	1	1	LnC	LnD	MSM <--> USB connector	LnB	LnA	Bridge <--> MOD	Yes
1	1	1	LnB	LnA	MSM <--> MOD	LnC	LnD	Bridge <--> USB connector	Yes

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FL730 AND FL731 ARE PLACEHOLDERS FOR TCM0605T SERIES

APPROVALS

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PROJECT NAME

84016562001 06d.cpm

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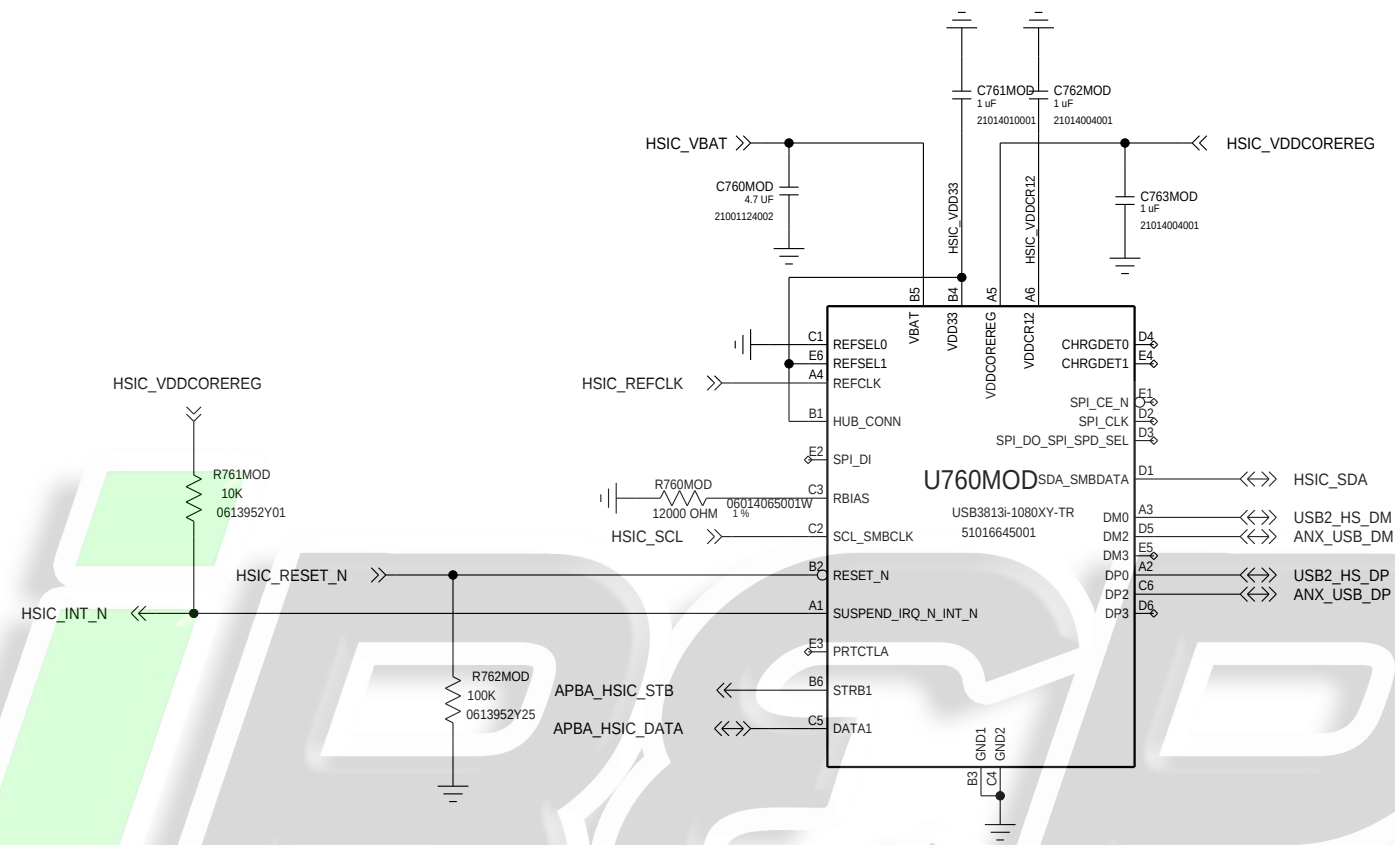
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2

A

MODS: HSIC



APPROVALS		MOTOROLA MOBILE DEVICES	
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LAST MODIFIED BY: Tue Feb 23 12:19:17 2016		PROJECT NAME B 84016562001_06d.cpm	
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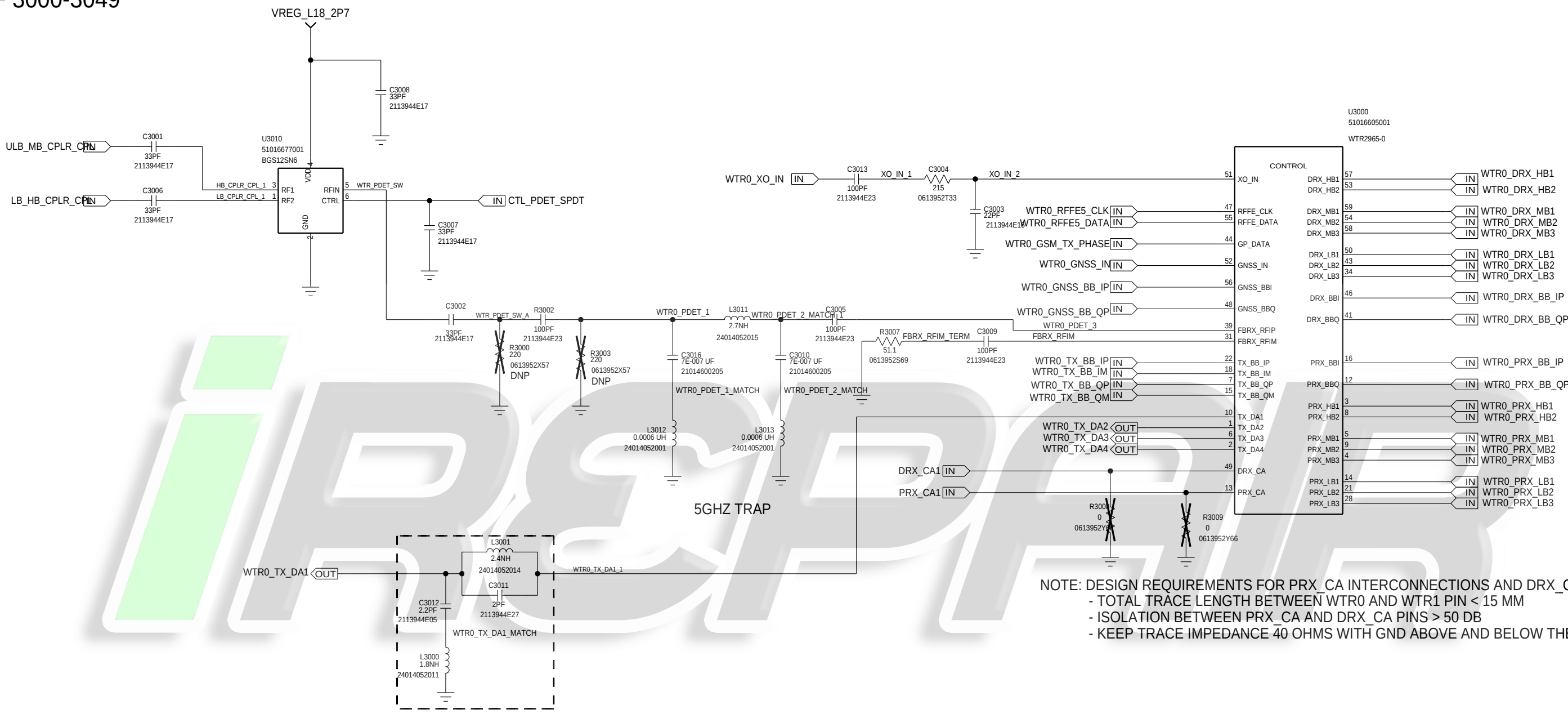
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8	7	6	5	4	3	2	
---	---	---	---	---	---	---	--

WTR2965 BB RF

REF 3000-3049



NOTE: DESIGN REQUIREMENTS FOR PRX_CA INTERCONNECTIONS AND DRX_CA INTERCONNECTIONS:

- TOTAL TRACE LENGTH BETWEEN WTR0 AND WTR1 PIN < 15 MM
- ISOLATION BETWEEN PRX_CA AND DRX_CA PINS > 50 DB
- KEEP TRACE IMPEDANCE 40 OHMS WITH GND ABOVE AND BELOW THESE TRACES

NOTE: THIS LC NOTCH IS REQUIRED TO MEET 4FMOD SPECS IN B13 BAND.

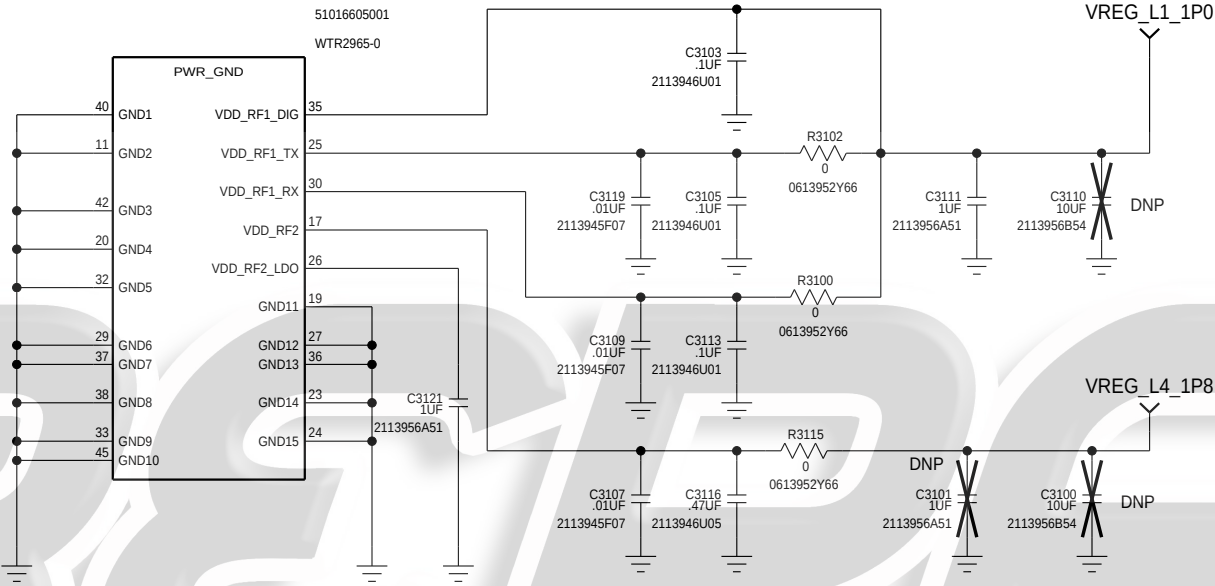
BASED ON 80-NP237-43 REV. A

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LAST MODIFIED BY: Thu Mar 17 12:11:41 2016		BLOCK TITLE: blk_wtr2965		PROJECT NAME: 84016562001_06d.cpm	
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WTR2965 POWER

REF 3100-3149

U3000



NOTE: C3119, C3109, C3107 ARE PRIORITY 1 AND SHOULD BE PLACED CLOSEST TO VDD PINS.

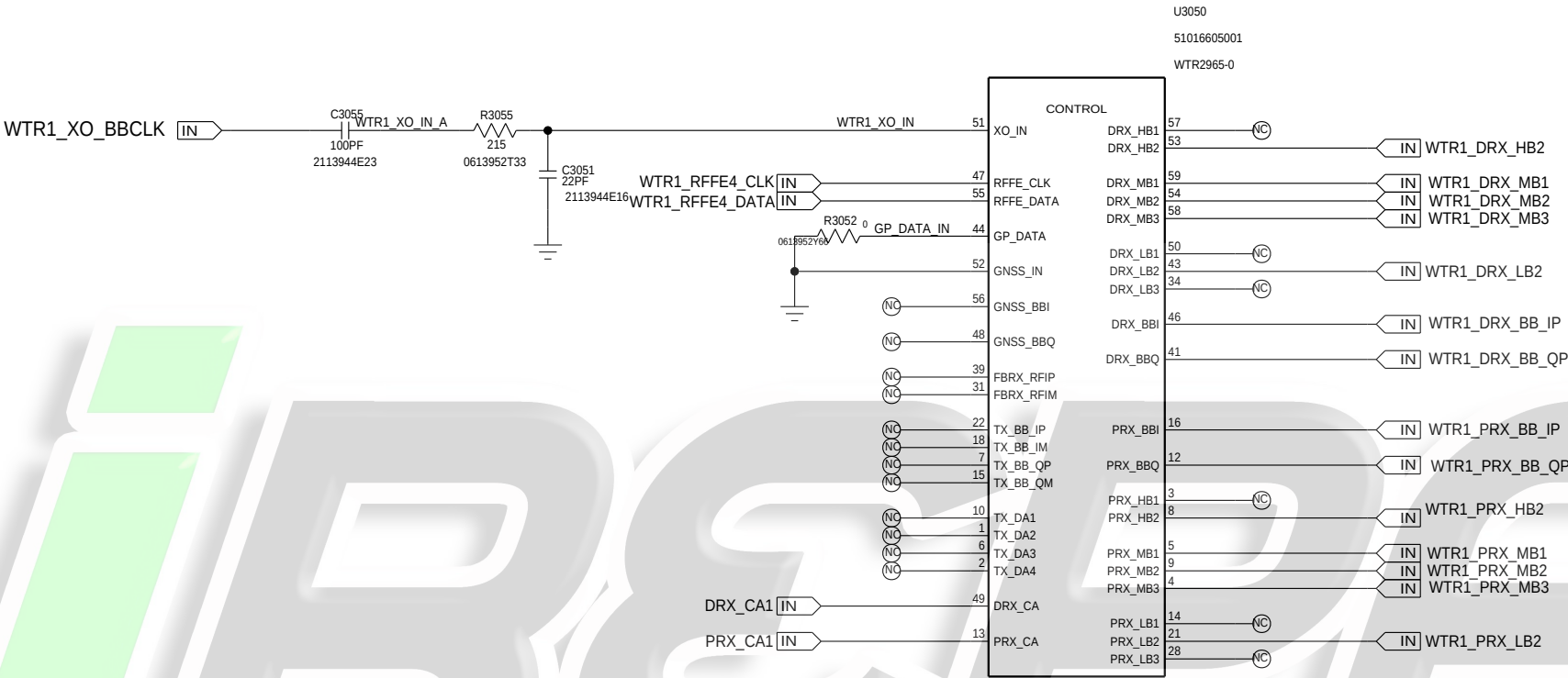
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		REV 01	
		CAD FILE NAME:	
		PAGE 59 OF 61	

CA_WTR2965 BB RF

REF 3050-3099



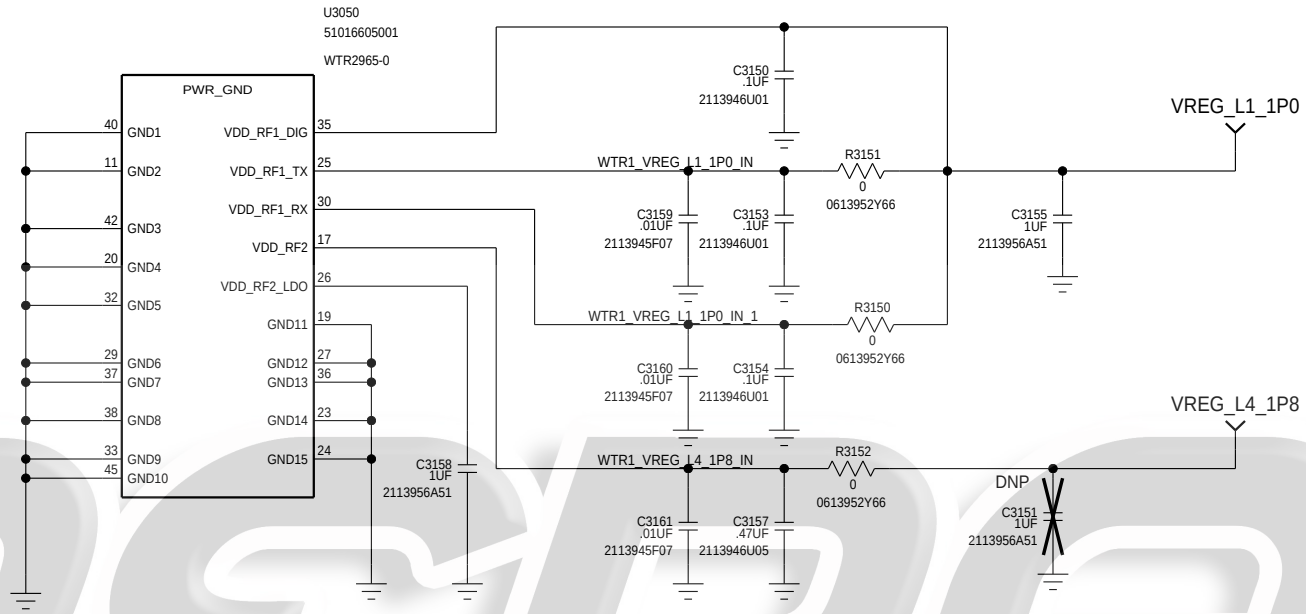
NOTE: DESIGN REQUIREMENTS FOR PRX_CA INTERCONNECTIONS AND DRX_CA INTERCONNECTIONS:

- TOTAL TRACE LENGTH BETWEEN WTR0 AND WTR1 PIN < 15 MM
- ISOLATION BETWEEN PRX_CA AND DRX_CA PINS > 50 DB
- KEEP TRACE IMPEDANCE 40 OHMS WITH GND ABOVE AND BELOW THESE TRACES

APPROVALS		 MOTOROLA MOBILE DEVICES	
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		B 84016562001_06d.cpm	REV 01
		CAD FILE NAME:	
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CA_WTR2965 POWER

REF 3150-3199



NOTE: C3159, C3160, C3161 ARE PRIORITY 1 AND SHOULD BE PLACED CLOSEST TO VDD PINS.

APPROVALS		 MOTOROLA MOBILE DEVICES	
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		CAD FILE NAME:	
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